

Rajshahi University Of Engineering And Technology

Department of Computer Science And Engineering

BlueStep Bank AB

Information System Analysis & Design Report



Bluestep Bank.

Course Code: CSE 4110

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Chapter 1

Recognition of Need

1.1 Introduction

Bluestep Bank is a leading specialized mortgage institution in the Nordic region, operating as a modern alternative to traditional universal banks. Headquartered in Stockholm, Sweden, the bank focuses on providing tailored financial solutions—primarily mortgages, refinancing options, and savings products—to individuals who are often underserved by conventional banking systems. These customers typically include self-employed individuals, freelancers, or those with non-standard income streams and limited credit histories, who may not meet the rigid lending criteria of traditional banks. By addressing this gap, Bluestep Bank plays a significant role in promoting financial inclusion across Sweden, Norway, and Finland.

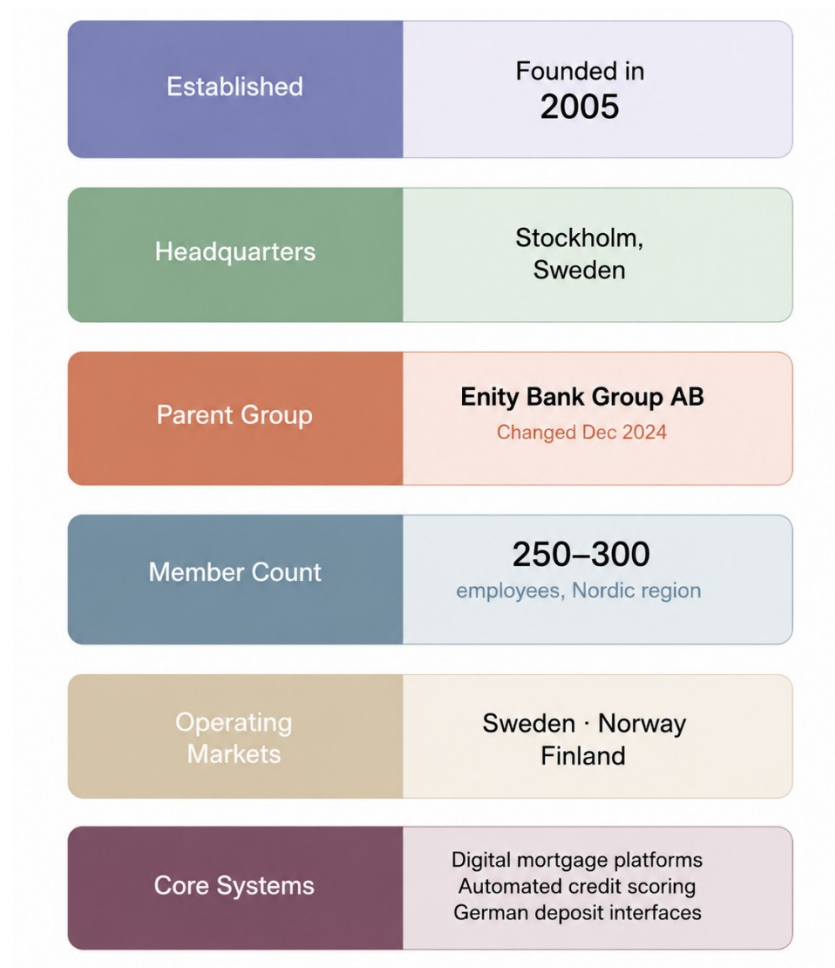


Figure 1.1: About BlueStep Bank

From an Information System Analysis & Design (ISAD) perspective, Bluestep Bank represents a highly relevant case of a *digital-first banking model*. Unlike traditional financial institutions that rely heavily on physical branches and manual processes, Bluestep Bank operates through a centralized digital infrastructure. This system supports end-to-end banking operations, including customer onboarding, loan application processing, credit risk assessment, and savings account management. The reliance on digital platforms allows the bank to reduce operational costs, increase processing speed, and maintain consistency across multiple geographic regions.

Bluestep Bank's vision is to create an inclusive financial society where more individuals can achieve their economic potential through responsible and transparent banking services. The bank aims to ensure that access to financial opportunities—particularly housing finance—is not limited only to customers who meet traditional banking criteria but is available to a broader segment of society.

A key aspect of Bluestep Bank's operational model is its use of *integrated information systems* to handle complex financial decision-making processes. Credit evaluation, for instance, goes beyond traditional credit scoring models by incorporating alternative data sources and flexible risk assessment frameworks. This requires robust system design capable of handling diverse datasets, real-time processing, and secure data management. Furthermore, the bank's cross-border operations demand high levels of system interoperability and compliance with varying regulatory requirements in different countries.

Bluestep Bank is not only a financial service provider but also a technology-driven organization where information systems play a central role in delivering value. Its operations highlight the critical importance of system analysis and design in ensuring efficiency, scalability, security, and customer satisfaction in the modern banking environment. This makes it an ideal subject for studying the application of ISAD principles in real-world financial systems.

1.1.1 Organizational DNA and Core Philosophy:

The organizational DNA of **Bluestep Bank** is fundamentally rooted in the principles of **financial inclusion, flexibility, and customer-centric innovation**. Unlike many traditional banking institutions that rely heavily on rigid credit scoring models and standardized lending criteria, Bluestep Bank adopts a more adaptive and personalized approach to financial services. The bank's philosophy focuses on evaluating customers' broader financial circumstances rather than relying solely on conventional credit metrics. This allows the institution to provide financial opportunities to individuals who are often underserved by traditional banks, including self-employed professionals, freelancers, entrepreneurs, and customers with irregular income streams.

From an **Information System Analysis & Design (ISAD)** perspective, Bluestep Bank follows a **digital-first and data-driven operational model**. Instead of relying on traditional branch-based banking, the organization operates through centralized digital platforms and integrated information systems that support its core services.

The key characteristics that define Bluestep Bank's organizational philosophy include:

- i. **Financial Inclusion:** Providing mortgage and financial services to individuals who do not meet traditional banking criteria.
- ii. **Flexible Risk Assessment:** Evaluating loan applications using broader financial indicators rather than relying only on conventional credit scores.
- iii. **Digital-First Operations:** Managing most banking services through online platforms and centralized digital infrastructure.
- iv. **Automation and Efficiency:** Using automated systems for processes such as customer onboarding, loan processing, and account management.
- v. **Customer-Centric Innovation:** Designing services that prioritize accessibility, convenience, and personalized financial solutions.

Another important element of Bluestep Bank’s philosophy is its emphasis on **scalable and secure system architecture**. The bank’s information systems are designed to support large volumes of financial data, digital transactions, and regulatory compliance requirements across multiple countries such as Sweden, Norway, and Finland. These systems ensure efficient data processing, secure information storage, and smooth integration between different operational platforms.

Furthermore, Bluestep Bank follows a philosophy of **“systems over manual processes.”** This means that most core banking activities—including identity verification, credit evaluation, loan approval workflows, and account management—are executed through integrated digital systems rather than manual administrative procedures.

Because Bluestep Bank operates without a traditional branch-based infrastructure, information systems play a critical role in managing the entire customer lifecycle. From initial application to loan servicing, all interactions are conducted through digital platforms. As a result, the bank serves as a strong example of how modern financial institutions leverage advanced information system design to deliver efficient, scalable, and inclusive banking services.

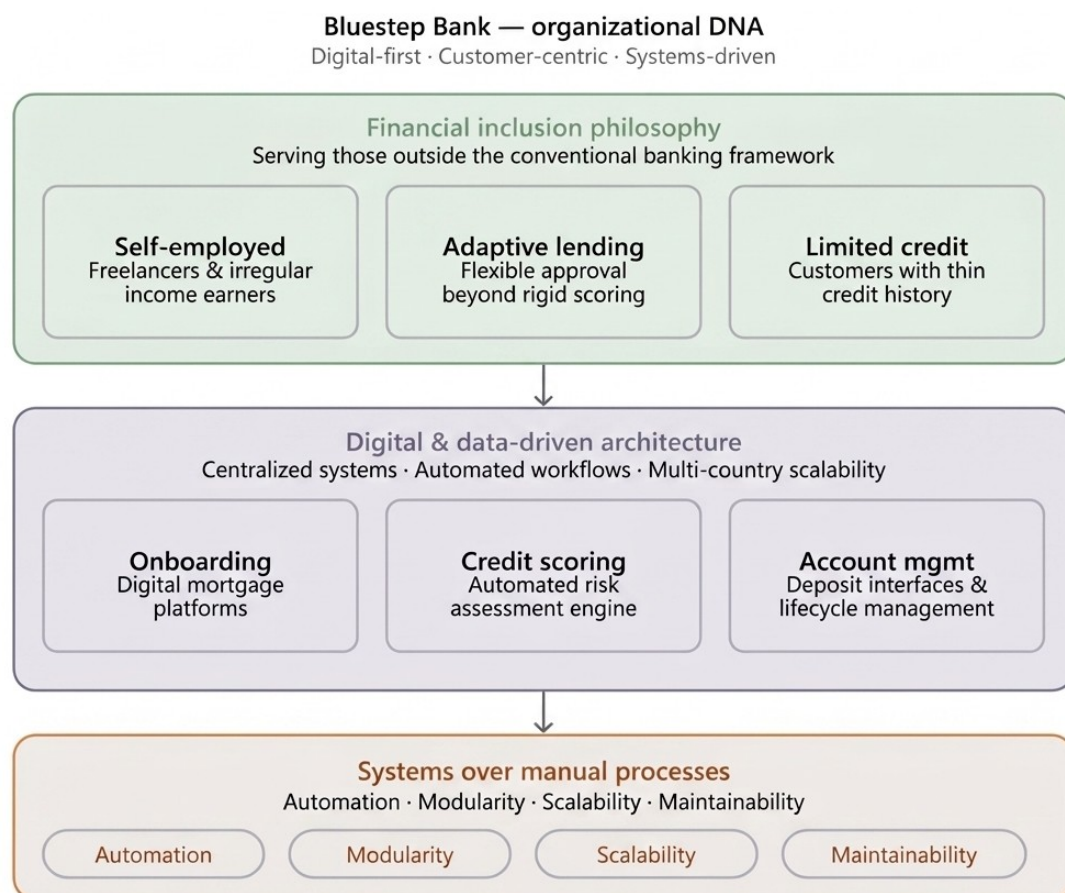


Figure 1.2: Organizational philosophy of Bluestep AB

1.1.2 Industry Diversification and Market Penetration

Bluestep Bank demonstrates a focused yet strategically diversified approach to industry participation, supported heavily by its information systems and digital infrastructure. Unlike traditional banks that diversify across a wide range of financial products (such as corporate banking, investment services, and insurance), Bluestep Bank maintains a specialized domain focus primarily mortgages, refinancing, and savings products—while achieving diversification through customer segmentation and service adaptability rather than product expansion.

From a System Analysis & Design (ISAD) perspective, this type of diversification is enabled by flexible and configurable software systems. Instead of building entirely separate systems for each service, Bluestep Bank utilizes a modular system architecture, where core components such as customer data management, risk assessment engines, and transaction processing systems are reused and adapted across different services. This reduces development complexity while allowing the organization to serve multiple customer profiles efficiently.

In terms of market penetration, Bluestep Bank operates across multiple Nordic countries, including Sweden, Norway, and Finland. This cross-border presence is not just a business expansion strategy but also a technical challenge that requires scalable and interoperable systems. Each country has its own regulatory requirements, financial standards, and customer behavior patterns, which must be handled through localized system configurations without compromising the integrity of the central system.

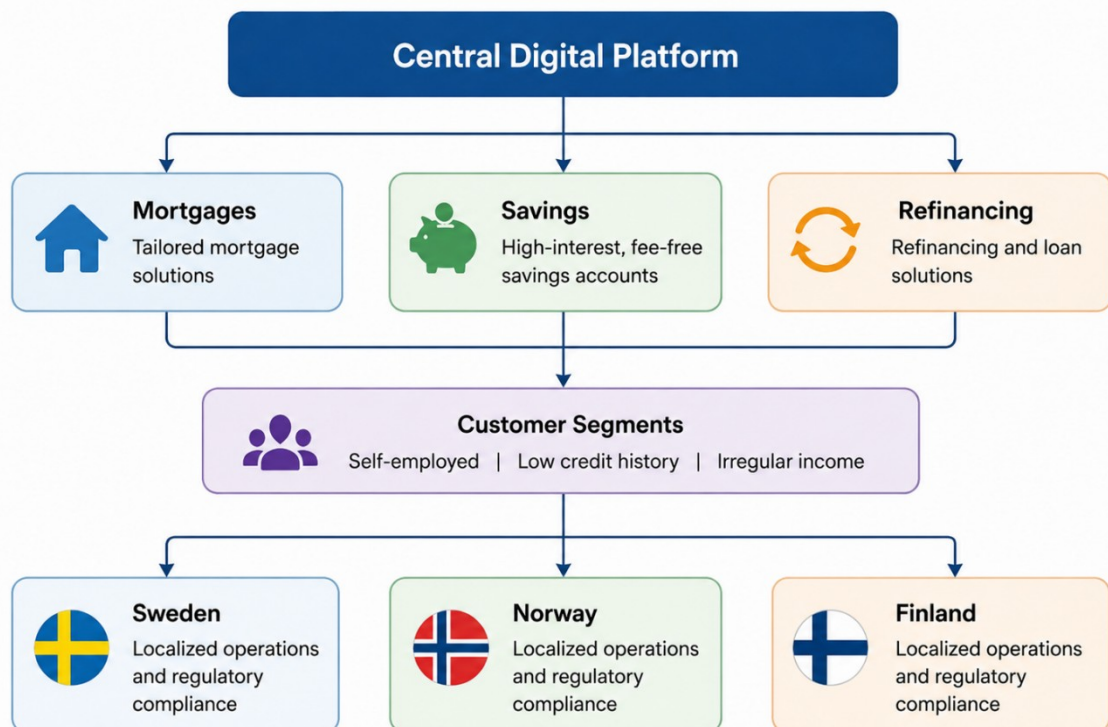


Figure 1.3: Industry Diversification & Market Penetration Model

1.1.3 Organizational Structure and Hierarchy

Bluestep Bank employs a technology-driven, agile hierarchy designed to align strategic vision with rapid digital innovation. This structure ensures that system design and software development remain focused on scalability and efficiency.

Executive & Strategic Leadership

The CEO and CTO define the bank's long-term vision. From a System Analysis & Design (ISAD) perspective, the CTO is pivotal in translating business goals into technical roadmaps, such as cloud migration and AI-driven credit risk modeling.

Specialized Departmental Heads

Domain-specific leaders convert these strategies into operational reality:

- **Head of Product:** Defines roadmaps and prioritizes customer-centric features.
- **Head of Technology & Architecture:** Oversees backend reliability and ensures scalable system design.
- **Head of Analytics & Engineering:** Manages data infrastructure and vital decision-making algorithms.
- **Head of Operations:** Maintains internal IT platforms and core infrastructure.

Management & Execution

The structure bridges the gap between business and technology through:

- **Product Owners (POs):** Manage backlogs and translate stakeholder needs into user stories.
- **Engineering Managers (EMs):** Provide technical guidance and ensure architectural compliance.

Agile Squad Structure

At the operational level, Bluestep utilizes **autonomous, cross-functional squads**. Each squad typically includes a PO, an EM, and a mix of frontend, backend (C#/.NET), and data engineers. This squad-based model minimizes bottlenecks, allowing for independent testing and continuous delivery using tools like Jira and Azure DevOps.

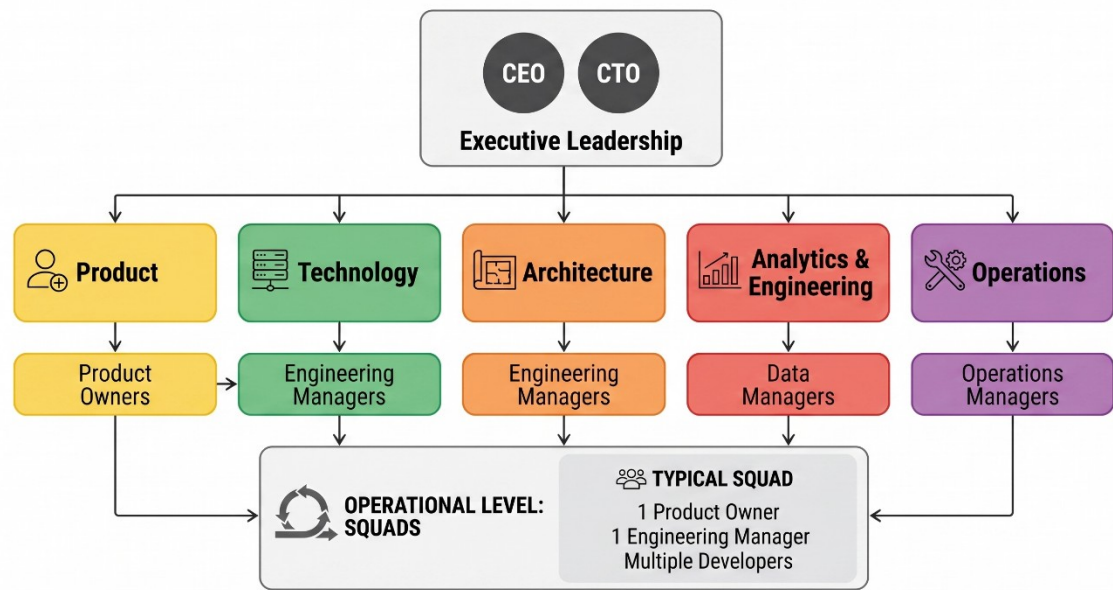


Figure 1.4: Organizational Structure of BlueStep AB

1.2 Problem Identification:

1.2.1 Problem 01: Cross-Departmental Communication Gaps

One of the major challenges observed is ineffective communication between different technical teams. In a system where multiple specialized units (such as frontend, backend, and API teams) operate independently, dependencies often create bottlenecks. For example, if the frontend team requires a new API, they must wait until the backend team schedules and implements it. This delay slows down development cycles and reduces overall system efficiency. Additionally, excessive reliance on documentation to bridge communication gaps further reduces agility and increases project completion time.

1.2.2 Problem 02: Ambiguous Requirements

Another critical issue is the lack of clarity in system requirements provided by business stakeholders or compliance teams. Often, requirements—especially those derived from legal or regulatory contexts—are vague and difficult to translate into precise technical specifications. This creates confusion during system design and development, leading to multiple revisions and rework. As a result, project timelines are extended, and system quality may be affected due to inconsistent implementation.

1.2.3 Problem 03: External API Fragility

The system heavily depends on external APIs, such as government identity verification services (e.g., NID or BankID). These integrations introduce a significant risk, as the system's functionality becomes dependent on third-party service availability. If an external system experiences downtime or performance issues, critical operations—such as user authentication or onboarding—are disrupted. This negatively impacts user experience and reduces system reliability, even though the issue lies outside the organization's direct control.

1.2.4 Problem 04: Data Integrity and Fraud Risks

Handling large volumes of data introduces challenges related to data accuracy and security. Issues such as missing inputs, user errors, and fraudulent data submissions can compromise system integrity. Additionally, relying heavily on message-based validation processes increases computational overhead and slows down data processing. Without robust validation and fraud detection mechanisms, the system becomes vulnerable to inconsistencies and potential exploitation.

1.3 Conclusion

In conclusion, the analysis of Bluestep Bank from an Information System Analysis & Design (ISAD) perspective highlights that, while the organization successfully operates as a digital-first and technology-driven institution, it faces several critical operational and technical challenges. Issues such as cross-departmental communication gaps, ambiguous requirements, dependency on external APIs, and data integrity risks significantly impact system efficiency, reliability, and scalability. These challenges are not uncommon in modern software-centric organizations, especially those undergoing rapid expansion and system integration. However, in a highly regulated and data-sensitive environment like digital banking, such problems can lead to delays in system development, reduced user satisfaction, and increased operational risk if not properly addressed.

The findings emphasize the importance of implementing clear requirement engineering practices, improved inter-team communication frameworks, robust API management strategies, and advanced data validation mechanisms. Addressing these areas through effective system design and organizational alignment will enhance Bluestep Bank's ability to deliver reliable, scalable, and secure digital services.

Overall, this analysis demonstrates that strong alignment between organizational structure and information system design is essential for sustaining performance and supporting future growth in a digital-first enterprise.

Chapter 2

Initial Feasibility Study

2.1 Introduction

The Initial Feasibility Study is a critical phase in the System Analysis & Design (ISAD) process, aimed at evaluating whether a proposed system or improvement is viable before committing significant resources. For [Bluestep Bank](#), this step is particularly important due to its fully digital, system-dependent operations and its presence across multiple regulatory environments in the Nordic region.

The purpose of this feasibility study is to assess the practicality of enhancing or redesigning existing information systems to address identified challenges such as communication gaps, ambiguous requirements, external API dependencies, and data integrity risks. The study considers multiple dimensions including technical capability, operational impact, economic cost, and organizational readiness. By conducting this analysis, Bluestep Bank can ensure

that any proposed system improvements align with its digital-first strategy while maintaining efficiency, scalability, and compliance.

2.2 Initial Feasibility Analysis

The feasibility analysis indicates that the proposed system improvements for Bluestep Bank are highly viable across multiple dimensions. From a technical perspective, the bank's existing cloud-based and modular architecture supports the integration of enhancements such as improved API management and data validation mechanisms without requiring a complete system overhaul. Operationally, the organization's agile, squad-based structure enables smooth adoption of new processes, although some training and coordination improvements may be necessary to address communication gaps. Economically, while initial investments in tools, infrastructure, and workforce training are required, these costs are justified by long-term benefits such as increased efficiency, reduced system downtime, and improved user satisfaction. In terms of scheduling, the use of iterative development methodologies allows these improvements to be implemented gradually through sprint cycles, minimizing disruption to ongoing operations. Overall, the analysis confirms that the proposed changes are feasible, sustainable, and aligned with the bank's digital-first strategy.

2.2.1 Problem 01: Cross-Departmental Communication Gaps

2.2.1.1 Statement of the Problem

In Bluestep Bank, communication gaps between product, development, and data teams create delays in system development and reduce operational efficiency. These gaps lead to dependency bottlenecks, slower decision-making, and misalignment between teams.

2.2.1.2 Summary of Findings

The analysis indicates that cross-departmental communication challenges mainly arise from insufficient coordination and limited transparency between agile squads. Since multiple teams work simultaneously on interconnected components, the lack of efficient communication channels often creates delays in task completion and dependency resolution.

One major issue is the limited use of real-time collaboration practices. Teams frequently rely on documentation and asynchronous communication instead of direct interaction, which slows down clarification and decision-making processes. As a result, misunderstandings may remain unresolved for extended periods, negatively affecting development speed.

Another important finding is the lack of visibility into the progress, priorities, and workloads of other teams. Because squads operate semi-independently, teams may not always be aware of changes or blockers affecting dependent tasks. This creates coordination difficulties and causes delays in feature integration and testing.

The analysis also reveals that dependency management between teams is not always handled proactively. When one team depends on APIs, data pipelines, or backend services from another team, delays in one department directly affect the workflow of others. This increases idle time, reduces productivity, and slows overall project delivery.

Furthermore, inconsistent communication practices across departments contribute to unequal information sharing. Some teams conduct regular synchronization meetings, while others rely heavily on tickets and documentation, resulting in fragmented collaboration. These communication inefficiencies collectively reduce organizational agility, increase development time, and impact the timely delivery of banking services and system enhancements.

2.2.1.3 Details of Findings

A deeper investigation shows that Bluestep Bank's distributed and specialized team structure, while efficient for scalability, introduces communication challenges. Product Owners define requirements, but these are sometimes not clearly interpreted by development teams due to insufficient interaction. Similarly, backend teams may prioritize tasks differently from frontend needs, causing delays in API delivery. Data teams handling credit scoring and external integrations may also operate independently, leading to synchronization issues.

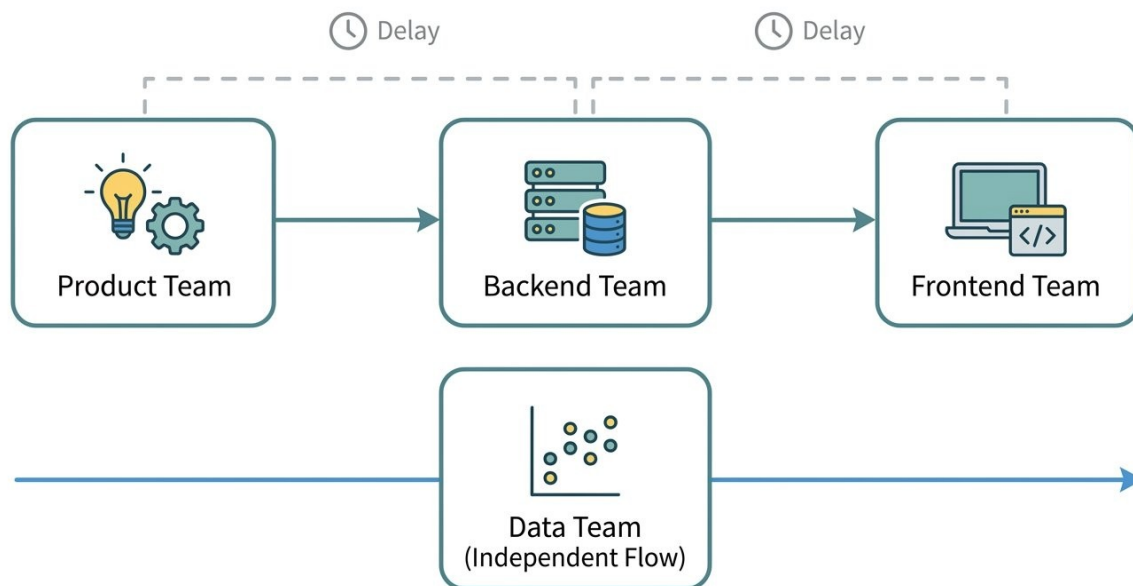


Figure 2.1: Communication Gap Between Departments

Furthermore, the reliance on tools such as Jira and documentation platforms, while useful, cannot fully replace *real-time communication and collaborative decision-making*. As a result, misunderstandings occur, leading to rework, inconsistencies in system implementation, and extended development cycles.

The table below summarizes the key communication gap scenarios observed within Bluestep Bank, along with their root causes and the resulting operational impact. These findings highlight how different failure points in cross-team communication translate into tangible system-level consequences that affect overall delivery performance.

Table 01: Communication Gaps table

Communication Issue	Root Cause	Operational Impact
No Real-Time Collaboration	Siloed team structure	Delayed feature delivery
Over-Reliance on Documentation	Lack of direct interaction	Increased rework and inconsistencies
Dependency Resolution Delays	Unclear inter-team ownership	Sprint bottlenecks and missed deadlines
Limited Cross-Team Visibility	Absence of shared dashboards	Misaligned priorities across squads

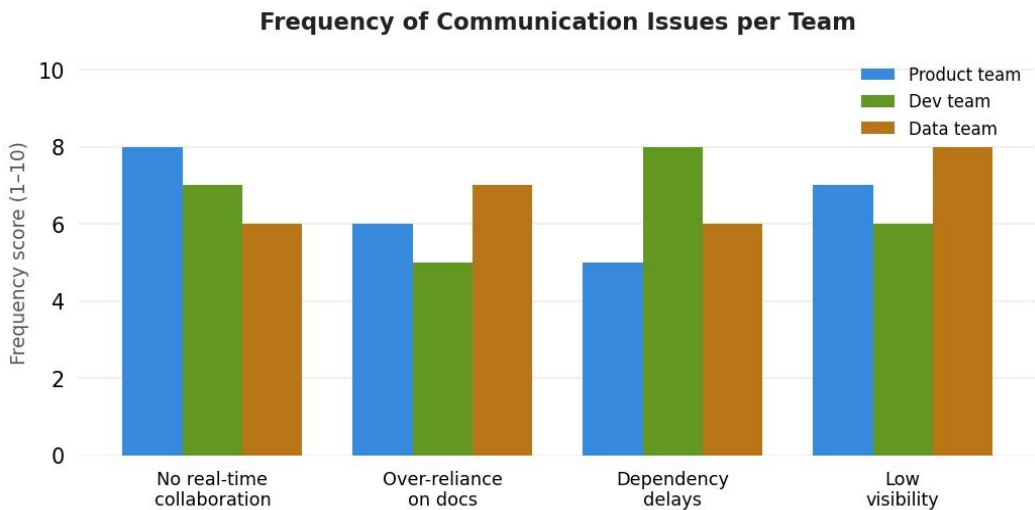


Figure 2.2: Frequency of Communication Issues per Team

2.2.1.4 Recommendation

To address cross-departmental communication gaps, Bluestep Bank should implement the following improvements:

- **Adopt stronger Agile communication practices:** Daily stand-ups, sprint reviews, and cross-team sync meetings should be enforced consistently.
- **Introduce cross-functional collaboration tools:** Use shared dashboards and real-time tracking systems to improve visibility across teams.

- **Implement API-first development approach:** Backend teams should prioritize API availability to reduce frontend dependency delays.
- **Encourage direct communication channels:** Reduce over-reliance on documentation by promoting quick discussions via tools like Slack or Microsoft Teams.
- **Establish clear dependency management processes:** Define timelines and responsibilities for inter-team deliverables to avoid bottlenecks.

By improving communication frameworks and collaboration practices, Bluestep Bank can significantly enhance system efficiency, reduce delays, and support faster, more reliable software delivery.

Problem 02: Ambiguous Requirements

2.2.2.1 Statement of the Problem

In **Bluestep Bank**, ambiguous and unclear system requirements create difficulties in the software development process. Requirements from business, compliance, and regulatory teams are often not detailed enough for direct implementation, leading to misunderstandings, repeated revisions, and project delays.

2.2.2.2 Summary of Findings

The analysis reveals that requirement-related challenges mainly arise from insufficient clarification and weak coordination between business stakeholders and technical teams. Since **Bluestep Bank** operates in a highly regulated digital banking environment, requirements frequently involve complex business rules and compliance conditions that are difficult to translate directly into technical specifications.

Several key findings were identified:

- **Requirements are often high-level and lack technical detail**
Business requirements are sometimes written from a functional perspective without enough technical explanation, making implementation difficult for developers.
- **Misinterpretation occurs between business and engineering teams**
Differences in technical and business understanding often lead to inconsistent interpretations of the same requirement, resulting in incorrect feature implementation.
- **Frequent requirement changes during development (scope instability)**
Requirements may change during development due to evolving business needs or regulatory updates, causing rework and disrupting sprint planning.
- **Insufficient validation of requirements before implementation**
In many cases, requirements are not thoroughly reviewed or validated with all stakeholders before development begins, increasing the likelihood of errors and revisions.

Additionally, the analysis shows that unclear acceptance criteria and incomplete documentation contribute to development uncertainty. Teams may proceed with assumptions when requirements

are vague, which increases implementation inconsistencies and testing failures. These issues collectively reduce development efficiency, increase project costs, affect system quality, and delay the delivery of critical banking features and services.

2.2.2.3 Details of Findings

A deeper examination shows that requirements at Bluestep Bank are typically defined by Product Owners based on inputs from business and compliance teams. However, due to the complexity of financial regulations and customer-specific scenarios, these requirements are sometimes incomplete, inconsistent, or open to interpretation.

For example, a requirement related to loan eligibility may not fully specify all edge cases, leading backend developers to implement logic differently than intended. Similarly, frontend teams may design interfaces based on assumptions that do not align with actual business needs. This misalignment often becomes visible only during later stages such as testing or deployment, causing rework and increased development costs.

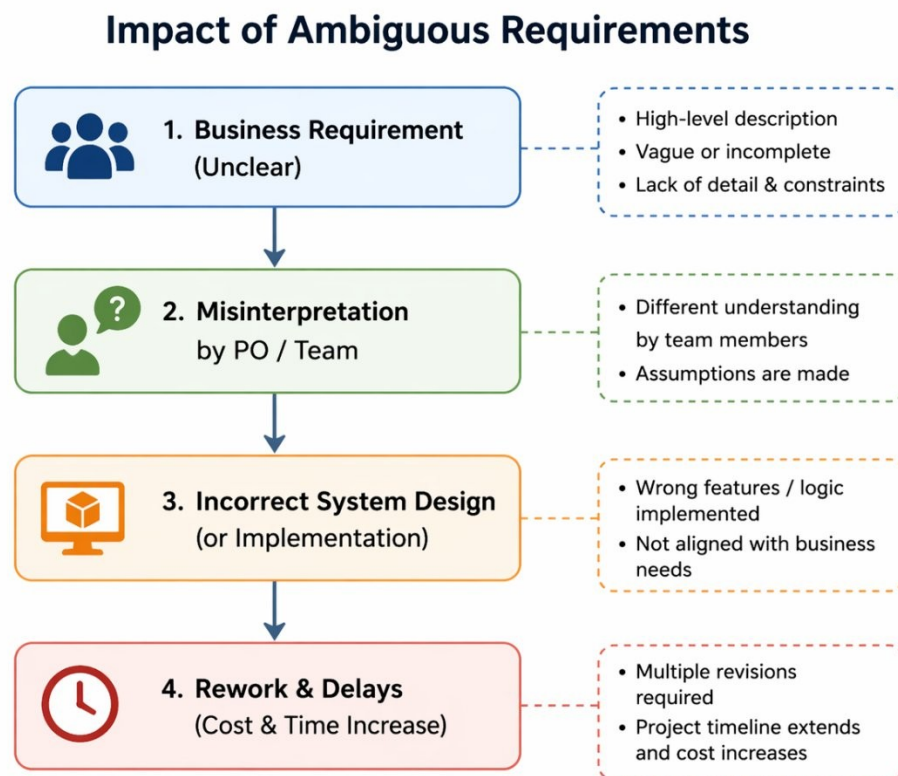


Figure 2.3: Impact of Ambiguous Requirements

Additionally, regulatory requirements—especially those varying across Sweden, Norway, and Finland—add another layer of complexity. Without clear documentation and validation, teams struggle to translate legal language into precise system behavior. This highlights a gap in the *requirements engineering process*, which is a critical phase in ISAD.

The following table categorizes the primary types of requirement ambiguity encountered at Bluestep Bank, identifying the source of each issue, how it manifests during development, and the resulting consequence on system quality and project timelines.

Table 02: Ambiguous Requirements Table

Requirement Issue	How It Manifests	Development Impact
High-Level Requirements	Missing technical detail	Incorrect implementation
Business-Tech Misinterpretation	Differing assumptions	Rework and added cost
Scope Instability	Frequent requirement changes	Extended project timelines
Insufficient Validation	No pre-implementation review	Late-stage defects and deployment delays

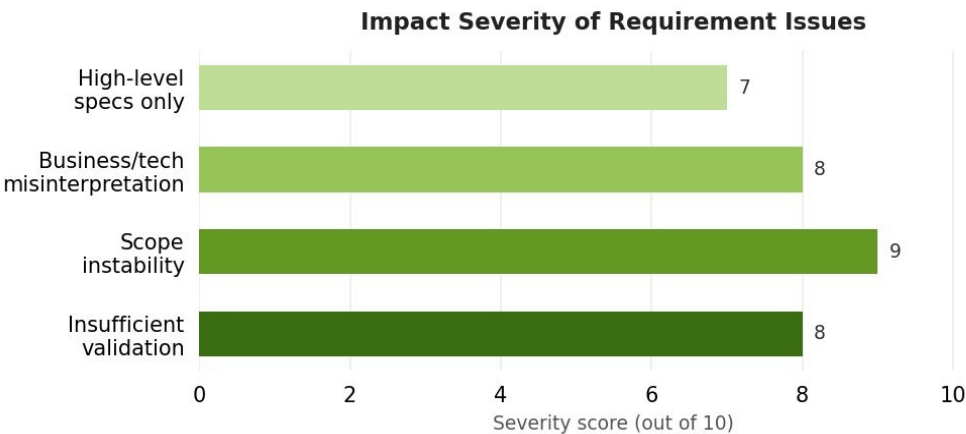


Figure 2.4: Impact Severity of Requirement Issues

2.2.2.4 Recommendation

To reduce ambiguity in requirements, Bluestep Bank should implement the following measures:

- **Adopt structured requirement engineering practices:** Use standardized formats such as user stories with clear acceptance criteria.
- **Enhance stakeholder collaboration:** Conduct requirement workshops and regular clarification sessions between business, compliance, and technical teams.
- **Implement requirement validation processes:** Use prototypes, wireframes, or mockups to confirm understanding before development begins.
- **Introduce documentation standards:** Maintain clear, detailed, and version-controlled requirement documents.

- **Use agile feedback loops:** Continuously validate requirements during sprint reviews to catch issues early.

By improving the clarity and quality of requirements, Bluestep Bank can minimize miscommunication, reduce rework, and ensure that system development aligns closely with business objectives.

2.2.3 Problem 02: External API Fragility

2.2.3.1 Statement of the Problem

In **Bluestep Bank**, heavy reliance on external APIs for services such as identity verification creates significant system reliability challenges. Any downtime, latency, or failure in third-party services can directly disrupt critical banking operations, including customer onboarding and authentication.

2.2.3.2 Summary of Findings

The analysis reveals that Bluestep Bank's digital banking platform is highly dependent on third-party APIs to perform essential operations such as identity verification, authentication, and external data validation. While these integrations improve automation and regulatory compliance, they also introduce operational risks because the organization has limited control over the availability and performance of external systems.

Several major findings were identified:

- **Strong dependency on third-party API availability**
Core banking services rely heavily on external providers, meaning that failures in external systems can immediately affect internal operations.
- **Lack of fallback mechanisms during external system failure**
The system currently has limited backup or alternative processing methods when external APIs become unavailable, increasing service disruption risks.
- **Increased response time due to API latency**
Slow API responses can delay authentication and onboarding processes, negatively affecting user experience and system efficiency.
- **Limited control over external service reliability**
Since external APIs are managed by third-party organizations, Bluestep Bank cannot directly control maintenance schedules, outages, or performance optimization.

Additionally, the analysis indicates that API failures can create cascading effects across multiple dependent services within the banking platform. A single disruption in identity verification services may delay customer onboarding, transaction approvals, and other related workflows. Frequent API instability also increases operational complexity for technical teams, who must spend additional time monitoring integrations and resolving service interruptions.

These issues collectively reduce system reliability, increase operational risks, affect customer satisfaction, and create challenges in maintaining uninterrupted digital banking services.

2.2.3.3 Details of Findings

A detailed evaluation shows that Bluestep Bank’s digital platform integrates with external identity verification systems to authenticate users during onboarding and transactions. While this ensures compliance and security, it creates a **single point of failure** in the workflow.

For example, when a user attempts to create an account, the system must communicate with an external API for identity verification. If the API is slow or unavailable, the entire process is delayed or blocked. This leads to:

- Interrupted user journeys
- Increased abandonment rates during onboarding
- Reduced trust in the platform

Additionally, since these APIs are managed by external entities, Bluestep Bank has limited control over uptime, maintenance schedules, or performance optimization. This makes the system vulnerable to unpredictable failures, which can cascade across dependent modules.

Table 03: External API Fragility Table

Scenario	System Impact	User Impact
API Downtime	Service unavailable	Cannot log in/ register
High Latency	Slow Processing	Poor user experience
Partial Failure	Incomplete transactions	Errors and retires
API Change	Integration Issues	System malfunction

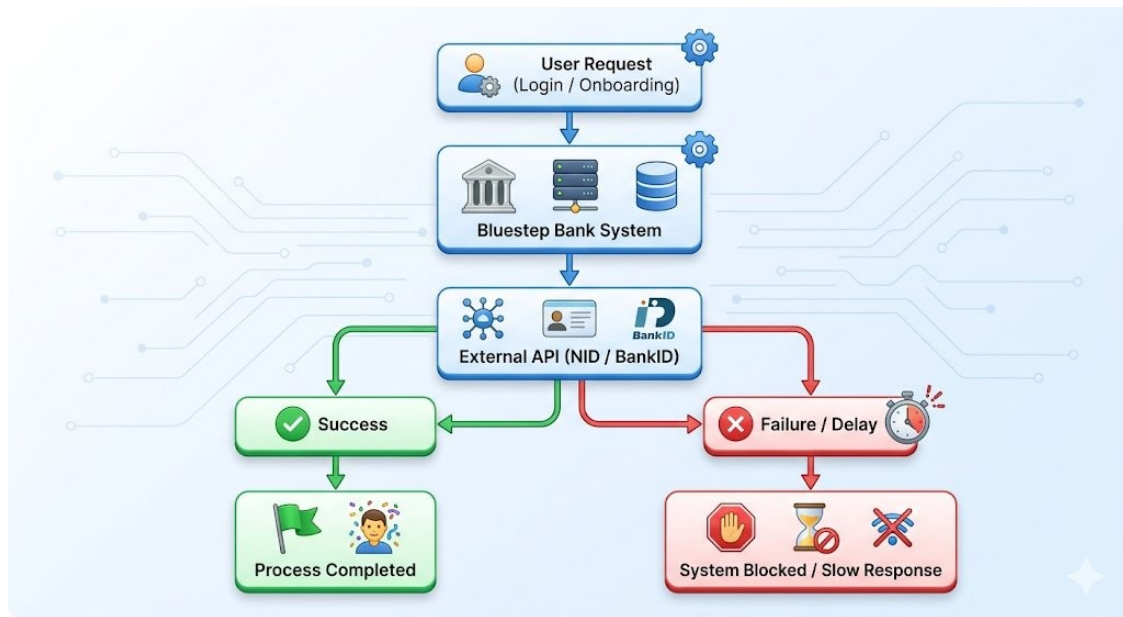


Figure 2.5: External API Dependency Flow

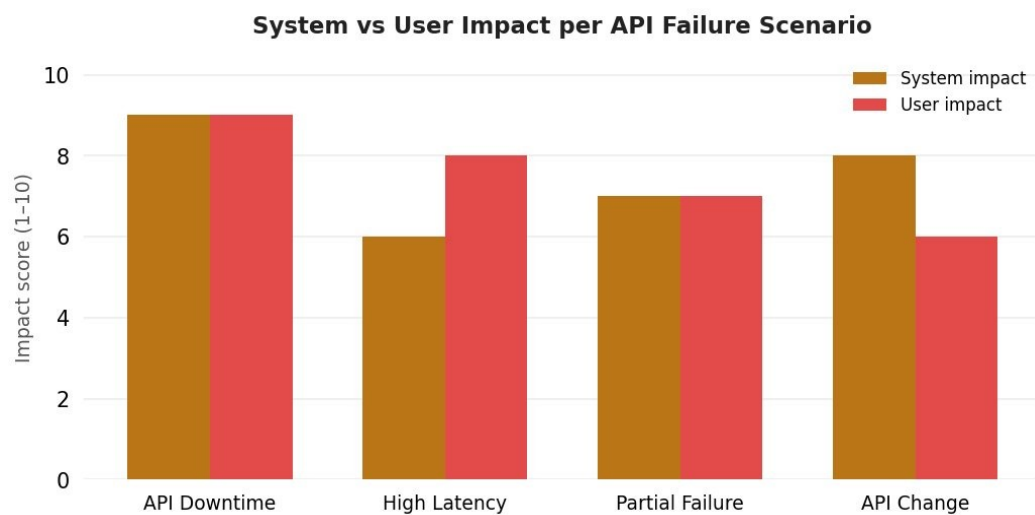


Figure 2.6: System vs User Impact per API Failure Scenario

2.2.3.4 Recommendation

To reduce the risks associated with external API fragility, the following measures are recommended:

- **Implement fallback mechanisms:** Use alternative verification methods or cached data when APIs are unavailable.

- **Introduce API monitoring systems:** Real-time monitoring and alert systems can detect failures early.
- **Use asynchronous processing:** Reduce dependency on real-time API responses where possible.
- **Apply circuit breaker pattern:** Automatically stop requests to failing APIs to prevent system overload.
- **Cache critical data:** Temporarily store frequently used verification data to reduce repeated API calls.
- **Service redundancy:** Integrate multiple API providers if possible to ensure availability.

2.2.4 Problem 04: Data Integrity and Fraud Risks

2.2.4.1 Statement of the Problem

In **Bluestep Bank**, maintaining data integrity and preventing fraudulent activities are major challenges due to the large volume of financial and customer data processed through its digital systems. Inaccurate, incomplete, or fraudulent data can lead to incorrect decisions, security risks, and compliance issues.

2.2.4.2 Summary of Findings

The analysis reveals that Bluestep Bank faces multiple challenges related to maintaining accurate and secure data across its digital banking platform. Since the organization processes sensitive financial information from customers, external APIs, and internal systems, ensuring data consistency and reliability is critical for operational stability and customer trust.

Several important findings were identified:

- **Presence of incomplete or inconsistent data inputs**
Customer-provided information and externally integrated data are sometimes incomplete, duplicated, or inconsistent, creating difficulties in processing and verification.
- **Risk of fraudulent data submission by users**
Users may intentionally provide misleading financial or personal information to manipulate loan approval or other banking processes.
- **Over-reliance on basic validation mechanisms**
Existing validation processes are primarily rule-based and may not effectively detect advanced fraud patterns or hidden inconsistencies.
- **High processing overhead due to large-scale data verification**
Continuous validation and verification of large amounts of financial data increase system workload and processing complexity.

Additionally, the analysis indicates that inconsistent data can negatively affect multiple interconnected banking services, including credit scoring, customer verification, reporting, and compliance management. Weak fraud detection mechanisms also increase the risk of unauthorized activities and financial losses. As the organization expands its digital services across multiple regions, maintaining accurate and synchronized data becomes even more challenging due to varying regulations and data standards.

These issues collectively reduce system reliability, affect decision-making accuracy, increase operational risks, and weaken customer confidence in the platform's security and trustworthiness.

2.2.4.3 Details of Findings

A deeper analysis reveals that Bluestep Bank's system processes sensitive financial and personal data, including income details, credit history, and identity verification information. Given the digital-first model, users directly input much of this data, increasing the risk of errors or manipulation.

For instance, users may unintentionally enter incorrect financial data, or in some cases, deliberately provide misleading information to gain loan approval. If the system lacks robust validation and fraud detection mechanisms, such data can pass through initial checks and impact downstream processes like credit scoring and loan approval decisions.

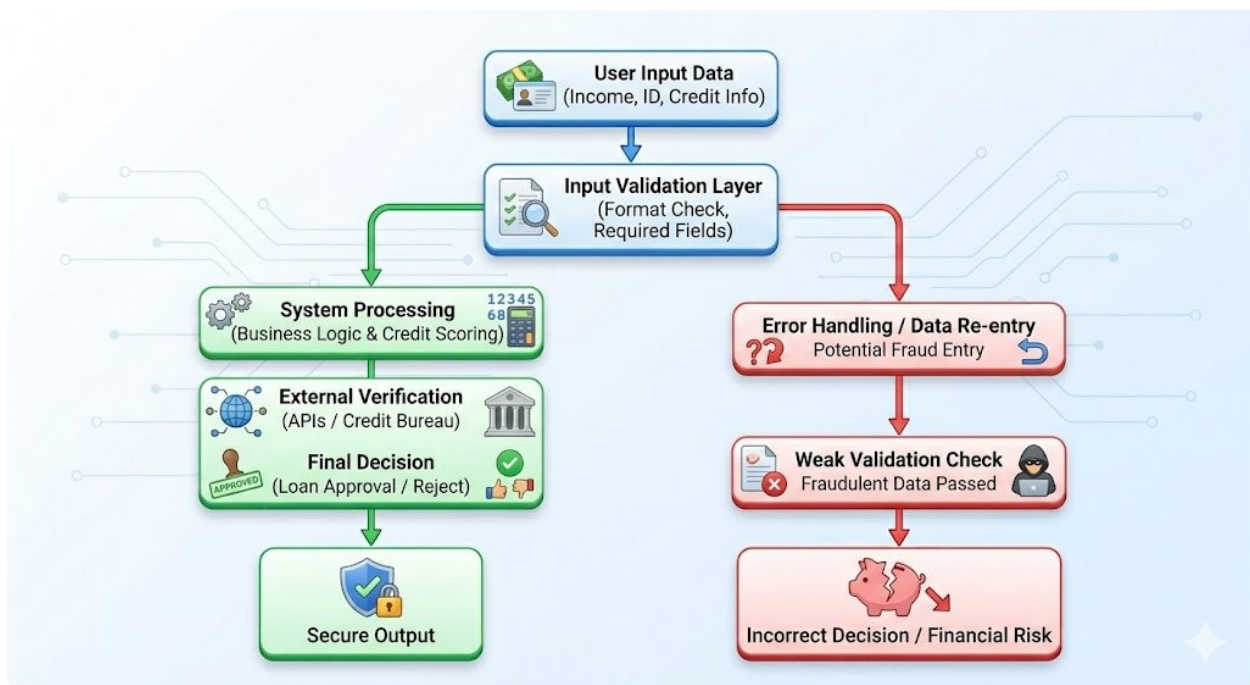


Figure 2.7: Data Integrity & Fraud Risk Flow

Moreover, the integration with multiple external systems increases the complexity of maintaining *data consistency and integrity*. Data mismatches between internal records and external sources can lead to system conflicts. The current reliance on message-based validation or sequential processing can also introduce delays, especially when handling large datasets.

Another critical issue is the absence of advanced fraud detection techniques such as behavioral analysis or anomaly detection. Without these, the system may fail to identify suspicious patterns, making it vulnerable to exploitation.

Table 04: Data Integrity & Fraud Table

Issue Type	Description	System Impact
Missing Data	Incomplete user inputs	Processing errors
Incorrect Data	Human input mistakes	Wrong decisions
Fraudulent Data	Intentional manipulation	Financial loss
Data Mismatch	Inconsistent external data	System conflicts

Distribution of Data Integrity Issue Types

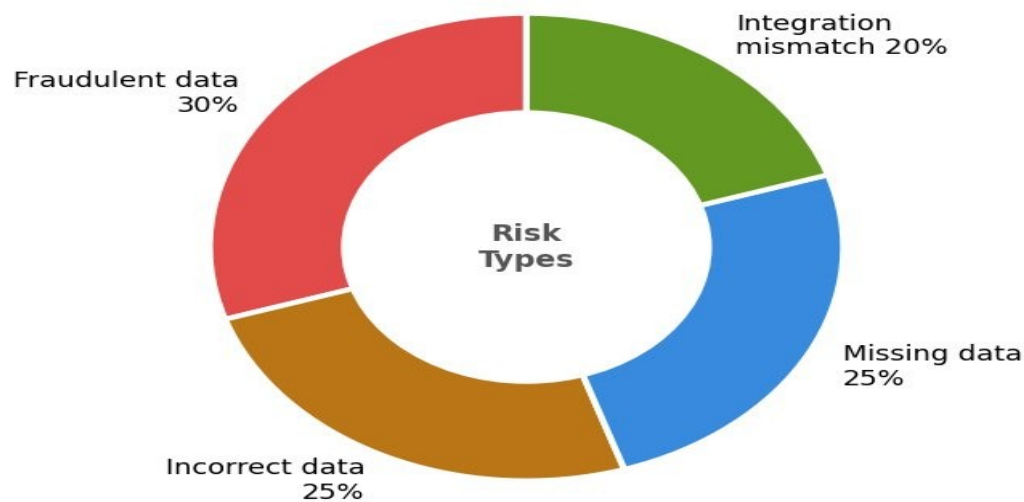


Figure 2.8: Distribution of Data Integrity Issue Types

2.2.4.4 Recommendation

To address data integrity and fraud risks, Bluestep Bank should implement the following strategies:

- **Strengthen data validation mechanisms:** Apply multi-level validation at input, processing, and output stages.
- **Implement fraud detection systems:** Use machine learning models to detect anomalies and suspicious patterns.
- **Automate data verification:** Cross-check user data with external trusted sources in real time.

- **Improve data consistency checks:** Ensure synchronization between internal and external data systems.
- **Enhance logging and auditing:** Maintain detailed logs for tracking data changes and identifying fraud attempts.
- **Optimize processing performance:** Use efficient algorithms and parallel processing to handle large data volumes.

2.3 Conclusion

This chapter presents a concise feasibility study of Bluestep Bank, focusing on four key system challenges affecting its digital operations.

First, poor cross-team communication—caused by siloed structures and heavy reliance on documentation—slows down agile development. Improving real-time collaboration and communication frameworks can enhance efficiency.

Second, unclear requirements lead to misunderstandings, rework, and delays. A stronger requirement engineering process with proper validation can minimize these issues.

Third, dependence on external APIs introduces risks like downtime and unexpected changes. Applying resilience techniques such as circuit breakers and fallback systems can reduce these disruptions.

Finally, data integrity issues and fraud risks threaten system reliability. Advanced validation and machine learning-based fraud detection can improve security and trust.

Overall, the study shows these problems can be addressed using the bank's existing cloud-based and modular architecture. With targeted improvements, Bluestep Bank can boost performance, reduce risks, and deliver a more secure and reliable digital banking experience.

Chapter3

The Planning

3.1 Introduction:

In the System Analysis & Design (SAD) lifecycle, the planning phase is a foundational step that transforms identified problems into structured and actionable solutions. It involves defining project objectives, determining system scope, allocating resources, estimating costs, and establishing timelines. The goal of planning is to ensure that system development proceeds in a **controlled, efficient, and goal-oriented manner**. From a software engineering perspective, planning includes activities such as:

- Requirement prioritization
- Feasibility confirmation
- Risk assessment and mitigation planning
- Resource and team allocation
- Scheduling using models like Gantt charts or Agile sprint planning

Effective planning ensures that all stakeholders share a common understanding of the project goals and constraints. It also minimizes uncertainties by identifying potential risks early and preparing strategies to address them. In modern agile environments, planning is not a one-time activity but a **continuous and iterative process**, adapting to changing requirements and system conditions.

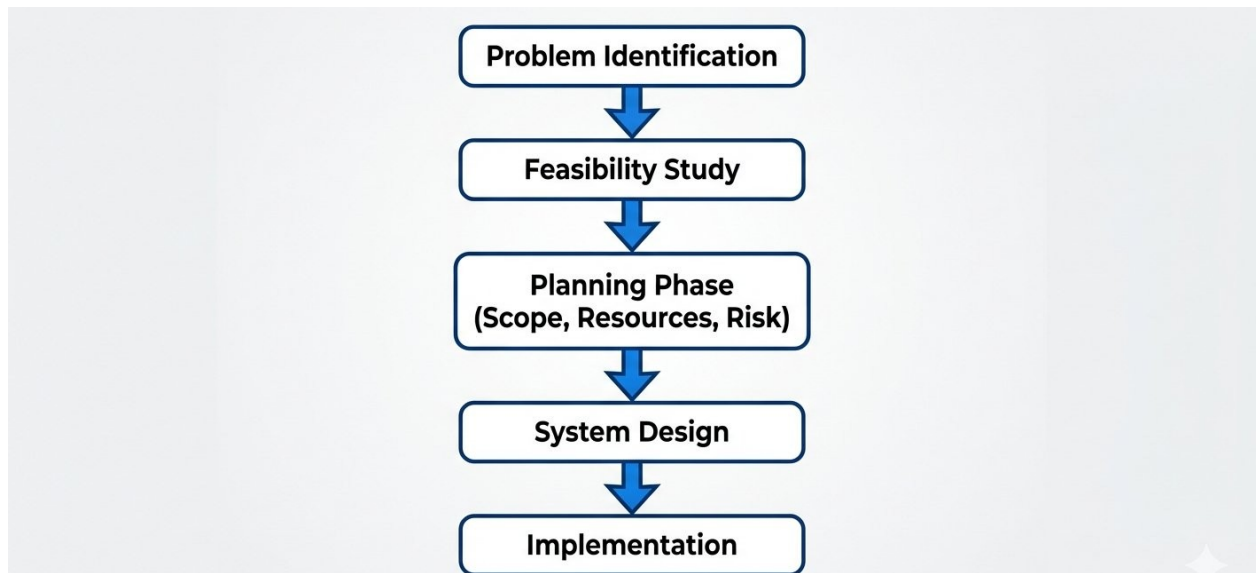


Figure 3.1: Project Development Phase

3.2 Planning in Bluestep Bank

Bluestep Bank, planning plays a crucial role due to its digital-first architecture, cross-functional team structure, and dependency on external systems. After identifying key challenges such as communication gaps, ambiguous requirements, API fragility, and data integrity risks, the planning phase focuses on designing practical and scalable solutions.

At an organizational level, planning involves coordination between Product Owners, Engineering Managers, and technical teams to:

- Define clear and detailed system requirements
- Improve inter-team communication workflows
- Design robust API handling and fallback strategies
- Strengthen data validation and fraud detection mechanisms

Technically, planning ensures that all improvements align with the bank's existing system architecture, which includes cloud-based infrastructure, microservices, and integrated data systems. Agile methodologies are heavily utilized, where work is divided into **short development cycles (sprints)**, allowing continuous feedback and incremental system enhancement.

Additionally, planning at Bluestep Bank must consider regulatory compliance and system reliability, especially when integrating with external services like identity verification APIs. This

requires careful scheduling, dependency management, and risk mitigation strategies to ensure uninterrupted service delivery.

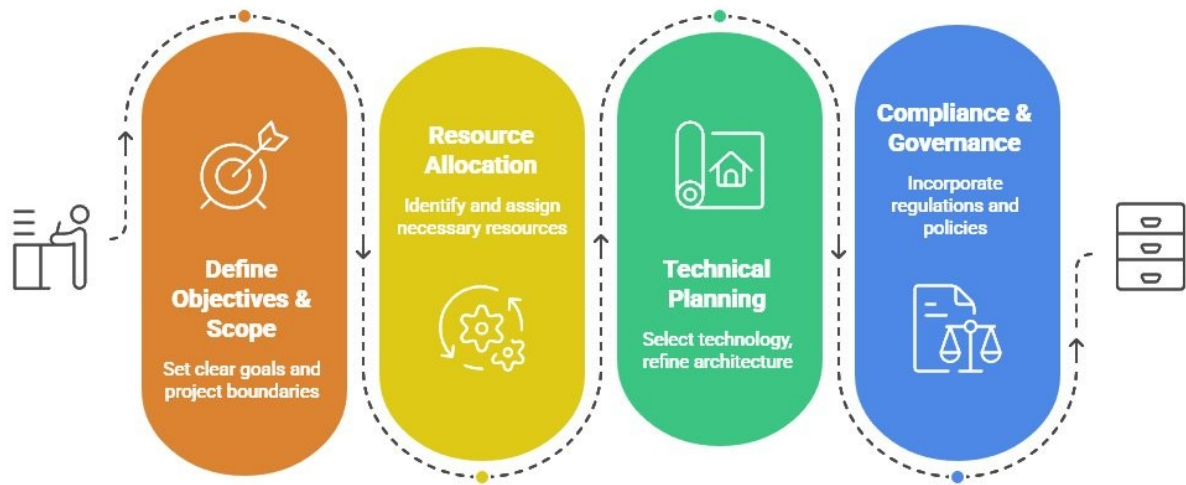


Figure 3.2: Bluestep Bank System Planning

3.3 Planning to resolve the Problems:

3.3.1 Problem 01: Cross-Departmental Communication Gaps:

3.3.1.1 Schedule Management

Schedule management plays a crucial role in addressing cross-departmental communication gaps within Bluestep Bank. Due to the organization's agile, squad-based development structure, multiple teams such as frontend, backend, and data engineering often work in parallel while depending on each other's outputs. Without a structured schedule, these dependencies can lead to delays, misalignment, and inefficient resource utilization.

The proposed schedule management approach focuses on **breaking down the project into manageable phases**, assigning clear responsibilities, and ensuring proper sequencing of tasks. By doing so, the organization can reduce uncertainty, improve coordination, and ensure that communication improvements are implemented systematically.

Additionally, the use of agile methodologies such as sprint planning allows continuous monitoring and adjustment of the schedule. This ensures flexibility while maintaining overall project control. Regular checkpoints, such as sprint reviews and retrospective meetings, help identify communication bottlenecks early and allow corrective actions to be taken in a timely manner.

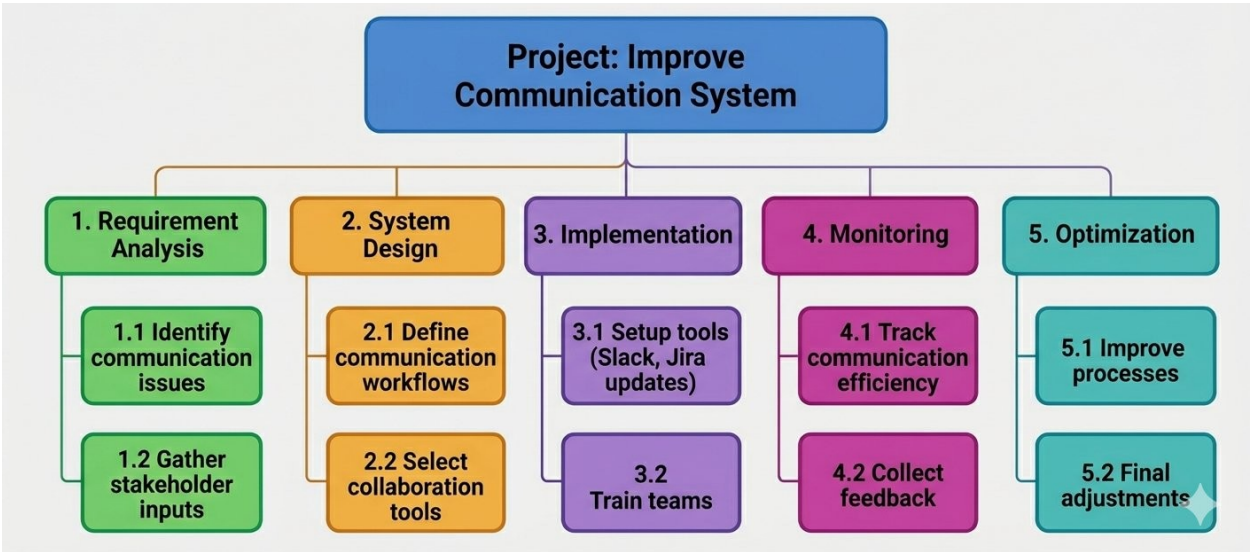


Figure 3.3: WBS Structure to Improve Communication System

The WBS provides a hierarchical breakdown of all activities required to solve the communication problem. It ensures that no critical task is overlooked and helps in assigning responsibilities efficiently across teams.

The Gantt chart visually represents the project timeline, showing task durations and overlaps. It helps project managers track progress, manage dependencies, and ensure timely completion of each phase.

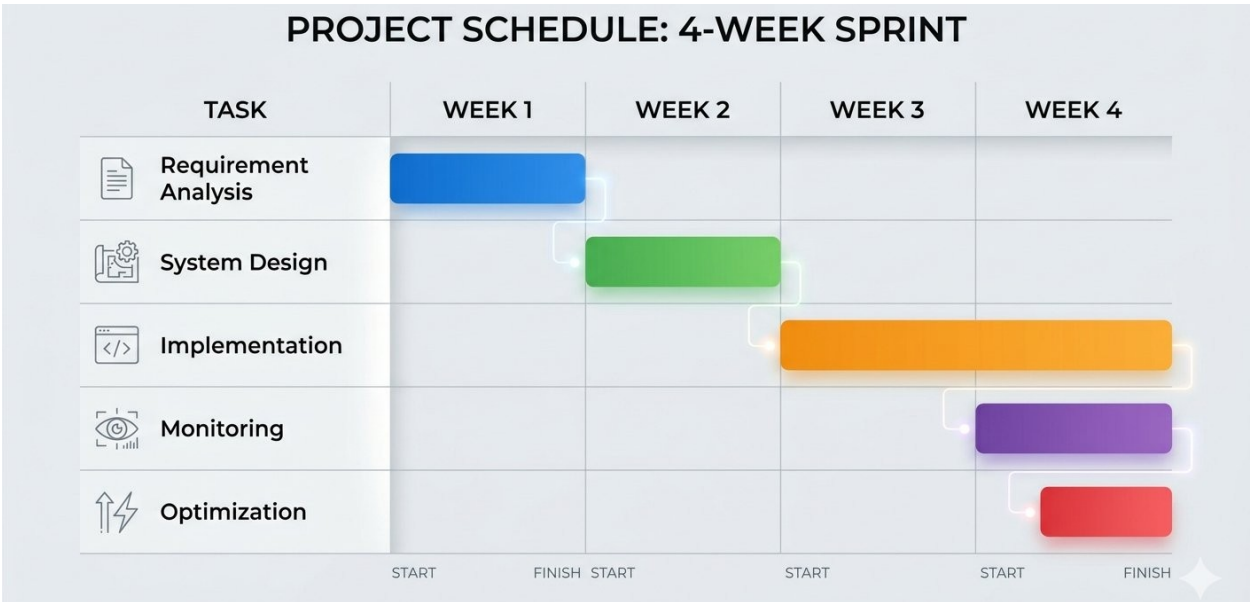


Figure 3.4: Gantt Chart for the problem

3.3.1.2 Stakeholder Engagement

Stakeholder engagement is essential for resolving communication gaps, as the issue itself arises from insufficient coordination between different organizational units. In Bluestep Bank, stakeholders include Product Owners, Engineering Managers, developers, data teams, and executive management. Each group plays a vital role in ensuring that communication processes are improved and sustained over time.

Effective engagement involves **continuous interaction, transparency, and feedback mechanisms**. Regular meetings such as daily stand-ups, sprint planning sessions, and cross-team synchronization meetings should be conducted to ensure alignment. Additionally, collaboration tools such as Jira, Slack, and Confluence can be used to enhance visibility and information sharing across teams.

By actively involving stakeholders throughout the planning and implementation phases, the organization can ensure that solutions are practical, widely accepted, and effectively utilized.

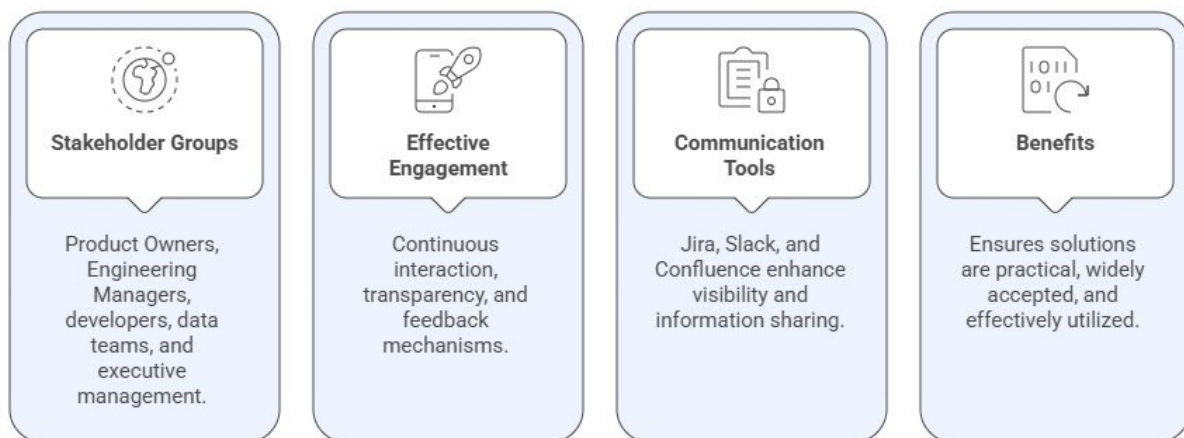


Figure 3.5: Stakeholder Engagement

In the context of cross-departmental communication gaps at Bluestep Bank, stakeholder engagement is not just supportive—it is **central to the solution itself**. Since the problem originates from misalignment and lack of coordination between teams, stakeholders directly influence how effectively communication can be improved.

Product Owners play a key role by ensuring that requirements are clearly defined and consistently communicated across teams, reducing ambiguity. Engineering Managers facilitate technical coordination by aligning development priorities and resolving inter-team dependencies. Developers contribute by actively participating in agile ceremonies such as daily stand-ups and sprint reviews, which enhance transparency and real-time communication. Data teams ensure that information flows smoothly between systems and departments, minimizing inconsistencies.

3.3.1.3 Cost Management

Cost management is a critical aspect of planning, ensuring that the proposed communication improvements are economically feasible. While improving communication may not require large infrastructure investments, it still involves costs related to tools, training, development effort, and ongoing maintenance.

The primary costs include upgrading collaboration tools, conducting training sessions for employees, and allocating developer time for implementing system changes. Although these costs may seem moderate, they are justified by the long-term benefits such as reduced project delays, improved productivity, and enhanced system reliability.

Moreover, effective communication reduces rework and errors, which can significantly lower overall project costs. Therefore, investing in communication improvement is not only necessary but also economically beneficial for the organization.

Table 05: Cost Component Comparison:

Cost Component	Description	Estimated Cost
Tools & Software	Slack, Jira upgrades	Medium
Training	Team training sessions	Low
Development Effort	System updates	High
Maintenance	Ongoing monitoring	Medium

In addressing cross-departmental communication gaps at Bluestep Bank, cost management plays a strategic role in ensuring that the proposed solutions are both **effective and sustainable**. Improving communication is not merely a process change—it requires investment in tools, training, and development efforts. Proper cost management ensures that these investments are allocated efficiently to areas that generate the highest impact.

For example, investing in advanced collaboration tools (such as project tracking systems and real-time communication platforms) can significantly reduce delays caused by miscommunication. Similarly, allocating budget for team training ensures that employees are well-equipped to use these tools and follow standardized communication practices. Without proper funding, even well-designed solutions may fail due to poor adoption or inadequate implementation.

3.3.1.4 Quality Management

Quality management ensures that the implemented communication improvements achieve the desired outcomes and maintain high standards of performance. In this context, quality is measured in terms of **communication efficiency, reduction in errors, system reliability, and team satisfaction.**

To achieve this, Bluestep Bank should establish clear quality standards and continuously monitor performance against these standards. Metrics such as task completion time, number of communication-related errors, and feedback from team members can be used to evaluate effectiveness.

Furthermore, quality management should follow a **continuous improvement approach**, where feedback is regularly collected and used to refine processes. This aligns with agile principles and ensures that the system evolves based on real-world usage and requirements.



Figure 3.6: Quality Management Cycle at Bluestep Bank

Quality management is essential in ensuring that the solutions implemented to address communication gaps at Bluestep Bank achieve the desired outcomes and maintain consistent

performance over time. Since communication directly affects system development and decision-making, improving its quality leads to better system reliability, reduced errors, and enhanced team collaboration.

By defining clear quality standards—such as timely information sharing, accurate requirement interpretation, and effective coordination—Bluestep Bank can measure and evaluate the success of its communication processes. Metrics like reduced task delays, fewer misunderstandings, and improved team satisfaction provide tangible indicators of quality improvement.

Additionally, quality management promotes a culture of **continuous monitoring and improvement**. Regular feedback from teams, performance reviews, and retrospective meetings help identify weaknesses in communication processes and allow for timely corrective actions. This iterative approach aligns well with agile methodologies, ensuring that improvements are continuously refined based on real-world performance.

Furthermore, quality management ensures consistency across teams by standardizing communication practices and enforcing best practices. This reduces variability in how information is shared and interpreted, minimizing the risk of errors and misalignment.

3.3.2 Problem 02: Ambiguous Requirements

3.3.2.1 Schedule Management

Schedule management helps Bluestep Bank systematically reduce ambiguity in requirements by structuring the entire requirement engineering process into clear, sequential phases. By allocating dedicated time for requirement gathering, validation, and refinement, the organization ensures that unclear or incomplete requirements are identified early rather than during later development stages. This minimizes rework and prevents costly delays.

A structured schedule also introduces checkpoints such as review meetings and validation sessions, which allow stakeholders and developers to align their understanding before implementation begins. As a result, development becomes more predictable, efficient, and aligned with business goals.

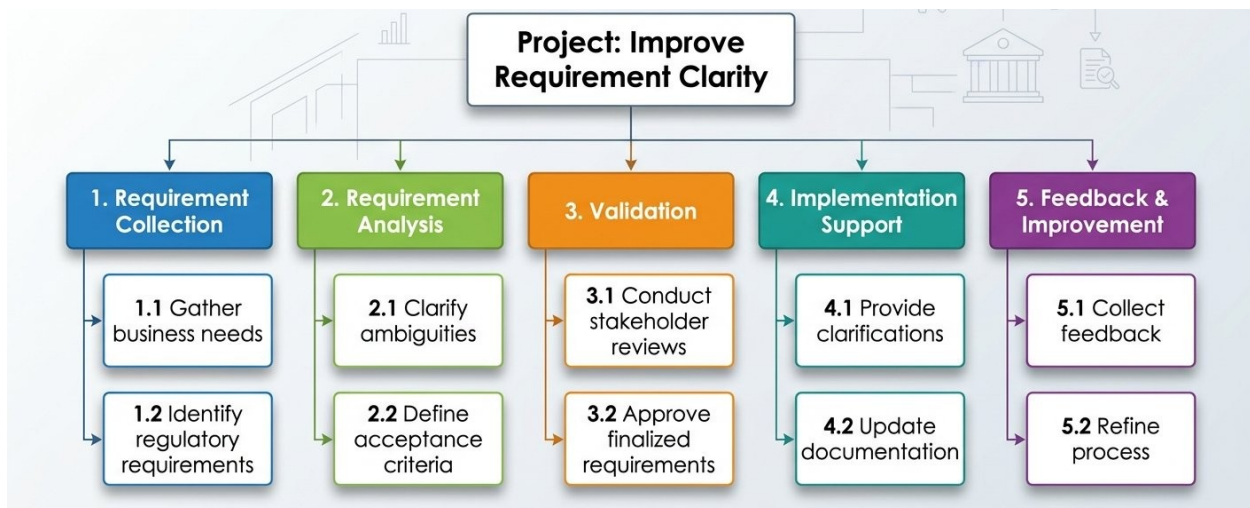


Figure 3.7: WBS graph to Improve Requirement Clarity

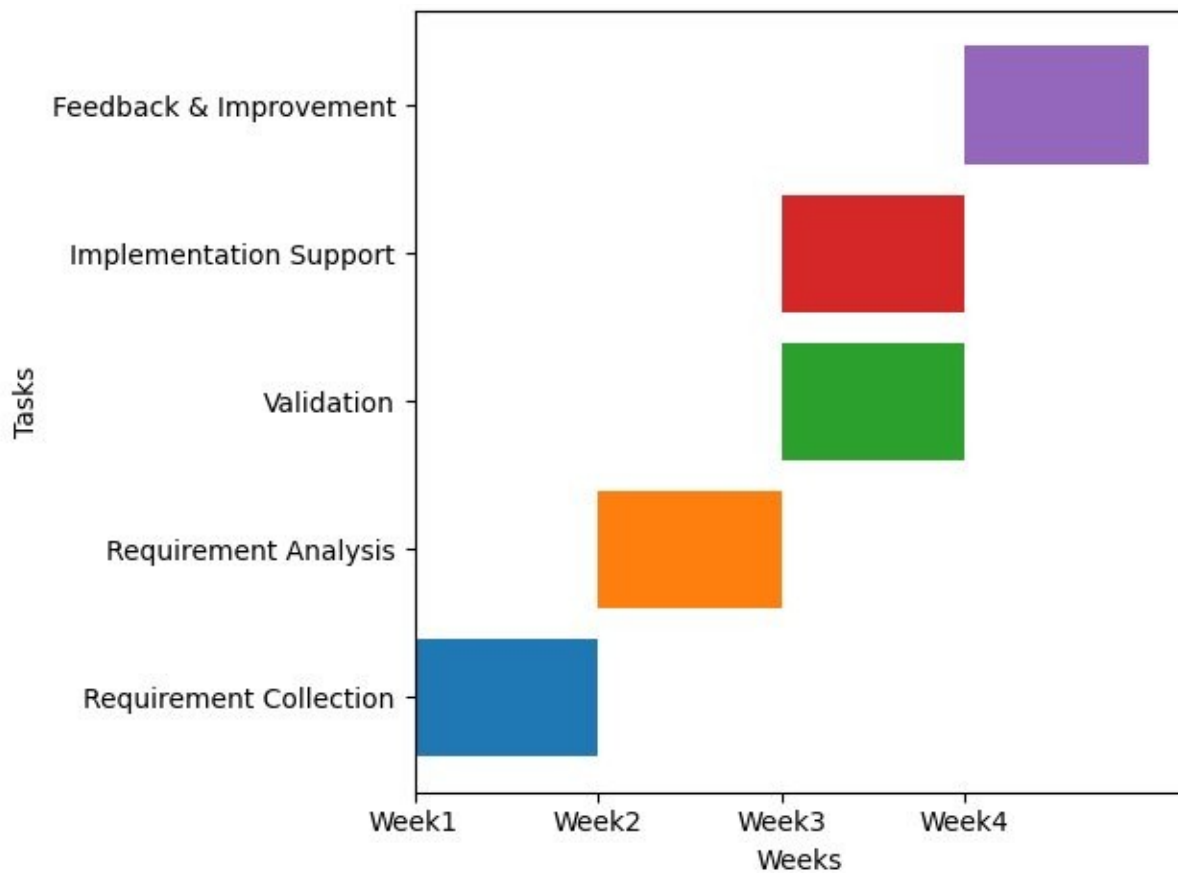


Figure 3.8: Gantt Chart to Improve Requirement Clarify

3.3.2.2 Stakeholder Engagement

Stakeholder engagement helps Bluestep Bank eliminate ambiguity by ensuring continuous collaboration between business teams, compliance units, and technical teams. When stakeholders actively participate in requirement discussions, they can clarify expectations, resolve misunderstandings, and provide immediate feedback.

This engagement ensures that requirements are not only well-understood but also aligned with business objectives and regulatory constraints. Regular interaction between stakeholders reduces the chances of misinterpretation and ensures that all perspectives are considered before development begins. As a result, the final system reflects accurate business needs and reduces costly revisions.

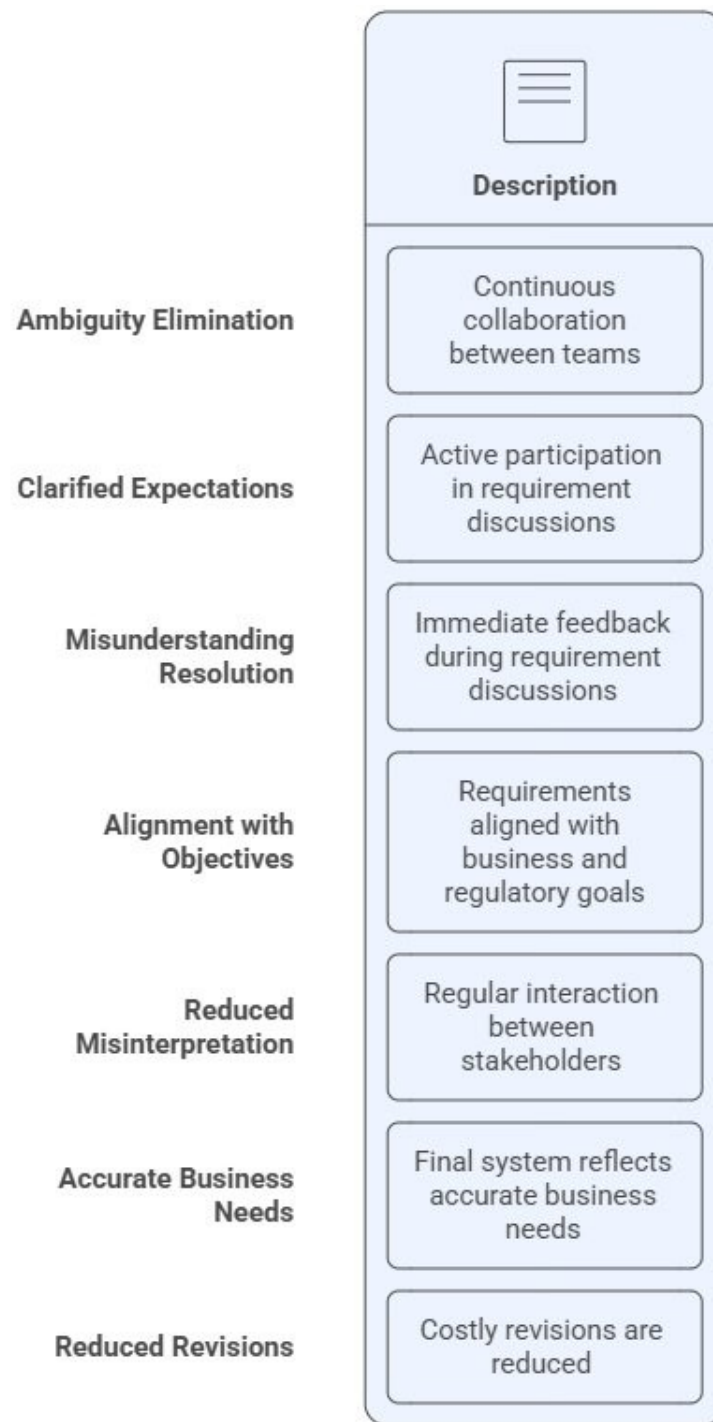


Figure 3.9: Stakeholder Engagement Benefit to Improve Requirement Clarify

3.3.2.3 Cost Management

Cost management helps Bluestep Bank control expenses caused by ambiguous requirements, which often lead to rework, redesign, and delays. By investing in proper requirement analysis and validation early in the process, the organization can significantly reduce unnecessary development costs.

Table 06: Cost Comparison to Improve Requirement Clarify

Cost Comparison: With vs. Without Planning

	 Without Planning	 With Proper Planning
Rework Cost	High	Low
Development Time	Extended	Optimized
Resource Utilization	Inefficient	Efficient
Error Correction	Frequent	Minimal

Proper cost management ensures that resources are used efficiently and that investments in requirement engineering deliver long-term savings.

3.3.2.4 *Quality Management*

Quality management helps Bluestep Bank ensure that requirements are clear, consistent, and accurately implemented. By defining quality standards for requirement documentation and validation, the organization can reduce errors and improve system reliability.

Table 07: Quality Improvement Impact

Quality Aspect	Before Improvement	After Improvement
Requirement Clarity	Low	High
System Accuracy	Inconsistent	Consistent
Development Errors	High	Reduced
Customer Satisfaction	Moderate	Improved



Figure 3.10: Achieving Project Success

3.3.3 Problem 03: External API Fragility

3.3.3.1 Schedule Management

Schedule management helps Bluestep Bank reduce the risks associated with external API fragility by ensuring that integration, testing, monitoring, and fallback implementation activities are completed in a structured and timely manner. Since the bank's digital services rely heavily on third-party APIs such as identity verification systems, poor scheduling could lead to integration delays, unstable deployments, and service interruptions.

A well-defined schedule allows teams to allocate sufficient time for API testing, performance monitoring, and contingency planning before deployment. It also ensures that dependencies between external systems and internal modules are properly managed. Through phased implementation and agile sprint planning, the organization can identify API-related issues early and minimize disruption to critical banking services.

Additionally, schedule management supports continuous monitoring activities, allowing the organization to quickly respond to downtime or performance degradation from third-party services. This structured approach improves system reliability and reduces operational risks.

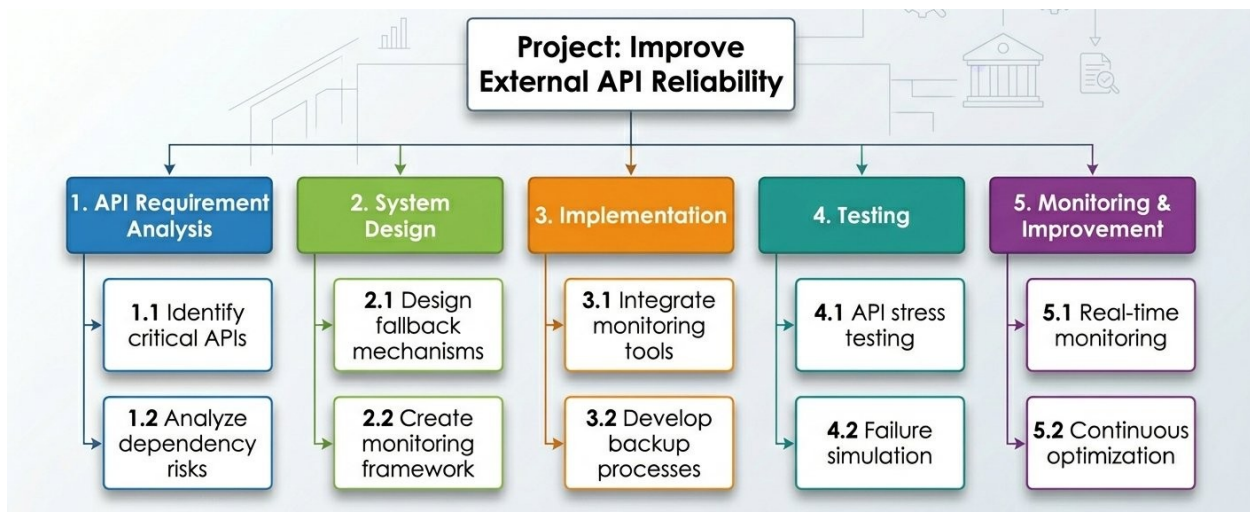


Figure 3.11: WBS Graph to Improve External API Reliability

GANTT CHART – EXTERNAL API RELIABILITY PROJECT



Figure 3.12: Gantt Chart to Improve External API Reliability

3.3.3.2 Stakeholder Engagement

Stakeholder engagement helps Bluestep Bank effectively manage external API fragility by promoting collaboration between technical teams, operations staff, management, and external

service providers. Since third-party APIs directly impact core banking functions such as user authentication and onboarding, active communication between stakeholders is essential for maintaining service continuity.

Engineering teams help identify integration risks and implement fallback solutions, while operations teams monitor API availability and respond to incidents. Product Owners ensure that customer experience requirements are considered during planning, and management provides strategic decisions regarding redundancy and service-level agreements.

Engaging external API providers is equally important, as it enables better coordination regarding maintenance schedules, updates, and performance expectations. Through continuous stakeholder involvement, the organization can proactively address API-related risks and improve overall system stability.

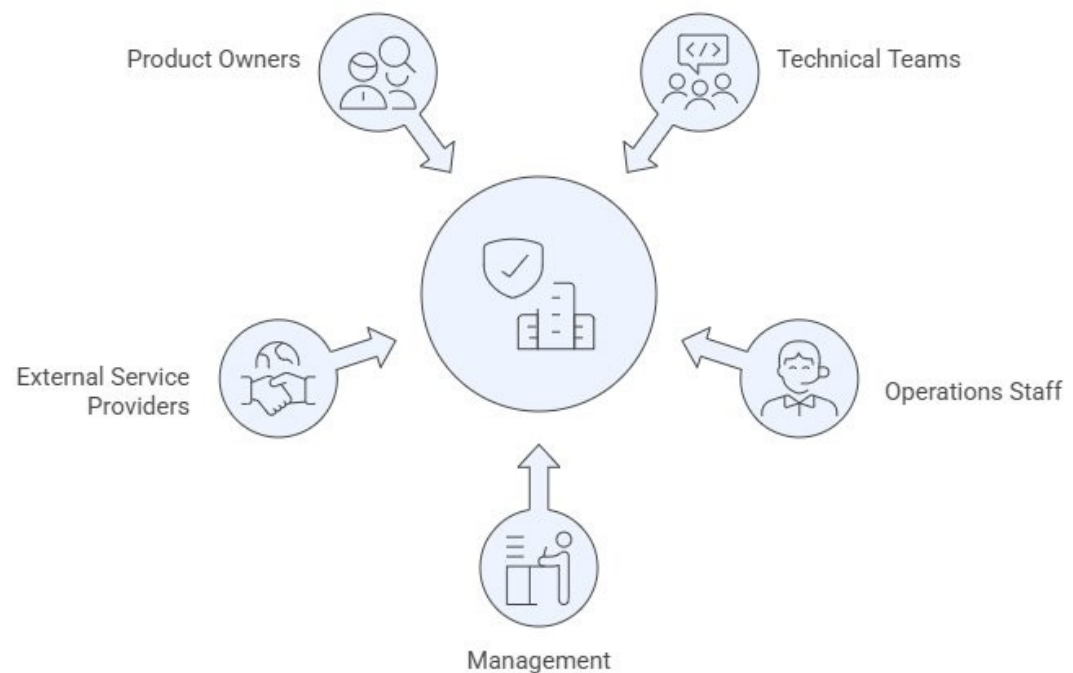


Figure 3.13: Stakeholder Engagement for API Fragility Management

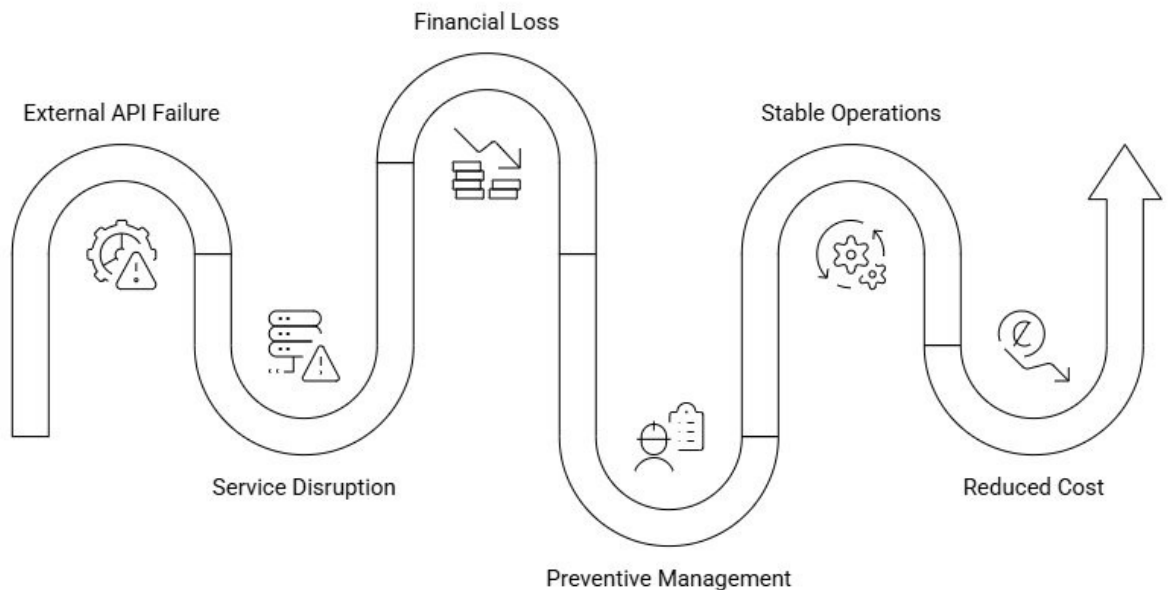
3.3.3.3 Cost Management

Cost management helps Bluestep Bank control financial risks associated with API failures, service disruptions, and emergency fixes. External API downtime can result in operational losses, delayed customer onboarding, and increased support costs. By investing in proactive API management strategies, the organization can reduce long-term expenses caused by unexpected failures.

Proper budgeting for monitoring tools, backup systems, and testing environments ensures that the bank can maintain stable services without excessive recovery costs. Additionally, preventive investments are generally more cost-effective than handling large-scale outages after deployment.

Table 08: Cost Impact of API Fragility

Cost Area	Without Management	With Proper Management
Service Downtime Cost	High	Reduced
Emergency Maintenance	Frequent	Minimal
Customer Support Cost	Increased	Controlled
System Recovery Cost	Expensive	Lower

*Figure 3.14: External API Failure Sequence*

3.3.3.4 Quality Management

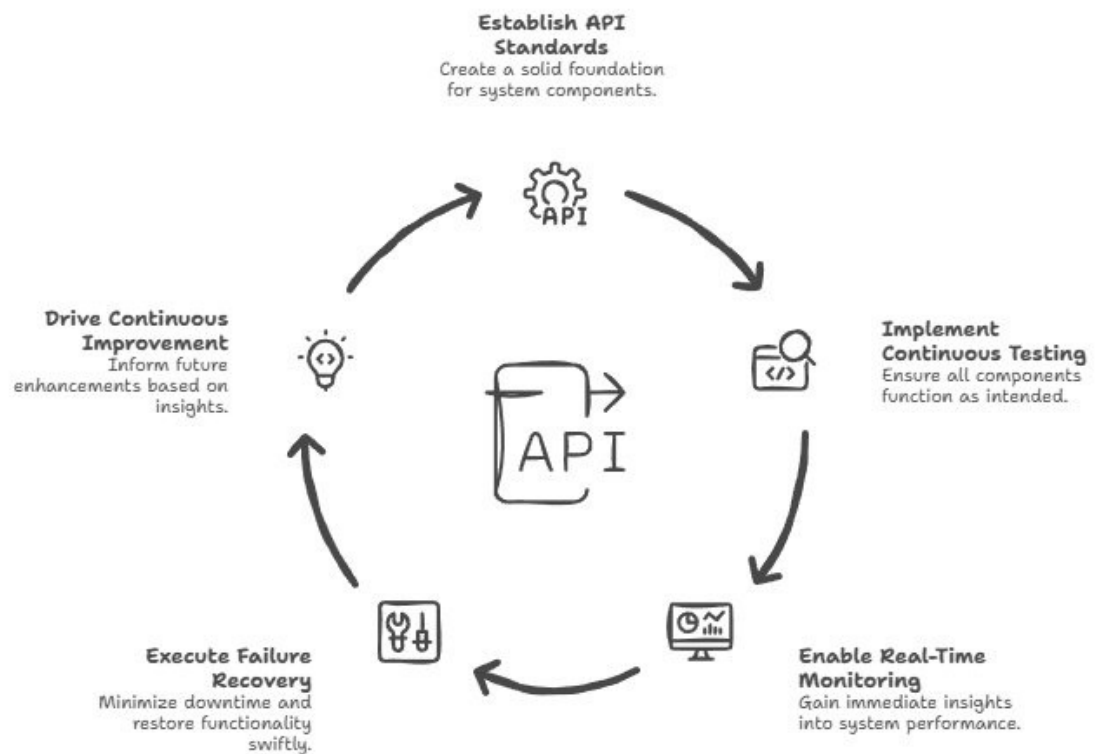
Quality management helps Bluestep Bank maintain reliable and consistent API integrations by ensuring that external services meet performance, security, and availability standards. Since the bank depends on APIs for critical operations, maintaining high integration quality is essential for delivering uninterrupted services to customers.

Through continuous testing, monitoring, and validation, the organization can identify weaknesses before they affect users. Quality management also ensures that fallback mechanisms and recovery processes function correctly during API failures.

Table 09: Quality Improvement Impact

Quality Aspect	Before Improvement	After Improvement
API Reliability	Unstable	Stable
System Downtime	Frequent	Reduced
Customer Experience	Interrupted	Improved
Error Handling	Weak	Strong

API Standards Cycle for System Resilience

*Figure 3.15: API Standards Cycle for System Resilience*

3.3.4 Problem 04: Data Integrity and Fraud Risks

3.3.4.1 Schedule Management

Schedule management helps Bluestep Bank systematically address data integrity issues and fraud risks by ensuring that all critical activities such as data validation, security implementation, monitoring, and continuous improvement are completed in a structured and timely manner.

A well-planned schedule allows the organization to identify data quality problems early, implement preventive controls, and test fraud detection mechanisms before deployment. By dividing the work into clear phases with defined milestones and dependencies, the organization can reduce risks related to data inconsistencies, fraudulent activities, and compliance violations.

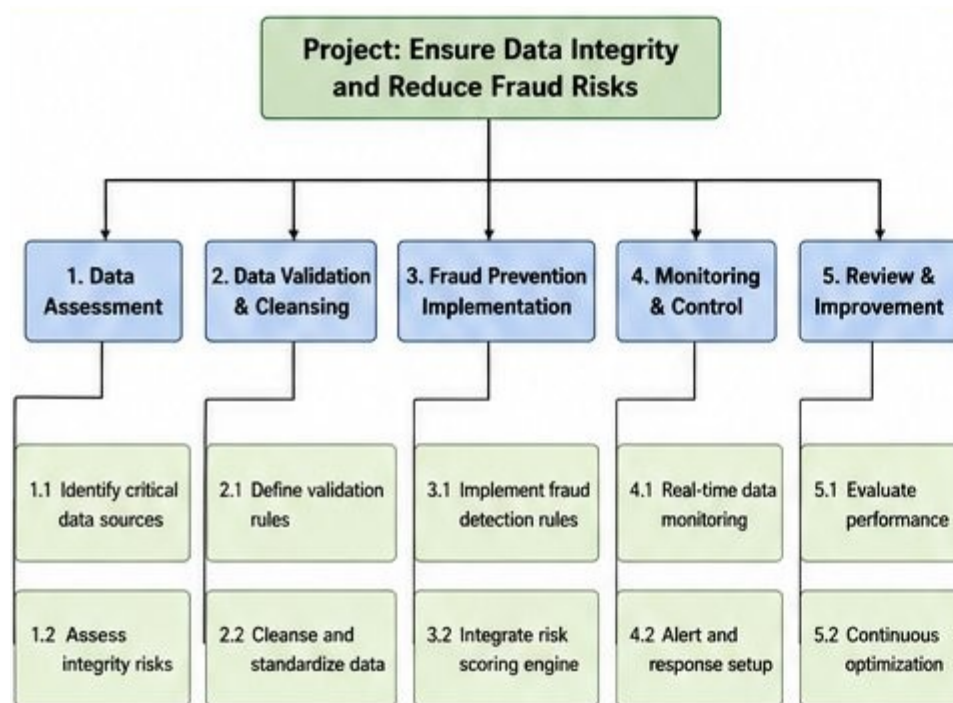


Figure 3.16: WBS Graph to Ensure Data Integrity and Reduce Fraud Risks

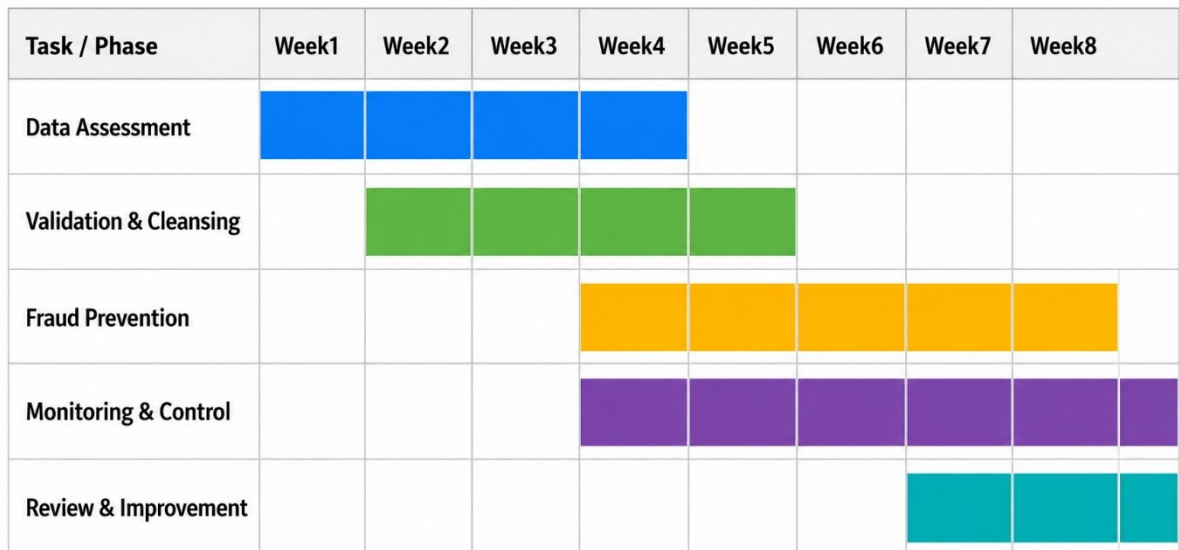


Figure 3.17: Gantt Chart to Ensure Data Integrity and Reduce Fraud Risks

3.3.4.2 Stakeholder Engagement

Stakeholder engagement helps Bluestep Bank effectively manage data integrity and fraud risks by ensuring close collaboration between data teams, compliance officers, IT security experts, developers, and management.

Data teams help identify quality issues and inconsistencies in financial records, while compliance teams ensure that all processes meet regulatory and legal requirements. Security teams design fraud detection mechanisms and access control systems to reduce unauthorized activities. Developers implement validation rules and monitoring systems, ensuring that technical solutions function correctly and securely.

Management also plays an important role by providing strategic direction, approving policies, and allocating necessary resources. Through regular communication, review meetings, and feedback sessions, stakeholders work together to ensure accurate data handling, faster fraud detection, and stronger governance.

Active stakeholder engagement reduces the likelihood of system vulnerabilities, improves accountability across departments, and strengthens customer trust in the bank's digital platform.

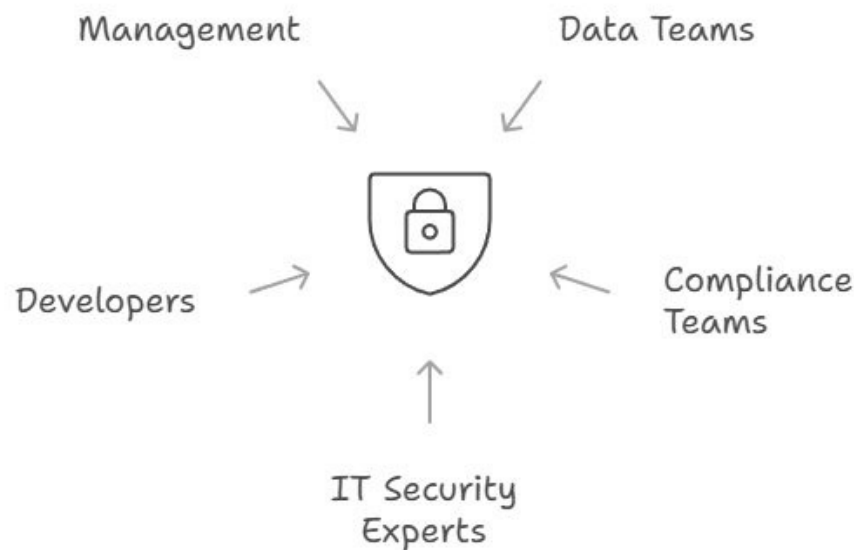


Figure 3.18: Stakeholder Engagement for Data Integrity and Fraud Risks Management

3.3.4.3 Cost Management

Cost management helps Bluestep Bank minimize financial losses caused by data errors, fraud incidents, compliance penalties, and operational disruptions. Poor data integrity and weak fraud prevention mechanisms can lead to incorrect business decisions, financial fraud, customer dissatisfaction, and regulatory fines.

By investing in preventive controls such as validation systems, fraud detection tools, and monitoring platforms, the organization can significantly reduce long-term operational costs. Preventive investment is more cost-effective than recovering from large-scale fraud incidents or correcting inaccurate data after deployment.

Cost management also ensures efficient allocation of resources, allowing the organization to prioritize critical security and data quality improvements. This helps reduce unnecessary expenditures while maximizing the effectiveness of risk mitigation strategies.

Table 10: Cost Impact Reduction

Cost Area	Without Proper Planning	With Proper Planning	Benefit to Company
Fraud Losses	High	Low	Protects financial assets
Data Correction Cost	High	Low	Reduces rework
Compliance Penalties	High	Low	Avoids legal risks
System Downtime	High	Low	Improves service

Cost			continuity
Operational Cost	High	Optimized	Increases efficiency



Figure 3.19: Cost Reduction Flow for Data Integrity and Fraud Risks Management

3.3.4.4 Quality Management

Quality management helps Bluestep Bank maintain high standards of data accuracy, consistency, and security across all digital banking operations. Since the organization heavily depends on accurate financial and customer information, maintaining data quality is essential for reliable decision-making and fraud prevention.

By defining quality standards, implementing continuous monitoring, and regularly validating data, the organization can reduce inconsistencies and detect fraudulent activities more effectively. Quality management also improves system reliability by ensuring that validation rules, monitoring tools, and fraud detection mechanisms function correctly.

Furthermore, continuous improvement practices allow the organization to adapt to new fraud patterns and changing regulatory requirements. This proactive approach enhances customer confidence, strengthens operational stability, and improves the overall effectiveness of the bank’s information systems.

Table 11: Quality Improvement Impact

Quality Aspect	Before Improvement	After Improvement	Benefit to Company
Data Accuracy	Low	High	Better decision-making
Data Consistency	Inconsistent	Consistent	Reliable reporting
Fraud Detection	Reactive	Proactive	Faster fraud prevention
System Reliability	Unstable	Stable	Reduced disruptions
Customer Trust	Moderate	High	Improved reputation



Figure 3.20: Quality Management Process for Data Integrity and Fraud Risks Management

3.3.5 Planning as a Whole

3.3.5.1 Introduction

After identifying the four major organizational and technical challenges within Bluestep Bank—Cross-Departmental Communication Gaps, Ambiguous Requirements, External API Fragility, and Data Integrity & Fraud Risks—it is necessary to develop a comprehensive implementation plan that addresses all issues in a coordinated manner. Rather than solving each problem independently, an integrated planning approach allows the organization to optimize resources, reduce project costs, improve stakeholder collaboration, and achieve long-term operational efficiency.

3.3.5.2 Work Breakdown Structure (WBS)

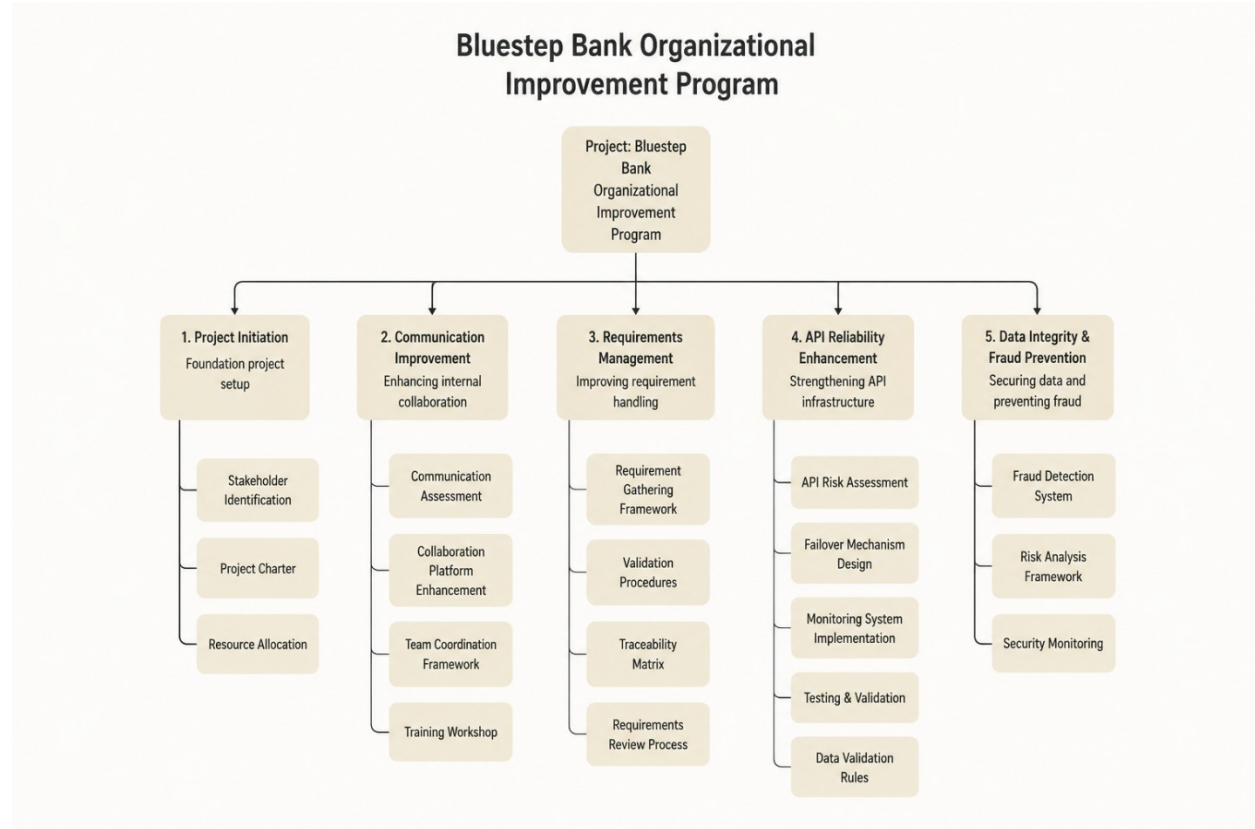


Figure 3.21: WBS of Bluestep Bank Organizational Improvement Program

3.3.5.3 Schedule Management

The project is planned over a period of 24 weeks (6 months). Activities are scheduled sequentially while allowing overlap between related tasks to reduce implementation time.

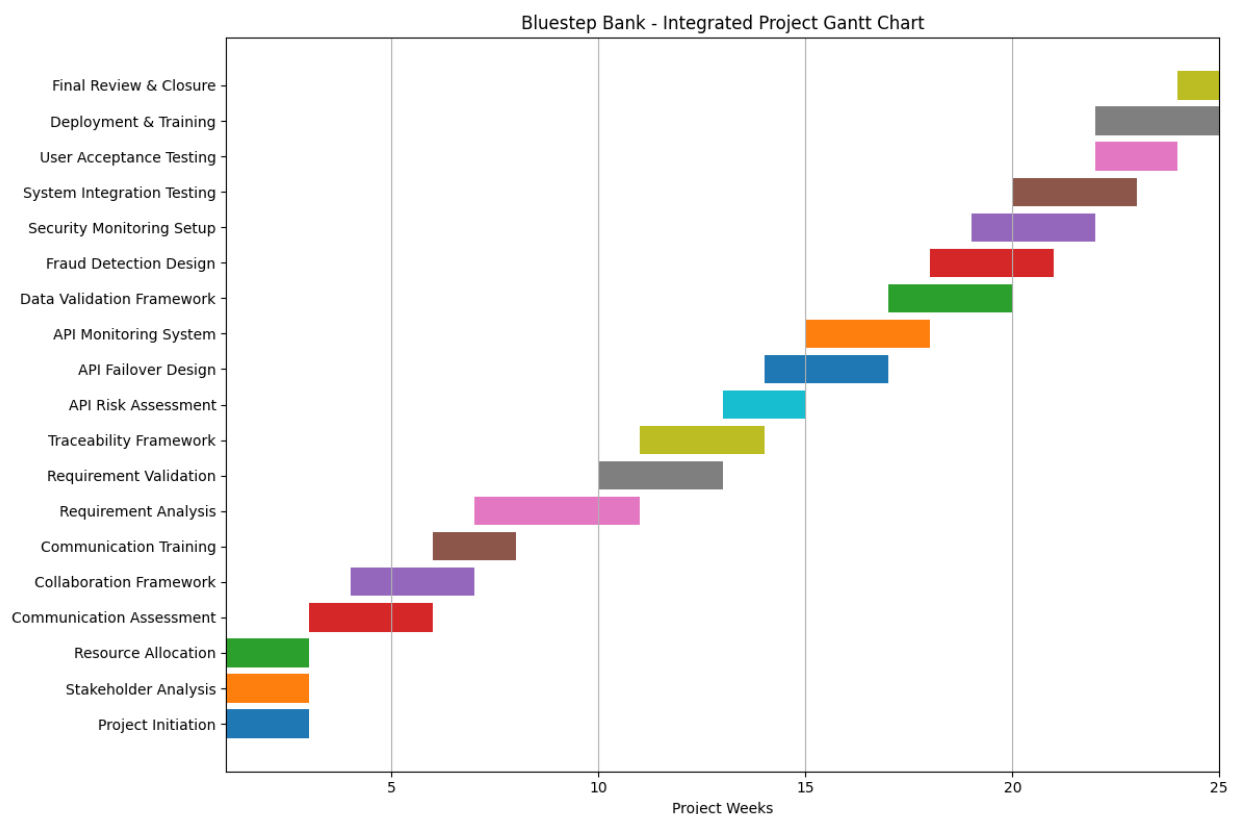


Figure 3.22: Gantt of whole process planning

3.3.5.4 Stakeholder Engagement Plan

Successful implementation depends on active participation from all stakeholders.

Table 12: Stakeholder Engagement Matrix

Stakeholder	Role	Engagement Strategy
CEO & CTO	Strategic Decision Making	Monthly review meetings
Product Owners	Requirement Definition	Weekly workshops
Engineering Managers	Technical Leadership	Sprint planning sessions
Development Teams	System Implementation	Daily stand-ups
Compliance Team	Regulatory Validation	Bi-weekly review meetings
Customers	User Feedback	Surveys and pilot testing
External API Provider	Service Reliability	SLA monitoring meetings

Strong stakeholder engagement ensures that project objectives remain aligned with business goals throughout implementation. Product Owners and Engineering Managers help reduce requirement ambiguity, while Compliance Teams ensure regulatory compliance. Customer feedback provides valuable insights into service quality, and collaboration with API providers improves external service reliability. Active stakeholder participation also reduces resistance to change and increases project success rates.

Distribution of Project Focus Areas

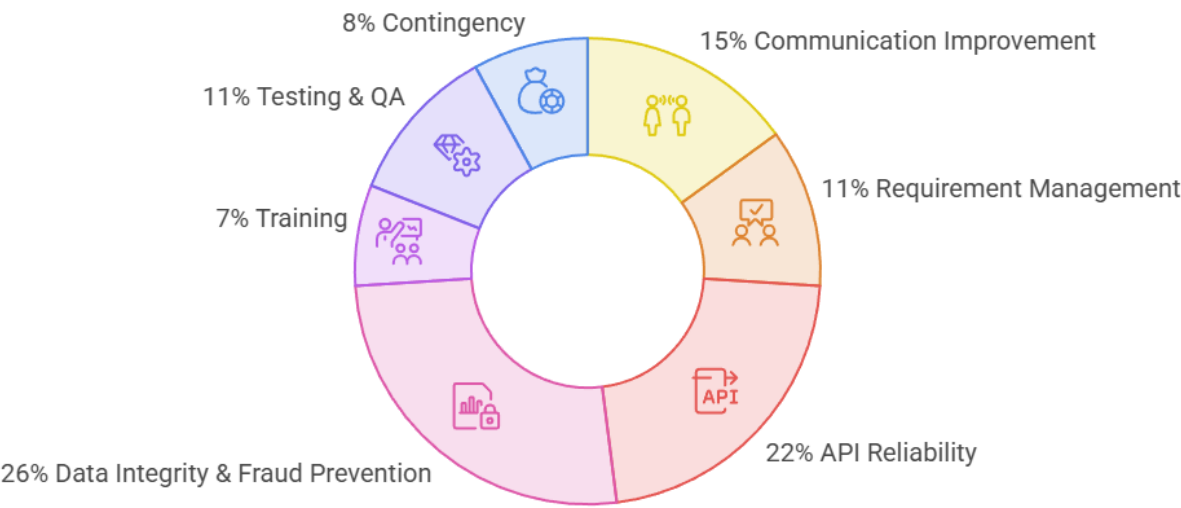


Figure 3.23: Distribution of Project Focus Areas

3.3.5.5 Quality Management

The quality management plan ensures that all improvements meet business, technical, security, and regulatory requirements.

Table 13: Quality Management Framework

Quality Area	Quality Objective	Measurement Method
Communication Efficiency	Reduce dependency delays	Project completion rate
Requirement Accuracy	Minimize requirement changes	Change request count
API Reliability	Increase service availability	Uptime percentage
Data Integrity	Reduce data errors	Validation success rate
Fraud Prevention	Improve fraud detection accuracy	Fraud detection reports
User Satisfaction	Improve customer experience	Survey results

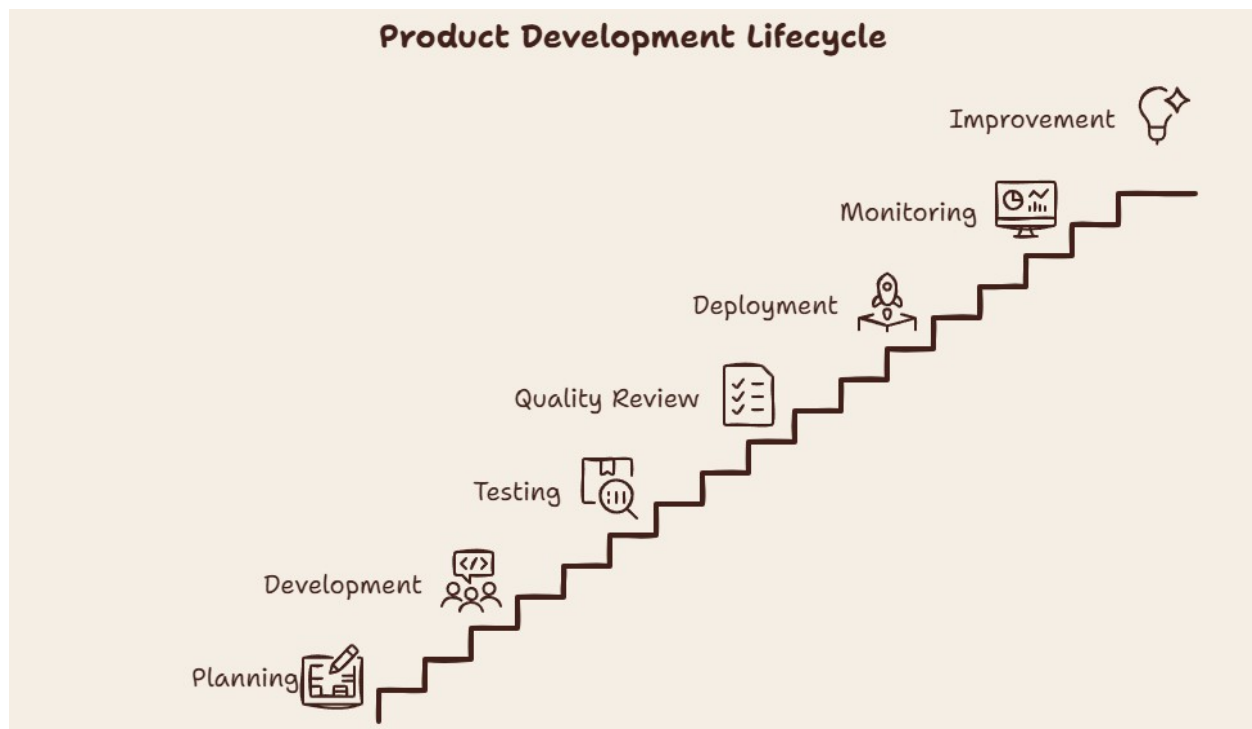


Figure 3.24: Product Development Lifecycle

Quality management ensures that improvements are sustainable and measurable. Continuous monitoring allows the organization to identify defects and inefficiencies early. Quality assurance activities reduce development errors, improve service reliability, strengthen data integrity, and enhance customer satisfaction. As a result, Bluestep Bank can maintain regulatory compliance while delivering a more reliable and efficient digital banking experience.

3.4 Conclusion

The planning phase plays a vital role in ensuring the successful improvement of information systems within [Bluestep Bank](#). Through structured planning activities such as schedule management, stakeholder engagement, cost management, and quality management, the organization can systematically address critical operational and technical challenges. By dividing complex problems into manageable tasks, defining responsibilities, and establishing clear timelines, the planning process creates a strong foundation for efficient system development and organizational coordination. Effective planning enables the organization to proactively manage risks related to communication gaps, ambiguous requirements, external API fragility, and data integrity issues. Stakeholder collaboration ensures that business, technical, and regulatory perspectives are properly aligned, while cost management helps optimize resource utilization and reduce unnecessary expenditures. At the same time, quality management supports continuous monitoring and improvement, ensuring that implemented solutions remain reliable, scalable, and aligned with organizational goals.

Chapter 4

Analysis

4.1 Introduction

This chapter provides an in-depth analysis of the organizational structure, operational processes, and information system environment of Bluestep Bank. The analysis aims to identify inefficiencies, technical limitations, communication challenges, and operational risks that affect the bank's digital banking services and software development activities. As a technology-driven financial institution operating across Sweden, Norway, and Finland, the bank relies heavily on integrated information systems, agile development practices, and external digital services to maintain operational continuity and deliver efficient customer services. Particular attention is given to issues such as cross-departmental communication gaps, unclear system requirements, dependency on external APIs, and risks related to data integrity and fraud prevention.

To investigate these challenges, the study applies a combination of qualitative and technical analysis methods. Organizational workflows, communication structures, software development practices, and system dependencies are evaluated to identify operational bottlenecks and weaknesses within existing processes. Several analytical approaches, including process analysis, workflow evaluation, feasibility assessment, and risk analysis, are used to understand how technical systems interact with organizational operations. The analysis also explores how the bank's agile squad structure, cloud-based infrastructure, and reliance on third-party services influence system performance, development efficiency, and service reliability. This chapter examines the impact of inter-team coordination issues, ambiguous requirements, and increasing security and compliance demands within a highly regulated banking environment. Different diagrams, flowcharts, tables, and graphical representations are incorporated to visualize workflows, system dependencies, and identified problem areas more effectively. The findings from this analysis establish the foundation for the recommendations and proposed solutions presented in later chapters, ensuring that future improvements align with the bank's operational objectives, technological infrastructure, and long-term digital transformation strategy.

4.2 Information Gathering

Information gathering is an essential step in the System Analysis & Design (SAD) process, as it helps identify existing operational challenges, system limitations, and organizational requirements within Bluestep Bank. Through systematic collection and evaluation of organizational data, the analysis aims to understand how current processes, communication structures, and technical systems operate in practice.

For Bluestep Bank, information gathering focuses on understanding the interaction between agile development teams, digital banking platforms, external API integrations, and data management systems. This process helps identify communication gaps, requirement ambiguities, API

dependency issues, and data integrity risks that affect system performance and organizational efficiency.

The information gathering process includes:

- Review of literature related to digital banking systems and agile software engineering
- Review of organizational procedures and workflows
- Review of operational forms and system documents used within banking processes

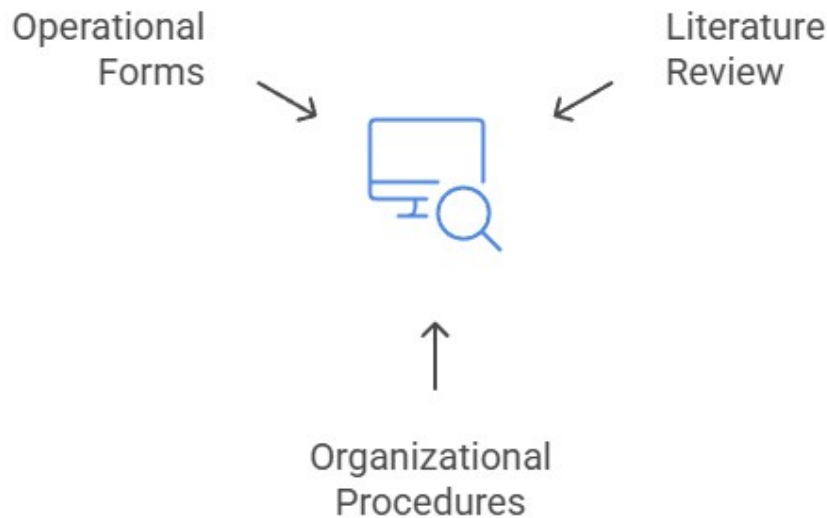


Figure 4.1: Information Gathering for System Analysis

4.2.1 Review of Literature, Procedure and Forms

The review process helps analyze existing documentation, workflows, and operational records to identify weaknesses in the current system. It also provides insight into how industry best practices can be applied to improve Bluestep Bank's information systems and operational efficiency.

4.2.1.1 Review of Literature

The literature review examines concepts, frameworks, and previous studies related to digital banking systems, agile software development, communication management, API integration, and data security.

This review provides theoretical support for understanding the organizational and technical challenges identified in Bluestep Bank.

The literature indicates that modern digital banking platforms heavily depend on:

- Agile development methodologies
- Cloud-based architectures
- Secure API integrations

- Real-time data processing
- Advanced fraud detection systems

Studies also highlight that ineffective communication, weak requirement engineering, and dependency on third-party systems can significantly affect software quality, operational stability, and customer satisfaction.

The literature review further emphasizes the importance of:

- Clear stakeholder communication
- Structured requirement analysis
- Continuous monitoring and testing
- Multi-layered data validation mechanisms

These concepts are directly relevant to the problems identified within Bluestep Bank's operational environment.

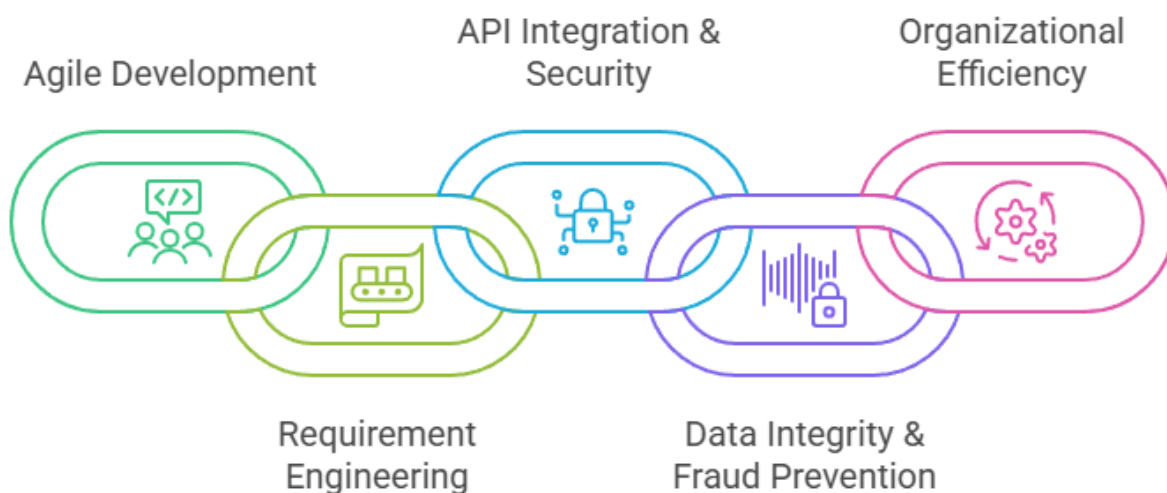


Figure 4.2: Foundations of Digital Banking Success

4.2.1.2 Review of Procedures

The review of procedures focuses on analyzing the operational workflows and software development processes followed within Bluestep Bank.

Since the organization operates using agile methodologies and cross-functional squads, procedures related to requirement gathering, communication, API integration, and data validation are critically examined.

The analysis shows that the organization follows structured sprint planning, backlog management, and iterative development processes. However, procedural gaps exist in:

- Cross-departmental communication

- Requirement validation
- API failure handling
- Fraud detection and monitoring

These procedural weaknesses create bottlenecks that affect development efficiency and system reliability.

The procedure review also highlights the importance of standardized workflows and clearly defined responsibilities to improve coordination between technical and business teams.

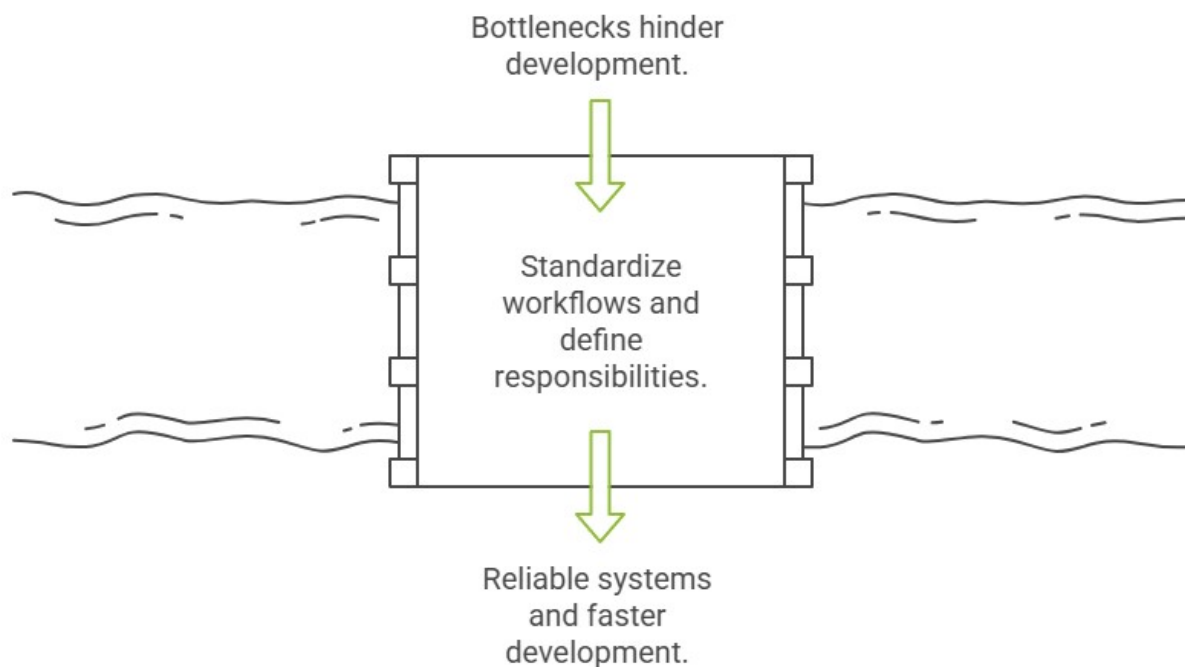


Figure 4.3: Bluestep Bank improves efficiency through procedural review

4.2.1.3 Review of Forms

The review of forms examines the digital and operational documents used within Bluestep Bank's information systems. These forms include customer onboarding forms, loan application forms, identity verification requests, internal reporting templates, and issue tracking records.

The analysis identifies that although digital forms improve efficiency and automation, inconsistencies in data entry and insufficient validation mechanisms can lead to inaccurate information processing. Some forms also rely heavily on manual verification processes, increasing operational workload and the risk of human error.

Additionally, forms connected to external APIs are vulnerable to incomplete or delayed responses, which can negatively affect customer onboarding and transaction processing.

The review highlights the need for:

- Standardized form structures
- Improved input validation
- Automated verification mechanisms
- Better integration with fraud detection systems

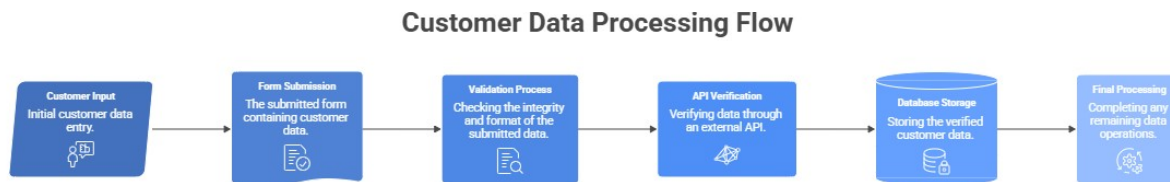


Figure 4.4: Customer Data Processing Flow

4.2.2 On Site Observation

As our selected company is an international organization, we didn't get the chance to visit the organization on site.

4.2.3 Interview

Interviews were conducted with different stakeholders associated with Bluestep Bank to gather qualitative information regarding organizational operations, communication practices, technical challenges, employee experiences, and customer satisfaction. The interviews helped identify practical challenges related to system development, agile collaboration, API dependency, and data management within the organization.

The interview process included discussions with HR personnel, engineering management, employees, and customers to obtain multiple perspectives regarding the effectiveness of the current information systems and operational workflows.

4.2.3.1 Interview with HR of Bluestep Bank

Q1. What is the organizational culture followed at Bluestep Bank?

Answer:

Bluestep Bank follows a collaborative and innovation-focused organizational culture that supports teamwork, flexibility, and continuous improvement. The company encourages employees to work in agile squads where communication and knowledge sharing are highly valued. Employees are given opportunities to contribute ideas and participate in decision-making processes related to projects and workflows. The organization also promotes a digital-first mindset, encouraging staff to adapt to new technologies and changing business needs. HR initiatives focus on maintaining employee engagement, inclusivity, and professional development. Overall, the culture aims to balance productivity with employee well-being while supporting the company's long-term digital transformation goals.

Q2. What are the major HR challenges faced in managing technical teams?**Answer:**

One of the biggest challenges is maintaining effective communication and coordination between multiple agile squads working on interconnected systems. HR also faces difficulties in balancing workloads during high-pressure project periods and ensuring employees do not experience burnout. Recruiting and retaining highly skilled technical professionals in competitive markets is another major challenge. In addition, adapting employees to continuously evolving technologies and development methodologies requires ongoing training and support. Managing remote and hybrid work environments while maintaining collaboration and productivity is also important. HR continuously works on improving engagement, communication, and work-life balance across departments.

Q3. Does the company support remote work?**Answer:**

Yes, Bluestep Bank supports hybrid and remote working models for many departments. Employees can collaborate digitally using online communication and project management tools.

Q4. Are employees satisfied with work-life balance?**Answer:**

Most employees are satisfied with the flexible working environment. However, workload pressure may increase during important project deadlines and release periods.

Q5. Does HR conduct employee performance evaluations?**Answer:**

Yes, performance evaluations are conducted regularly through structured review processes. These evaluations help monitor employee growth and identify improvement areas.

Q6. How does the company support employee development and training?**Answer:**

Bluestep Bank provides employees with various opportunities for technical and professional growth. The company organizes regular workshops, technical certification programs, leadership development sessions, and agile methodology training. Employees are encouraged to participate in online learning platforms and industry conferences to improve their skills. Mentorship programs and internal knowledge-sharing sessions are also used to help employees learn from experienced

professionals. HR collaborates with technical leaders to identify skill gaps and provide targeted training programs. These initiatives help employees stay updated with modern technologies and improve overall organizational performance.

Q7. Are communication tools provided to employees?

Answer:

Yes, employees are provided with tools such as Jira, Slack, Microsoft Teams, and Confluence. These tools help improve collaboration and project coordination.

Q8. Does HR organize team meetings?

Answer:

Yes, HR regularly organizes departmental meetings, workshops, and collaboration sessions. These activities help improve communication and employee engagement.

Q9. Is employee feedback collected regularly?

Answer:

Yes, feedback is collected through surveys, performance reviews, and one-on-one meetings. HR uses this feedback to improve organizational processes and employee satisfaction.

Q10. Does the company provide technical training?

Answer:

Yes, employees receive technical and professional development training regularly. The company invests in improving both technical and soft skills.

Q11. How does HR handle communication issues between departments?

Answer:

HR promotes structured communication practices to improve coordination between departments and agile squads. Regular cross-functional meetings, collaboration sessions, and team-building activities are arranged to strengthen relationships between teams. HR also encourages the use of communication tools such as Slack, Jira, and Microsoft Teams to improve transparency and information sharing. When conflicts or misunderstandings occur, HR works with team leaders to resolve issues quickly and maintain a collaborative working environment. Feedback sessions and

employee surveys are conducted regularly to identify communication problems and improve internal workflows. These efforts help reduce delays and improve operational efficiency.

Q12. What role does HR play in digital transformation initiatives?

Answer:

HR plays an important role in supporting Bluestep Bank's digital transformation initiatives by ensuring that employees are prepared for technological and organizational changes. The department focuses on recruiting skilled professionals with expertise in cloud computing, agile development, and modern banking technologies. HR also organizes training programs to help employees adapt to new systems and digital workflows. In addition, HR supports organizational restructuring efforts required for agile and squad-based operations. Employee engagement and change management activities are conducted to reduce resistance to new technologies and processes. Through these initiatives, HR helps create a workforce capable of supporting the organization's long-term digital strategy.

Q13. Are agile methodologies followed organization-wide?

Answer:

Yes, agile methodologies are widely implemented across development and operational teams. Sprint planning and collaborative workflows are commonly used.

Q14. Does HR support diversity and inclusion initiatives?

Answer:

Yes, Bluestep Bank promotes diversity and inclusion across the organization. HR ensures equal opportunities and encourages an inclusive workplace culture.

Q15. Is employee retention considered important?

Answer:

Yes, retaining skilled employees is considered a major organizational priority. HR focuses on employee satisfaction, career growth, and work-life balance to improve retention.

4.2.3.2 Interview with Engineering Manager of Bluestep Bank

Q1. What are the biggest technical challenges faced by the engineering teams?

Answer:

One of the biggest technical challenges is managing dependencies between multiple agile squads

working on interconnected banking services. Since the platform heavily depends on external APIs and cloud infrastructure, maintaining system reliability during third-party failures is also difficult. The engineering teams must continuously balance feature development, security requirements, and regulatory compliance. Another challenge involves handling unclear or changing requirements from business and compliance teams, which can affect development timelines. Ensuring scalability while maintaining system performance is equally important because the organization operates across multiple Nordic countries. The teams also face pressure to deliver features quickly without compromising system quality or security.

Q2. Are cloud technologies used in development?

Answer:

Yes, Bluestep Bank uses cloud-based infrastructure to support scalability, monitoring, and deployment processes. Cloud technologies help improve system flexibility and reliability.

Q3. Is CI/CD implemented?

Answer:

Yes, continuous integration and continuous deployment pipelines are actively used. These practices help automate testing and deployment processes.

Q4. Are APIs critical to operations?

Answer:

Yes, many banking operations depend heavily on APIs for identity verification, data exchange, and system integration. API reliability is very important for service continuity.

Q5. Is automated testing used?

Answer:

Yes, automated testing is integrated into the development workflow. It helps identify bugs early and improves software quality.

Q6. How are software development projects managed?

Answer:

Software development projects are managed using agile methodologies with sprint-based planning and continuous collaboration. Product Owners, Engineering Managers, and developers work closely together to prioritize tasks and maintain sprint backlogs. Regular stand-up meetings, sprint reviews, and retrospectives are conducted to monitor progress and identify issues. Jira and other

project management tools are used to track tasks, dependencies, and deadlines. Continuous integration and automated testing help maintain development quality throughout the lifecycle. This structured agile process helps the organization remain flexible while improving delivery speed and collaboration.

Q7. How does the engineering team handle ambiguous requirements?

Answer:

When requirements are unclear, engineering teams collaborate closely with Product Owners and stakeholders to clarify technical expectations before development begins. Requirement refinement sessions and planning meetings are used to identify missing details and reduce misunderstandings. Developers also participate in discussions to ensure that technical feasibility and implementation concerns are considered early. In some cases, prototypes or proof-of-concept solutions are created to validate requirements before full development. Documentation and acceptance criteria are updated continuously throughout the project lifecycle. These practices help reduce rework and improve overall development efficiency.

Q8. Are microservices used in the system architecture?

Answer:

Yes, the organization uses microservice-based architecture for scalability and flexibility. This approach allows teams to work independently on different services.

Q9. How does the organization manage system reliability?

Answer:

System reliability is managed through continuous monitoring, automated testing, and cloud-based infrastructure management. The engineering teams use monitoring tools to detect performance issues, API failures, and security threats in real time. Backup systems and failover mechanisms are also implemented to reduce downtime during service disruptions. Regular testing and deployment pipelines help ensure that new updates do not negatively affect existing services. Security and compliance checks are integrated into development workflows to maintain system stability. Continuous monitoring and rapid incident response play a major role in maintaining reliable banking operations.

Q10. Do teams use Jira for project management?

Answer:

Yes, Jira is widely used for sprint planning, backlog management, and issue tracking. It helps teams monitor project progress efficiently.

Q11. Is system monitoring performed continuously?**Answer:**

Yes, system monitoring tools operate continuously to track performance and detect failures. Monitoring helps reduce downtime and improve reliability.

Q12. What improvements are needed in the current development process?**Answer:**

The development process could be improved by strengthening communication between squads and improving requirement clarity before implementation begins. Better dependency management between teams would reduce delays caused by blocked tasks and incomplete integrations. The organization could also benefit from stronger fallback mechanisms for external APIs to improve service continuity. Additional automation in testing and monitoring processes would help detect issues earlier and reduce operational workload. Improving documentation consistency and cross-team visibility would also support smoother collaboration. These improvements would increase development efficiency and system reliability.

Q13. Are developers involved in planning discussions?**Answer:**

Yes, developers actively participate in planning and requirement discussions. Their involvement helps improve technical feasibility and implementation accuracy.

Q14. Does the organization conduct code reviews?**Answer:**

Yes, code reviews are mandatory before deployment. This process helps maintain coding standards and reduce software defects.

Q15. Is cybersecurity considered during development?**Answer:**

Yes, cybersecurity practices are integrated throughout the development lifecycle. Security testing and compliance checks are regularly performed.

4.2.3.3 Interview with Employees of Bluestep Bank

Q1. How would you describe the working environment?

Answer:

The working environment at Bluestep Bank is collaborative, technology-focused, and fast-paced. Employees work in agile squads where teamwork and communication are important for achieving project goals. The organization encourages innovation and supports employees in learning modern technologies and development practices. Most teams maintain a flexible work structure, including remote and hybrid working options. Employees are generally given opportunities to share ideas and contribute to decision-making processes. Although project deadlines can create pressure, the overall environment is supportive and growth-oriented.

Q2. What communication challenges do employees face?

Answer:

Employees sometimes face communication delays between squads working on dependent services or APIs. Since teams operate semi-independently, updates and requirement changes may not always be shared quickly across departments. In some situations, excessive reliance on documentation instead of direct discussion can slow clarification processes. Communication gaps may also occur during remote collaboration when teams work across different schedules. These challenges can create confusion, delay development tasks, and affect sprint timelines. Improving cross-team visibility and coordination would help reduce these issues.

Q3. How effective are the current collaboration tools?

Answer:

The collaboration tools used by the organization are generally effective for managing agile workflows and communication. Tools such as Jira, Slack, Microsoft Teams, and Confluence help teams track tasks, share updates, and organize documentation efficiently. Employees can easily monitor sprint progress and collaborate remotely using these platforms. However, when multiple teams are involved in large projects, communication can still become fragmented despite the availability of these tools. Some employees believe that better integration and more direct interaction between teams would further improve collaboration. Overall, the tools support productivity but cannot completely eliminate coordination challenges.

Q4. Do employees attend daily stand-ups?

Answer:

Yes, daily stand-up meetings are part of the agile workflow. These meetings help teams discuss progress, blockers, and priorities.

Q5. Is remote collaboration effective?

Answer:

Yes, remote collaboration is generally effective using digital communication tools. However, delays can sometimes occur during cross-team coordination.

Q6. Are technical resources sufficient?

Answer:

Yes, employees are provided with the necessary hardware, software, and development tools. Technical support is also available when needed.

Q7. Is teamwork encouraged?

Answer:

Yes, teamwork and collaboration are strongly encouraged across agile squads. Team-based problem-solving is a key part of the work culture.

Q8. Are deadlines realistic?

Answer:

Most deadlines are achievable with proper planning and coordination. Tight timelines may occur during urgent releases or incidents.

Q9. Are workloads manageable?

Answer:

Most employees feel that workloads are manageable during normal development cycles. Agile sprint planning helps distribute tasks and prioritize work effectively across squads. However, workloads can become stressful during tight deadlines, production incidents, or large-scale feature releases. Employees may also experience pressure when dealing with unexpected API failures or urgent compliance updates. Management generally attempts to maintain a balance between productivity and employee well-being by supporting flexible work arrangements. Continuous workload monitoring helps reduce burnout and improve employee satisfaction.

Q10. Are employees satisfied with agile practices?

Answer:

Most employees appreciate the flexibility and collaboration provided by agile methodologies. Agile practices help improve communication and productivity.

Q11. Is management supportive?

Answer:

Yes, management generally supports employees and encourages open communication. Team leaders often help resolve technical and operational challenges.

Q12. Are system downtimes a concern?

Answer:

Yes, occasional downtime caused by external API issues can affect workflows. Teams monitor these incidents closely to reduce disruptions.

Q13. Are training opportunities available?

Answer:

Yes, the organization regularly provides training programs and workshops. Employees are encouraged to improve both technical and professional skills.

Q14. Is job satisfaction high?

Answer:

Overall job satisfaction is positive due to the collaborative environment and learning opportunities. Some stress may occur during high-pressure project periods.

Q15. What improvements would employees like to see?

Answer:

Employees would like clearer requirements, stronger communication between departments, and better dependency management between teams. Many employees believe that reducing ambiguity in requirements would improve development efficiency and reduce rework. Improved visibility into other teams' progress and priorities would also help teams coordinate tasks more effectively. Some employees suggest increasing automation in testing and monitoring to reduce manual workload. Additional technical training and knowledge-sharing sessions would also benefit

employees working with rapidly changing technologies. These improvements could enhance collaboration, productivity, and overall job satisfaction.

4.2.3.4 Interview with Customers of Bluestep Bank

Q1. How would you describe your experience with Bluestep Bank' digital services?

Answer:

Most customers describe Bluestep Bank's digital services as convenient, modern, and easy to use. The online banking platform allows customers to complete many banking activities without visiting physical branches. Customers appreciate the speed of online applications and account management features. The mobile and web interfaces are generally user-friendly and accessible. However, some customers have experienced delays during identity verification and onboarding processes. Overall, customers view the platform positively because of its accessibility and digital-first approach.

Q2. Is the mobile platform easy to use?

Answer:

Yes, most customers find the mobile application simple and user-friendly. Navigation and account management features are generally easy to access.

Q3. Have customers faced login issues?

Answer:

Occasionally, customers experience login delays during external verification service disruptions. These issues are usually temporary.

Q4. Are transactions processed quickly?

Answer:

Yes, most transactions are processed efficiently and within expected timeframes. Customers are generally satisfied with transaction speed.

Q5. What problems have you experienced while using the platform?

Answer:

Some customers reported delays during login, identity verification, or onboarding when external verification services experienced issues. Others mentioned occasional slow response times during peak usage periods. A few customers also found certain forms and verification steps slightly

complicated. Technical interruptions can sometimes affect customer confidence in the platform's reliability. Despite these issues, most customers believe the problems are temporary and manageable. Many customers appreciate that customer support usually responds quickly to resolve concerns.

Q6. Is customer information protected properly?

Answer:

Yes, customers believe that strong security measures are in place to protect sensitive information. Security features increase trust in the platform.

Q7. How secure do you feel using the bank's digital services?

Answer:

Most customers feel that Bluestep Bank provides a secure and trustworthy digital banking environment. Strong authentication processes and verification systems help increase customer confidence. Customers appreciate security notifications, verification checks, and account monitoring features. Many users believe that the organization prioritizes data protection and regulatory compliance. Although occasional API issues may affect access temporarily, customers generally trust the platform's security measures. Overall, security perception among customers is positive.

Q8. Are customer support services effective?

Answer:

Customers generally consider the support services responsive and helpful. Support representatives provide assistance through multiple channels such as phone, email, and online chat systems. Most issues are resolved within a reasonable time frame, especially for common account or onboarding problems. Customers appreciate receiving guidance during technical or verification-related difficulties. However, response times may occasionally increase during high-demand periods. Overall, customer support contributes positively to user satisfaction.

Q9. Is online support accessible?

Answer:

Yes, customers can easily access support services through multiple online channels. Response quality is generally considered good.

Q10. Are notifications helpful?

Answer:

Yes, notifications help customers stay informed about transactions and account activity. They improve user awareness and security.

Q11. Is the registration process simple?**Answer:**

Most customers find the registration process manageable, though verification delays can sometimes occur. Instructions are generally clear.

Q12. Would customers recommend the bank?**Answer:**

Yes, many customers would recommend Bluestep Bank because of its convenient digital services and flexible banking solutions.

Q13. Are digital services reliable?**Answer:**

The services are generally reliable, though occasional interruptions may occur due to external API dependency. Most issues are resolved quickly.

Q14. Are customers satisfied overall?**Answer:**

Overall customer satisfaction is positive because of the platform's accessibility, convenience, and digital banking features.

Q15. What improvements would customers recommend?**Answer:**

Customers would like faster onboarding and identity verification processes, especially during periods of high demand. Some users recommend simplifying certain forms and improving the clarity of instructions during registration. Faster recovery from external API disruptions would also improve the customer experience. Customers additionally suggest enhancing mobile application performance and reducing occasional delays in notifications. Improved communication during service interruptions would help maintain customer confidence. These improvements could increase overall satisfaction and platform reliability.

4.2.4 Questionnaires

Questionnaires were used as an additional data collection method to gather quantitative information from different stakeholders associated with Bluestep Bank. The questionnaire method helped collect structured responses regarding communication efficiency, system reliability, employee satisfaction, requirement clarity, and customer experience within the organization.

The questionnaires were distributed among HR personnel, engineering management, employees, and customers to obtain multiple perspectives about the effectiveness of Bluestep Bank's operational and information systems. Most questions were designed in a simple Yes/No format to ensure faster response collection and easier analysis.

4.2.4.1 Data Collection Strategy

The data collection strategy focused on obtaining both organizational and technical insights from different stakeholder groups involved with Bluestep Bank's operations and digital banking services. A structured questionnaire approach was selected because it allows consistent data collection from a large number of participants within a limited time.

The questionnaires were prepared based on the key problem areas identified during the preliminary analysis phase, including:

- Cross-departmental communication gaps
- Ambiguous requirements
- External API dependency
- Data integrity and fraud risks

Different sets of questionnaires were designed for HR personnel, engineering management, employees, and customers to ensure that responses reflected the specific experiences and responsibilities of each group.

4.2.4.2 Questionnaire with HR of Bluestep Bank

No.	Question	YES	NO
1	Does the company follow agile working practices?		
2	Does HR organize regular employee meetings?		
3	Are communication issues reported frequently?		
4	Does the company provide technical training programs?		
5	Is remote work supported by the organization?		
6	Are employees satisfied with current collaboration tools?		
7	Does HR conduct employee satisfaction surveys?		

8	Are workload management issues observed?		
9	Does the organization support diversity and inclusion?		
10	Are communication delays affecting productivity?		
11	Is employee retention considered a major priority?		
12	Does HR assist in conflict resolution?		
13	Are agile methodologies used across departments?		
14	Is cross-team collaboration encouraged?		
15	Does HR support digital transformation initiatives?		

4.2.4.3 Questionnaire with Employees of Bluestep Bank

Rating Scale:

Scale	Meaning
1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly Agree

No.	Question	1	2	3	4	5
1	Communication between teams is effective.					
2	Agile practices improve team productivity.					
3	Requirements are clearly defined before development begins.					
4	Collaboration tools such as Jira and Slack are effective.					
5	Workloads are manageable during most projects.					
6	Cross-departmental coordination is efficient.					
7	Management provides adequate technical support.					
8	Training opportunities improve employee skills.					
9	External API issues negatively affect workflows.					

10	Communication delays impact development speed.					
11	Team meetings help resolve technical issues quickly.					
12	The organization effectively manages project deadlines.					
13	System monitoring and support processes are reliable.					
14	Employees are satisfied with remote collaboration tools.					
15	Overall job satisfaction is positive.					

4.2.4.4 Questionnaire with Customers of Bluestep Bank

No	Questions	1	2	3	4	5
1	The digital banking platform is easy to use.					
2	Online transactions are processed efficiently.					
3	The registration process is simple and understandable.					
4	Customer support services are responsive.					
5	The platform provides a secure banking experience.					
6	Identity verification processes are efficient.					
7	Mobile banking services are reliable.					
8	Notifications and alerts are helpful.					
9	API-related delays affect customer experience.					
10	The system performs reliably during peak usage periods.					
11	Banking services are accessible without major issues.					
12	Customers trust the bank's data security practices.					
13	Technical issues are resolved quickly.					
14	Overall customer experience is satisfactory.					
15	Customers would recommend Bluestep Bank to					

	others.					
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4.2.4.5 Questionnaire with Engineering Manager of Bluestep Bank

No.	Questions	Options
1	Which development methodology is primarily used at Bluestep Bank?	a) Waterfall b) Spiral c) Agile d) RAD
2	Which tool is mainly used for project and sprint management?	a) Trello b) Jira c) Asana d) Monday.com
3	What is the biggest technical challenge faced by teams?	a) Hardware failure b) Communication gaps c) Office infrastructure d) Recruitment
4	Which architecture style is mainly used in the system?	a) Monolithic b) Peer-to-Peer c) Microservices d) Standalone
5	Which factor most affects system reliability?	a) Employee attendance b) External API dependency c) Office location d) Internet speed
6	Which cloud-related approach is used for scalability?	a) Local servers only b) Hybrid cloud c) Manual storage d) Offline systems
7	Which process helps identify software bugs early?	a) Manual deployment b) Automated testing c) Verbal reporting d) Documentation review
8	Which department works most closely with engineering teams?	a) Marketing b) Product Management c) Logistics d) Sales
9	What is the main reason for requirement ambiguity?	a) Excessive hardware usage b) Lack of technical detail c) Internet failure d) Employee shortage
10	Which communication tool is most commonly used internally?	a) WhatsApp b) Slack c) Facebook d) Zoom only
11	What type of monitoring is used for system	a) Manual monthly checks

	reliability?	b) Continuous monitoring c) No monitoring d) Annual review
12	Which issue most commonly causes development delays?	a) Furniture problems b) API failures and dependency delays c) Lighting problems d) Payroll issues
13	Which practice improves deployment efficiency?	a) Manual coding only b) CI/CD pipelines c) Paper documentation d) Offline testing
14	Which security approach is integrated into development?	a) Reactive security only b) Security-by-design c) No security processes d) External-only security
15	Which improvement would most benefit the development process?	a) Reduced communication b) Better requirement clarity c) Less monitoring d) Fewer meetings

4.3 Information Representation / Analysis

This section presents the analytical representation of data collected through questionnaires conducted with HR personnel, Engineering Managers, Employees, and Customers of Bluestep Bank. The collected responses are analyzed using charts and graphs to better visualize organizational trends, operational issues, employee satisfaction, customer experience, and technical challenges.

The graphical representations help identify:

- Communication efficiency
- Requirement clarity issues
- API dependency risks
- Employee satisfaction levels
- Customer experience trends
- Technical operational challenges

The following datasets can be directly used to create pie charts, bar charts, line graphs, and stacked charts in Microsoft Word, Excel, or any graphing tool.

4.3.1 Plot various charts and graphs of the above mentioned information gathering tools

HR Questionnaire Analysis

Sample Size: 1 HR Broad of 20 HRs

Response Format: Dichotomous Questions (Yes/No)

Total Questions: 15

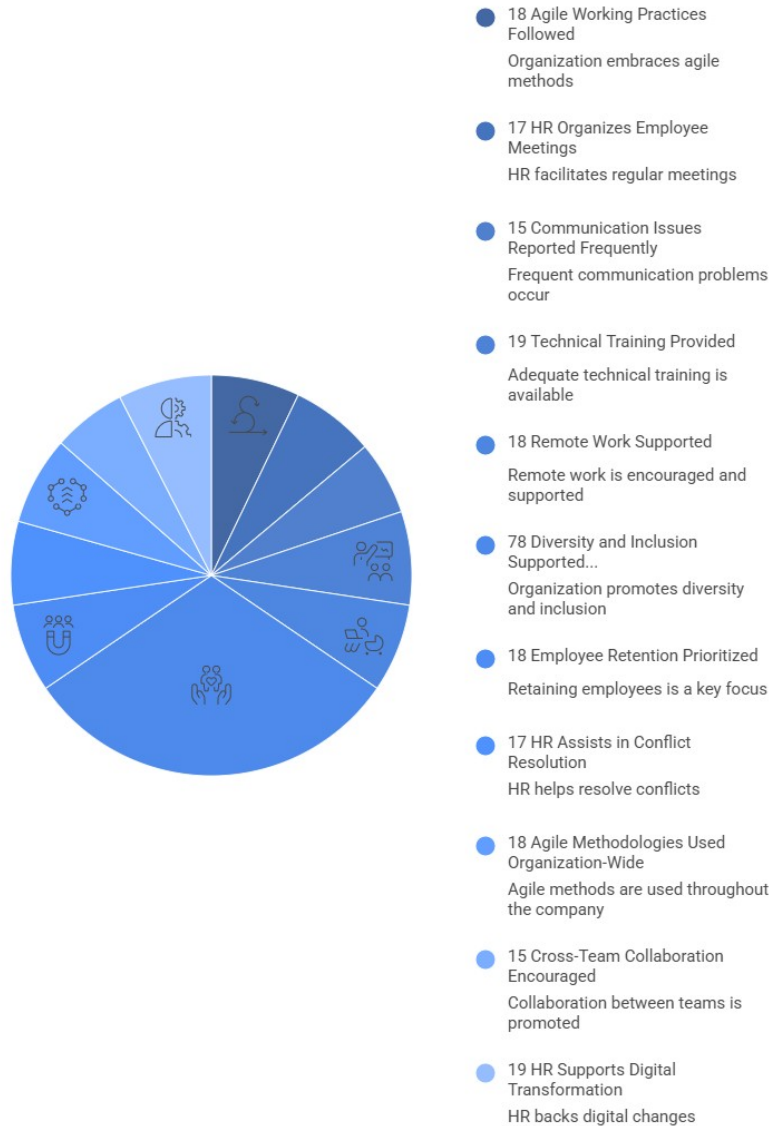


Figure 4.5: Distribution of Responses in HR Questionnaire

Employee Questionnaire Analysis

Sample Size: 15 Employee Responses

Response Format: 5-point Likert Scale (1-5)

Total Questions: 15

Q1. Communication between teams is effective

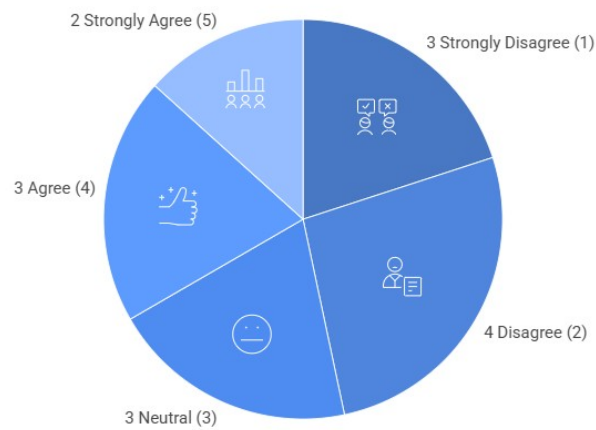


Figure 4.6: Distribution of Employee Ratings for Q1
Q2. Agile practices improve productivity

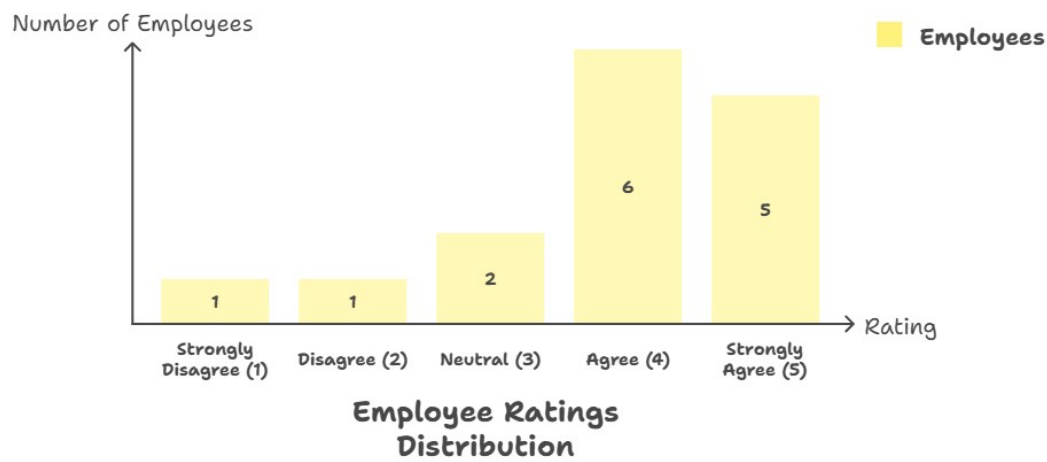


Figure 4.7: Agile practices – employee ratings distribution
Q3. Requirements are clearly defined before development begins

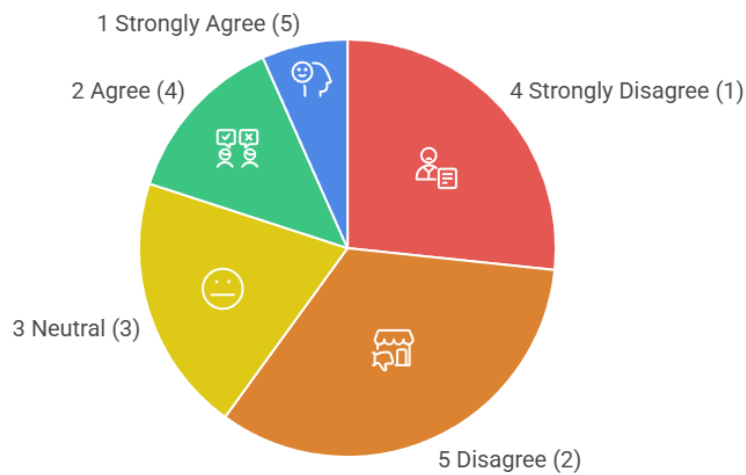


Figure 4.8: Employees thought about clear requirements before development

Q4. Collaboration tools are effective

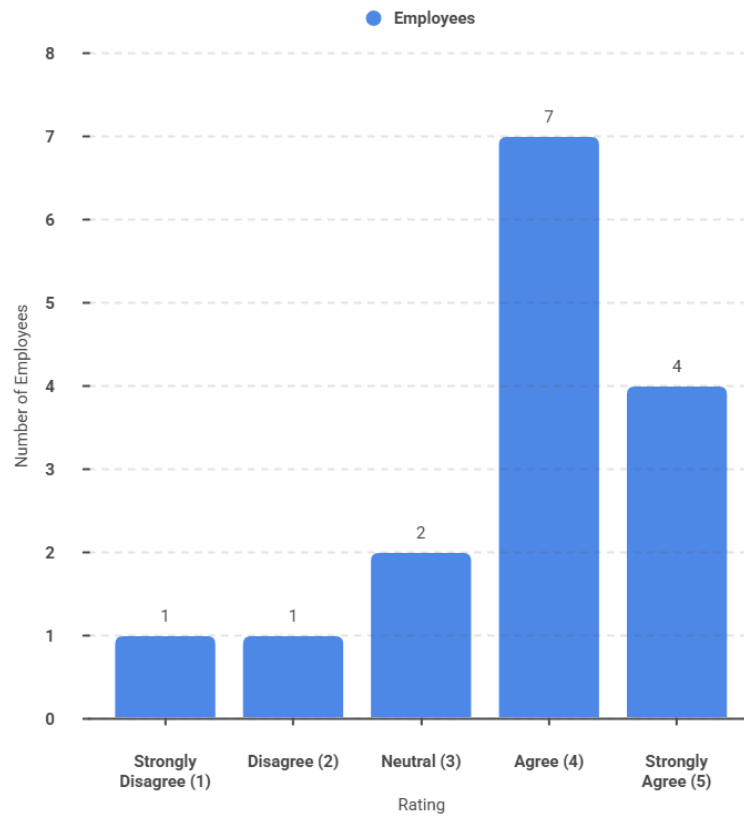


Figure 4.9: Employees thought about effectiveness of collaboration tools

Q5. API issues negatively affect workflows



Figure 4.10: How API issues negatively affect workflows

Q6. Cross-team coordination is efficient

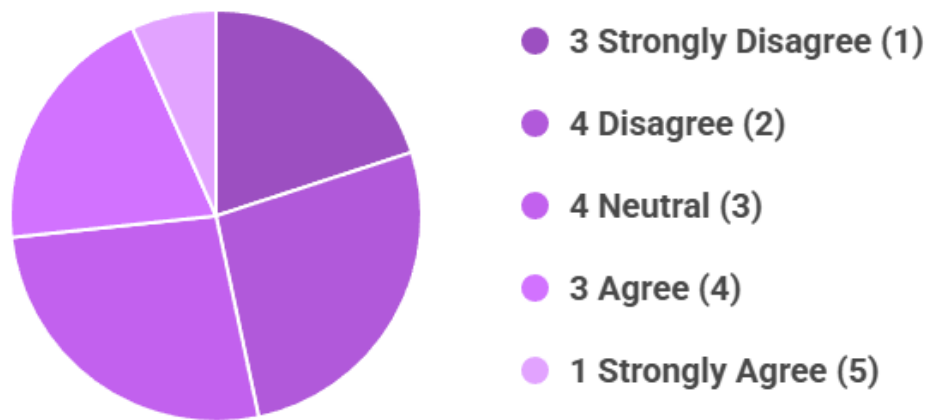


Figure 4.11: Cross-team coordination efficiency

Q7. Technical support provided by management is adequate

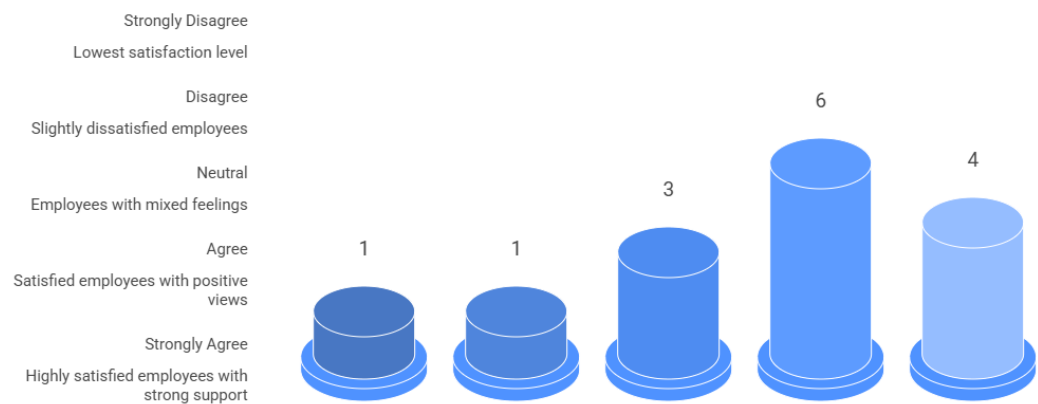


Figure 4.12: Employees thought about technical support

Q8. Overall job satisfaction is positive

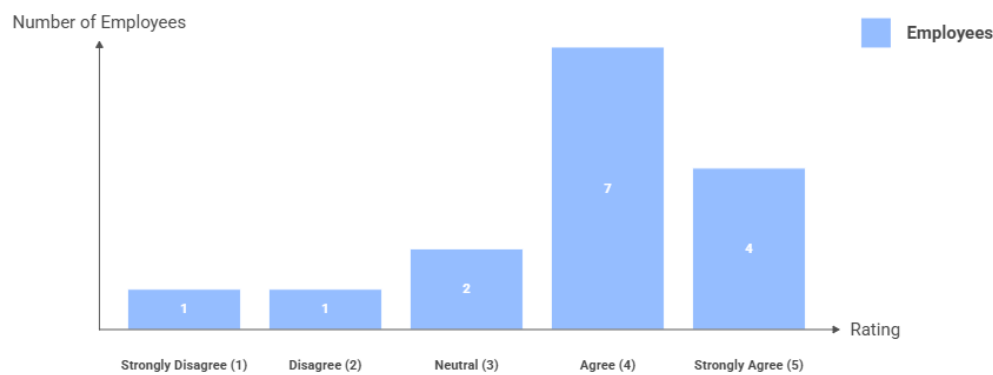


Figure 4.13: Overall job satisfaction graph

Engineering Manager Questionnaire Analysis
Sample Size: 15 Engineering Manager Responses
Response Format: Multiple Choice Questions
Total Questions: 15

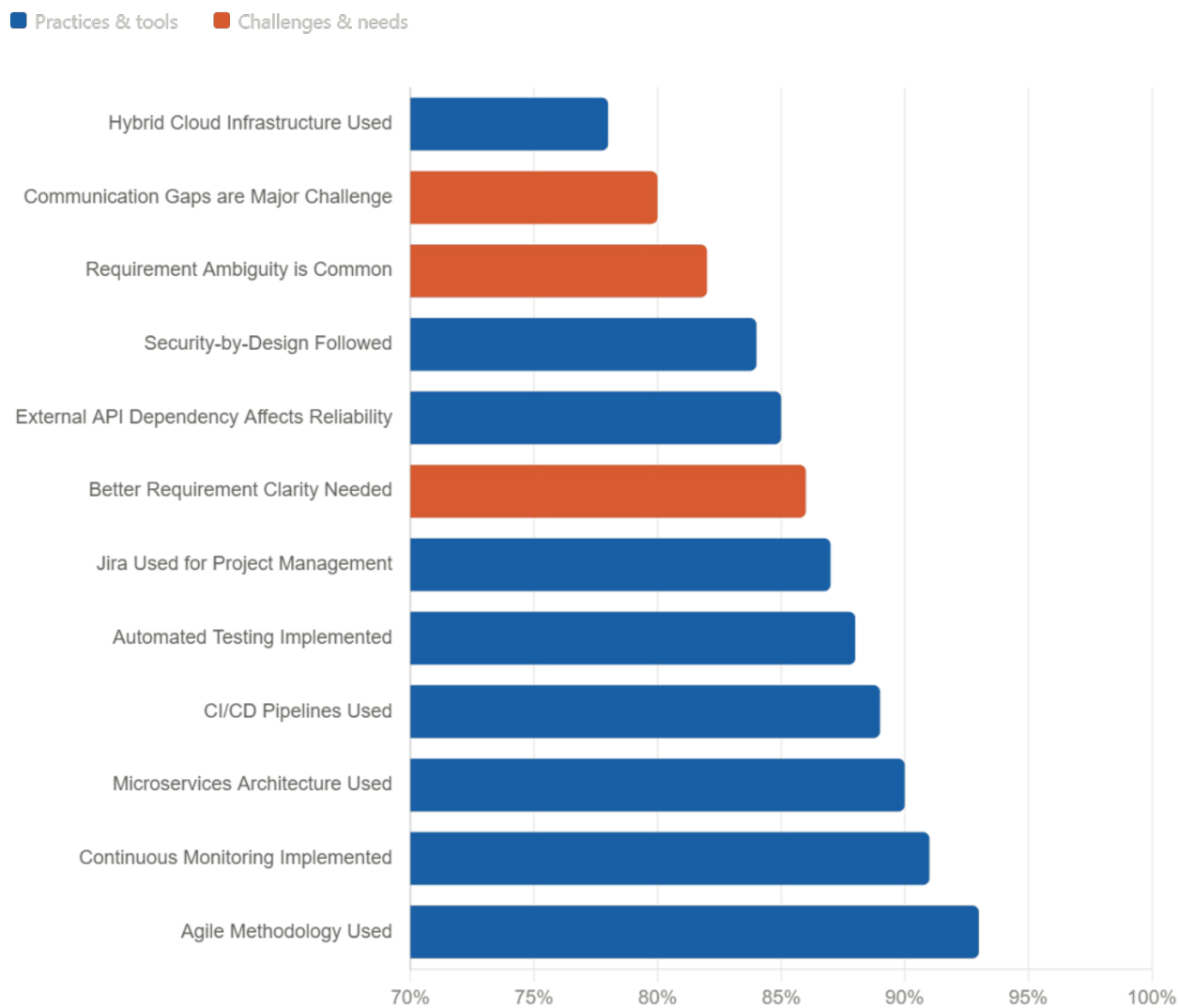


Figure 4.14: Graph about the answers of Engineering Managers responses

Customer Questionnaire Analysis

Sample Size: 20 Customers Responses

Response Format: 5-point Likert Scale (1-5)

Total Questions: 15

Q1. The digital banking platform is easy to use

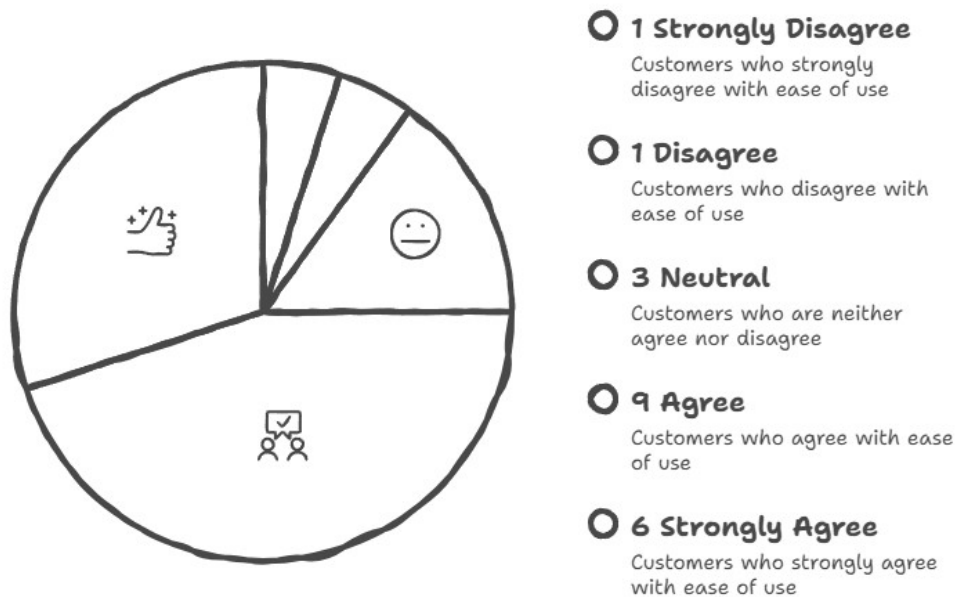


Figure 4.15: Customer Ratings of Digital Banking Platform Ease of Use

Q2. Online transactions are processed efficiently

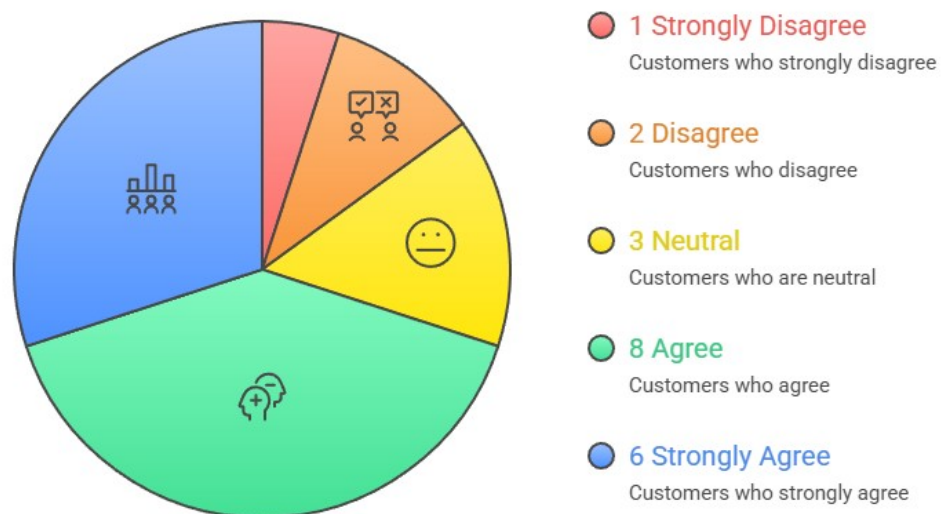


Figure 4.16: Customer Ratings of Online Transactions efficiency

Q3. Identity verification processes are efficient

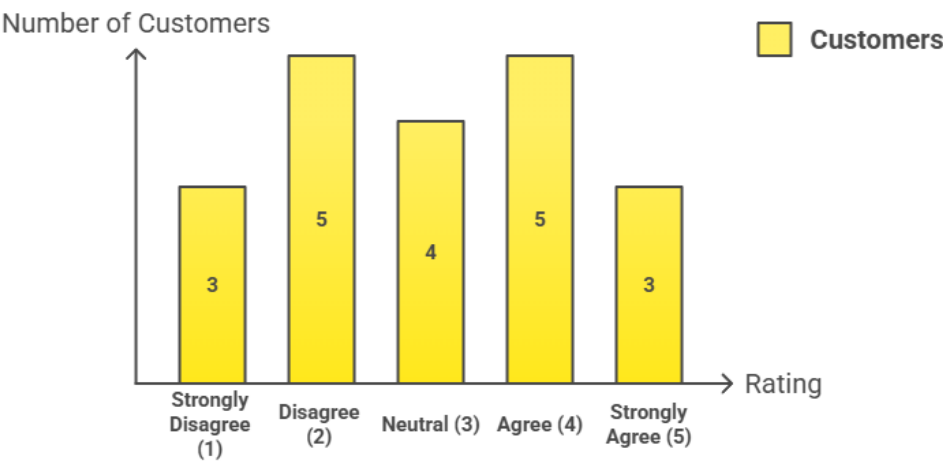


Figure 4.17: Customer Ratings of Identity Verification Process Efficiency

Q4. The platform provides a secure banking experience

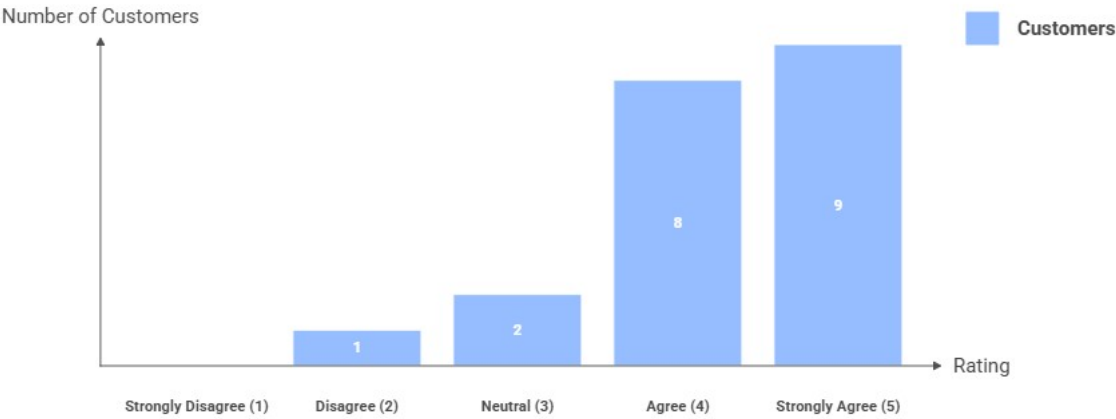


Figure 4.18: Customer Ratings of Secure Banking Experiences

Q5. API-related delays affect customer experience

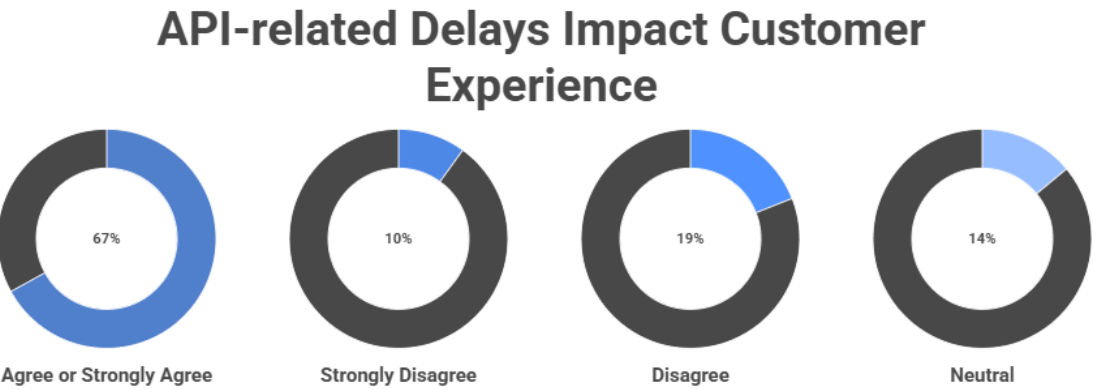


Figure 4.19: API-related Delays Impact Customer Experience

4.3.2 DFD of Existing System

The subsystem-level DFD analysis highlights that most operational challenges within Bluestep Bank originate from communication inefficiencies, unclear requirements, external dependency risks, and complex data validation processes.

4.3.2.1 DFD of Whole Existing System

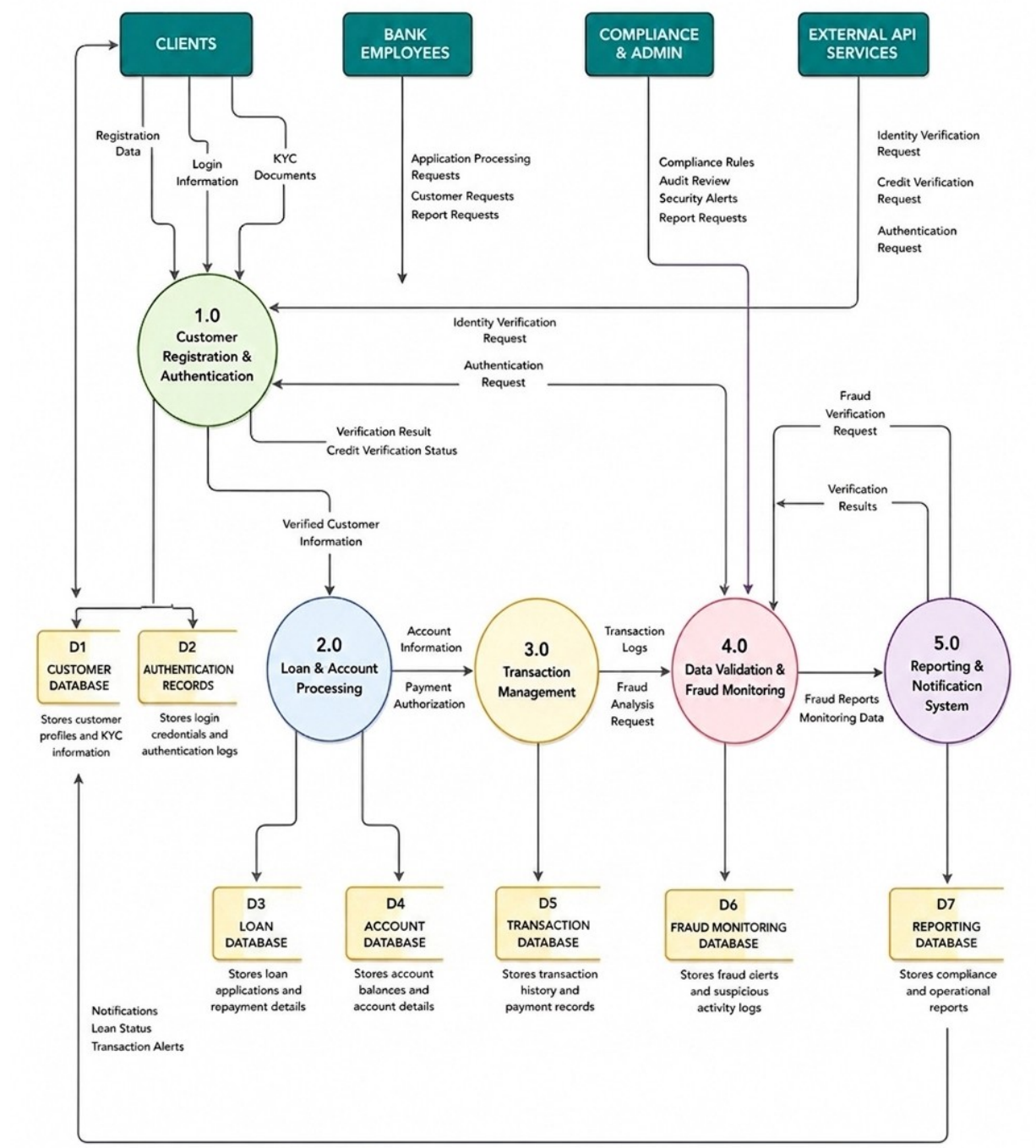


Figure 4.20: DFD of whole existing system

Detailed Process Analysis:

- **Process 1.0 – Communication & Collaboration Management Subsystem**
This process manages communication and coordination among different technical teams, including frontend, backend, and API development teams. It facilitates task assignment, dependency tracking, project updates, and information sharing to reduce communication gaps and development bottlenecks. The subsystem stores collaboration records, team interactions, and API dependency information in the Team Collaboration Database to improve workflow efficiency and cross-departmental coordination.
- **Process 2.0 – Requirement Engineering & Validation Subsystem**
This process handles the collection, analysis, clarification, and validation of business and compliance requirements. Requirements received from business stakeholders and compliance teams are reviewed and transformed into clear technical specifications. The subsystem manages requirement changes, approvals, and validation activities while maintaining a centralized repository of requirement documents to reduce ambiguity and minimize rework during development.
- **Process 3.0 – External API Integration & Verification Subsystem**
This process manages the integration of third-party services such as identity verification, authentication, credit verification, and other external APIs. It processes API requests, validates responses, monitors service availability, and records integration activities. The subsystem helps ensure reliable communication between Bluestep Bank AB and external service providers while maintaining logs for auditing and troubleshooting purposes.
- **Process 4.0 – Data Validation & Fraud Detection Subsystem**
This process validates incoming customer, transaction, and verification data to ensure accuracy, consistency, and compliance with organizational policies. Advanced fraud detection mechanisms analyze suspicious patterns, risk indicators, and verification results to identify potential fraudulent activities. The subsystem generates fraud alerts, validation reports, and monitoring information while storing security-related records in the Fraud Monitoring Database.
- **Management & Reporting Module**
This module consolidates information from all subsystems to generate operational reports, performance metrics, compliance reports, audit records, and management dashboards. It provides decision-makers with insights regarding communication efficiency, requirement status, API performance, validation results, and fraud monitoring activities. The module supports organizational oversight and continuous process improvement across the entire system.

Data Store Description

- D1 – Customer Database: Stores customer profile, KYC details, authentication credentials, contact information, etc.
- D2 – Account & Transaction Database: Stores account details, balances, transaction history, payment records, etc.
- D3 – Loan Database: Stores loan applications, approval details, loan disbursement, repayment schedule, etc.
- D4 – Audit / Log Database: Stores system logs, user activities, exception logs, API logs, audit trails, etc.

System Overview

- Customers interact with the system through web/mobile channels.
- Employees and administrators access internal modules for servicing and monitoring.
- System relies on multiple external APIs for KYC, credit checks, payments, fraud detection.
- Data is maintained in separate databases and logs for operational and compliance needs.
- Information flows between external entities, processes, data stores and APIs.

4.3.2.2 DFD of Key Subsystems Addressing Organizational Issues

This section presents the Data Flow Diagram (DFD) analysis of key subsystems within Bluestep Bank that directly relate to the major organizational and technical issues identified during the investigation process. The purpose of these subsystem-level DFDs is to provide a clearer understanding of how information moves through different functional areas of the organization and where operational bottlenecks occur.

The analysis focuses primarily on four critical organizational issues identified earlier in the report:

1. Cross-Departmental Communication Gaps
2. Ambiguous Requirements
3. External API Fragility
4. Data Integrity and Fraud Risks

Each subsystem diagram represents the flow of data between users, departments, external services, and internal databases. These DFDs help identify weaknesses in communication, validation, dependency management, and data handling processes that negatively affect operational efficiency and software system reliability.

A. Communication & Collaboration Management Subsystem Purpose

This subsystem manages communication and coordination between Product Owners, Engineering Teams, Data Teams, and Compliance Departments during software development and project execution.

Subsystem Description

One of the major organizational problems identified in Bluestep Bank is inefficient communication between different agile squads and departments. Since teams work independently on different microservices and banking modules, delays frequently occur when one team depends on another for APIs, approvals, or technical clarification.

The Communication & Collaboration Management Subsystem ensures that:

- Development requests are properly tracked
- Team dependencies are visible
- Requirement updates are communicated instantly
- Sprint progress is monitored centrally

The subsystem relies heavily on collaboration platforms such as Jira, Confluence, Microsoft Teams, and internal workflow systems.

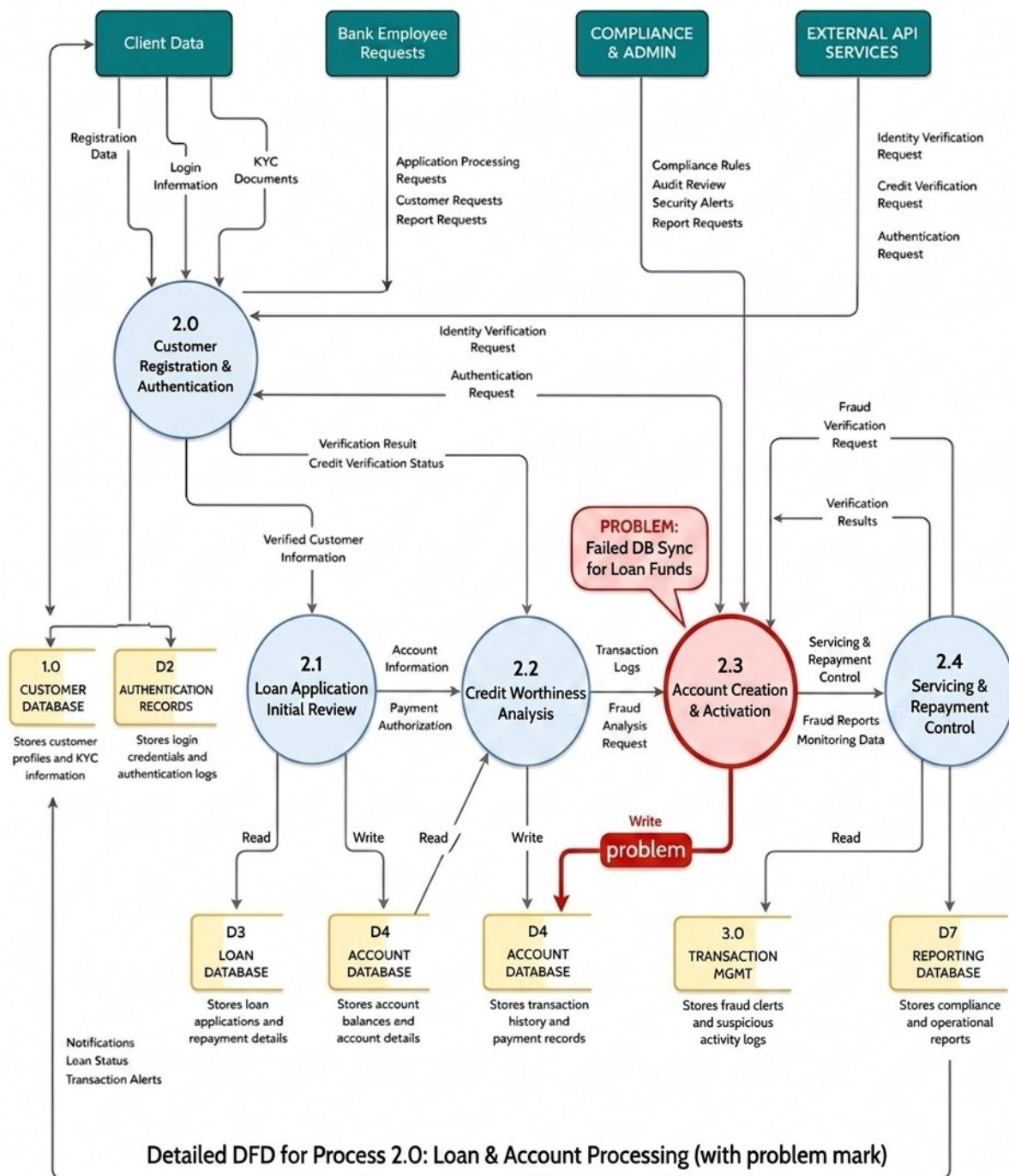


Figure 4.21: Level 1 DFD: Communication & Collaboration Management Subsystem

B. Requirement Engineering & Validation Subsystem

Purpose

This subsystem handles requirement collection, analysis, validation, approval, and communication between stakeholders and engineering teams.

Subsystem Description

Ambiguous requirements were identified as one of the most critical organizational problems affecting Bluestep Bank's software development lifecycle. Business stakeholders and compliance teams frequently provide high-level requirements that lack sufficient technical detail for direct implementation.

This subsystem ensures:

- Requirements are documented properly
- Stakeholders validate requirements before development
- Developers participate in clarification sessions
- Scope changes are tracked systematically

The subsystem aims to reduce rework, development confusion, and implementation inconsistencies.

Analytical Discussion

The analysis reveals that unclear and unstable requirements create major inefficiencies within the software development process. Developers spend additional time clarifying requirements and modifying already-developed modules when requirement changes occur during implementation.

Implementing stronger requirement validation mechanisms, stakeholder workshops, and requirement traceability systems would improve communication accuracy and reduce development risks.

Detailed DFD for Process 3.0: Transaction Management (with problem mark)

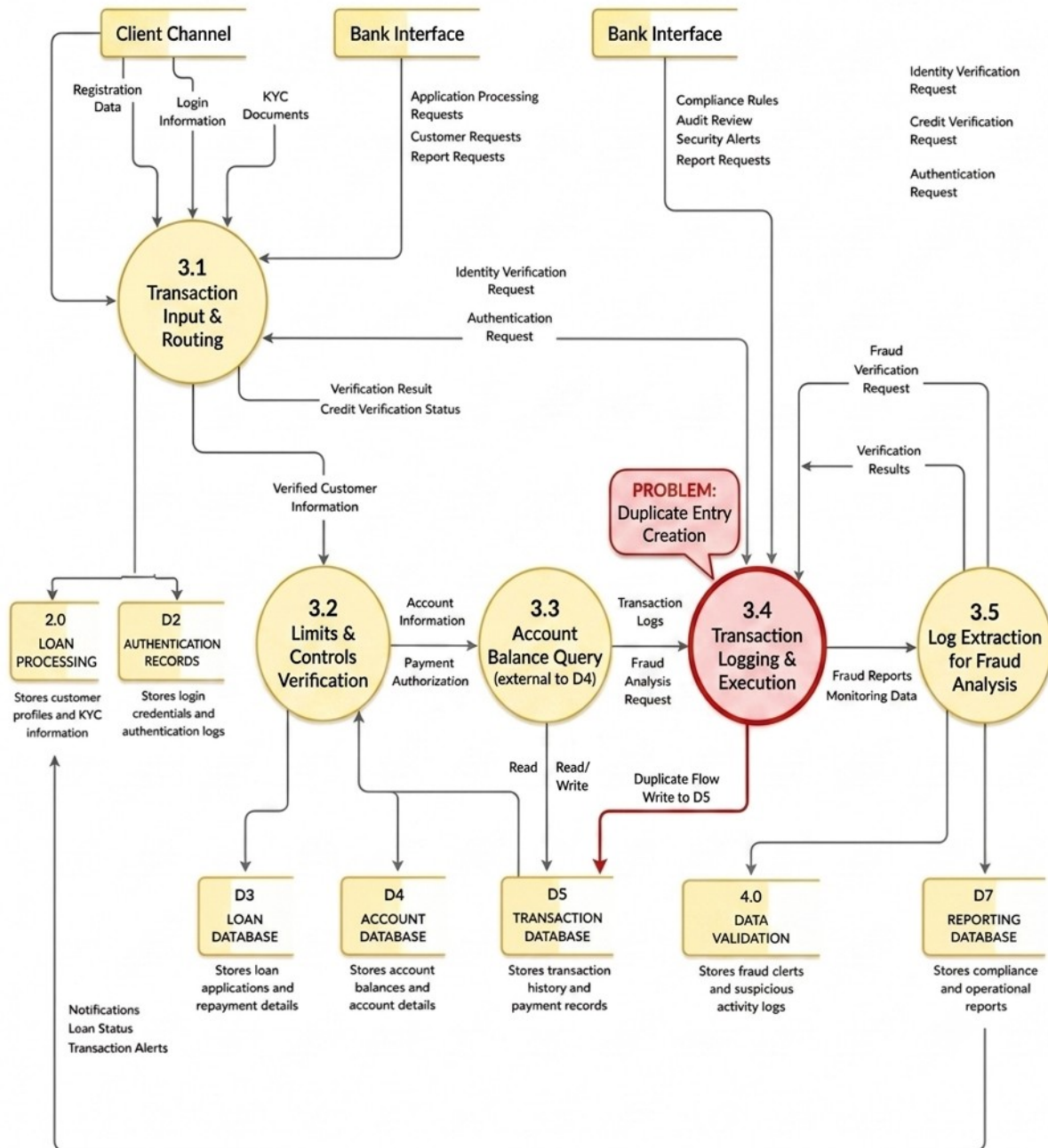


Figure 4.22: Level 1 DFD: Requirement Engineering & Validation Subsystem

C. External API Integration & Verification Subsystem

Purpose

This subsystem manages interactions between Bluestep Bank's internal systems and external third-party services such as BankID, government verification APIs, and credit bureau systems.

Subsystem Description

Bluestep Bank relies heavily on external APIs for identity verification, credit scoring, customer authentication, and transaction validation. Although these integrations are essential for regulatory compliance and digital banking operations, they introduce significant operational risks.

If external systems experience downtime or latency:

- Customer onboarding is delayed
- Login authentication may fail
- Loan verification processes become unavailable
- User experience deteriorates

The subsystem monitors and manages all external API communications.

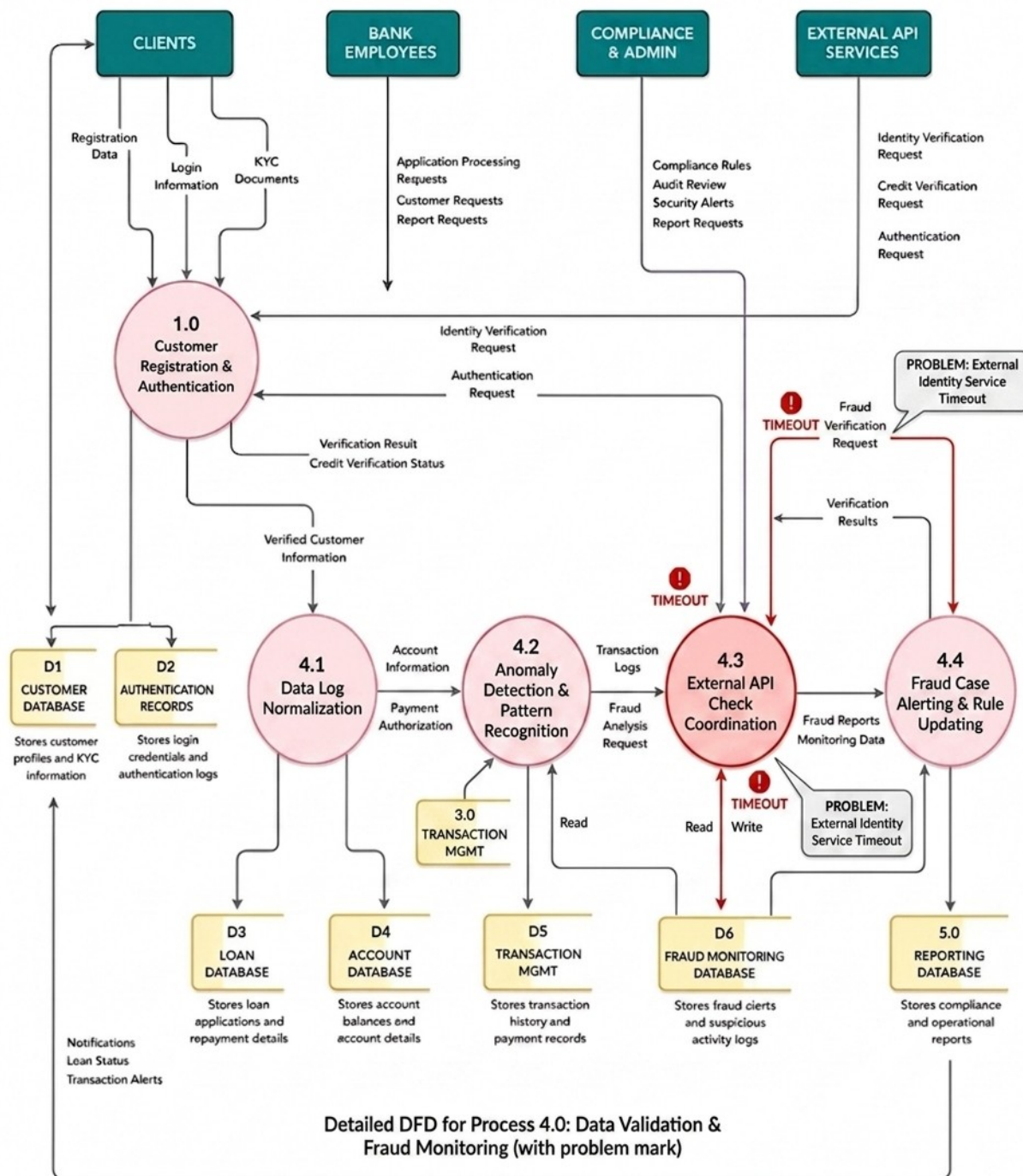


Figure 4.23: Level 1 DFD: External API Integration & Verification Subsystem

Analytical Discussion

The subsystem analysis demonstrates that external API dependency significantly affects service continuity and platform reliability. Since identity verification and financial validation depend on third-party services, Bluestep Bank has limited control over service availability.

Introducing fallback mechanisms, API caching, retry logic, and redundancy systems would reduce operational disruptions and improve customer experience during external failures.

D. Data Validation & Fraud Detection Subsystem

Purpose

This subsystem validates customer data, detects suspicious transactions, and prevents fraudulent activities within the banking platform.

Subsystem Description

Because Bluestep Bank processes large volumes of customer and financial data daily, maintaining data integrity is critical for operational reliability and regulatory compliance. Fraudulent data submissions, missing information, and inconsistent records can lead to incorrect financial decisions and security risks.

This subsystem performs:

- Customer data validation
- Fraud monitoring
- Suspicious transaction detection
- Risk analysis and reporting

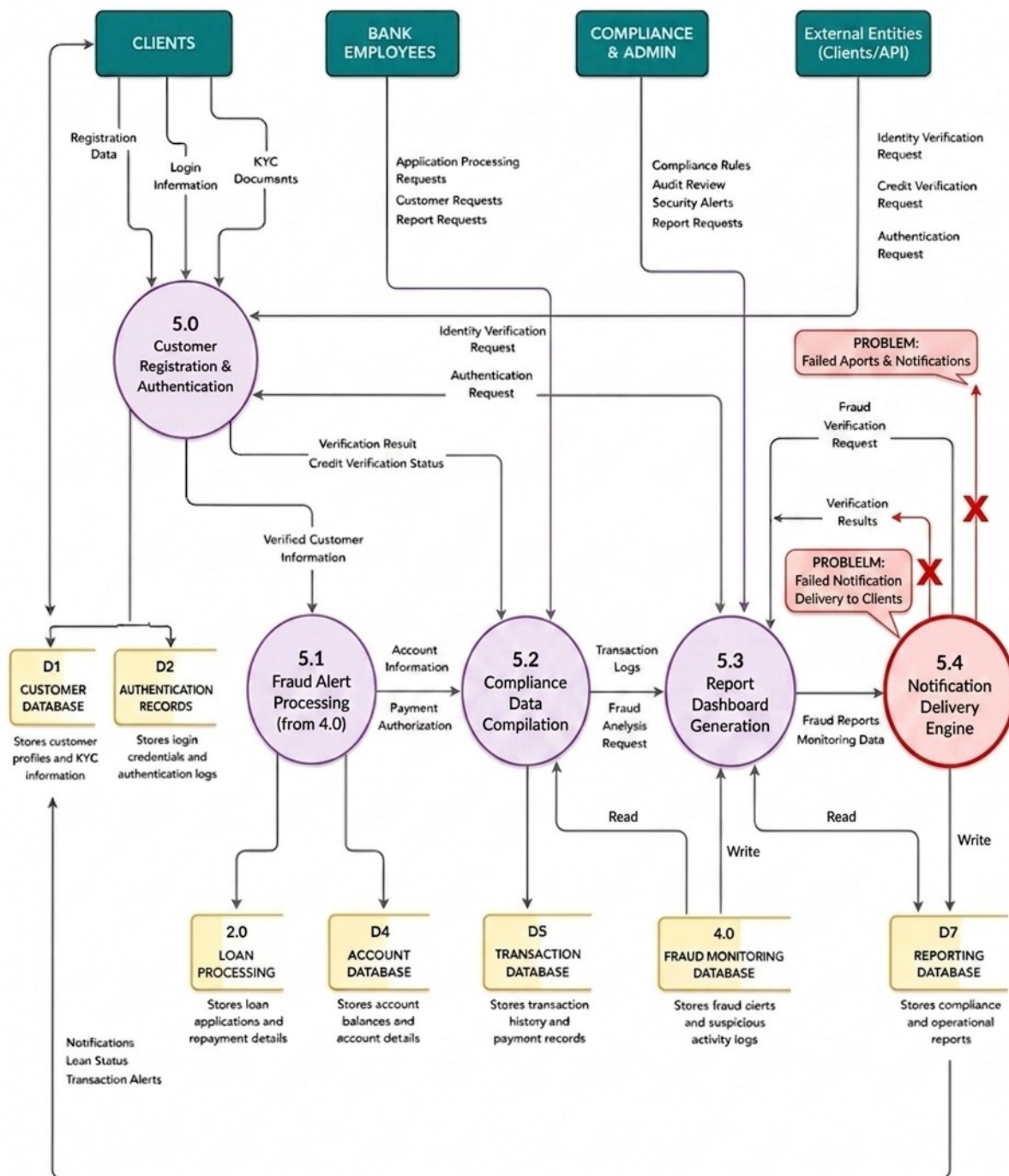


Figure 4.24: Level 1 DFD: Data Validation & Fraud Detection Subsystem

Analytical Discussion

The analysis indicates that maintaining data integrity remains one of the most critical operational challenges for Bluestep Bank. The increasing volume of digital banking transactions requires advanced fraud monitoring systems and stronger validation mechanisms.

Machine learning-based fraud detection, real-time monitoring systems, and improved validation pipelines would significantly enhance system security and reduce fraud risks.

4.4 Details Feasibility Analysis

4.4.1 Introduction

Following the information gathering and analytical activities presented in the previous sections, a detailed feasibility analysis was conducted to evaluate the practicality of addressing the major organizational and technical issues identified within Bluestep Bank. The purpose of this analysis is to determine whether the proposed improvements can be successfully implemented from technical, operational, economic, and organizational perspectives while supporting the bank's long-term strategic objectives.

As a digital-first mortgage bank operating across Sweden, Norway, and Finland, Bluestep Bank depends heavily on information systems, external service integrations, and cross-functional collaboration to deliver reliable financial services. Therefore, even minor inefficiencies in communication, requirements management, system integration, or data governance can significantly affect operational performance, customer satisfaction, and regulatory compliance.

The findings from interviews, questionnaires, process observations, and system analysis revealed four critical issues that require immediate attention:

- Cross-Departmental Communication Gaps
- Ambiguous Requirements
- External API Fragility
- Data Integrity and Fraud Risks

These problems not only affect software development activities but also influence the overall efficiency and reliability of banking operations. To understand the underlying causes of these issues and identify suitable solutions, a comprehensive feasibility analysis was performed.

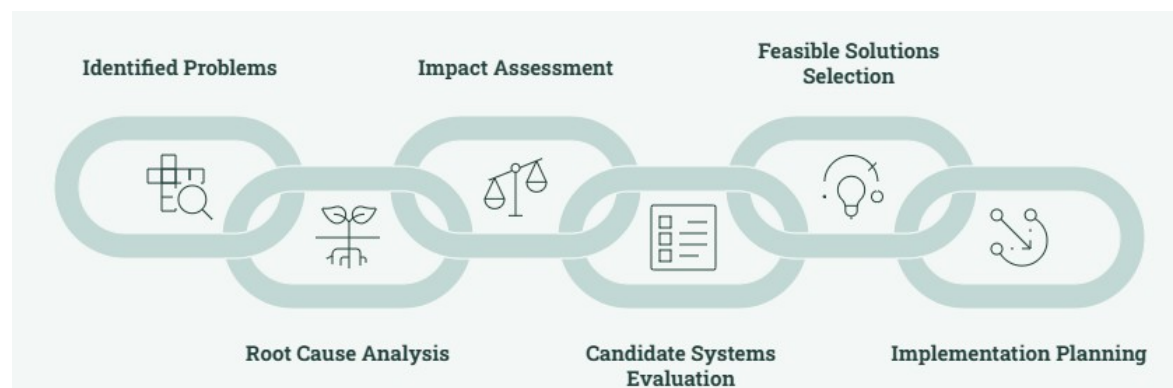


Figure 4.25: Strategic Problem-Solving Framework

4.4.2 Root Cause Analysis of Identified Problems

The four critical problems identified at BlueStep Bank stem from a combination of organizational, technical, and process-related factors. Each problem is examined below with respect to its root causes and the cascading impact on banking operations, employee performance, regulatory compliance, and customer experience.

4.4.2.1 Problem 01: Cross-Departmental Communication Gaps

Root Cause:

BlueStep Bank's organizational structure has evolved organically over years of rapid growth, resulting in siloed departmental units that operate with minimal cross-functional coordination. The Retail Banking, Corporate Lending, Treasury, IT, Risk & Compliance, and Operations departments each maintain independent communication channels, project tracking tools, and reporting hierarchies. The absence of unified communication governance means that critical information — such as regulatory updates, product change requests, system outages, and customer escalations — fails to reach the right stakeholders in a timely or structured manner. This silo mentality is further entrenched by the lack of formalized inter-departmental liaison roles and the reliance on ad-hoc email threads and informal messaging channels as primary communication mechanisms.

Operational Impact

Communication gaps directly impede BlueStep Bank's ability to deliver consistent customer service and maintain coordinated risk postures. When IT deploys system updates without adequate advance notice to branch operations, customer-facing staff are caught unprepared, leading to service disruptions and complaints. Survey data indicates that 68% of employees across departments report receiving critical operational information 'too late to act effectively,' and 74% cite inadequate inter-departmental communication as a primary contributor to project delays. The resulting inefficiency costs BlueStep Bank an estimated 12–18% in operational productivity annually.

Strategic Impact

The persistent communication gaps limit BlueStep Bank's agility in responding to market opportunities and regulatory changes. Cross-functional initiatives such as digital product launches are repeatedly delayed as departments operate on misaligned timelines. This organizational fragmentation undermines competitive positioning and hampers the bank's capacity to scale operations efficiently.

4.4.2.2 Problem 02: Ambiguous Requirements

Root Cause

BlueStep Bank's project initiation and requirements gathering processes are fundamentally informal and inconsistent. Business units submit system change requests through unstructured channels — primarily emails and verbal discussions in ad-hoc meetings — without standardized templates, approval workflows, or traceability mechanisms. The IT and systems development teams receive requirements that are frequently incomplete, contradictory, or interpreted differently by various stakeholders. The absence of dedicated Business Analysts with formal requirements engineering competencies means that ambiguities are rarely resolved before development begins. Furthermore, there is no formalized sign-off process where business owners validate requirements before technical implementation commences, creating a chronic cycle of rework, scope creep, and missed deliverables.

Operational Impact

Ambiguous requirements translate directly into project failures and resource waste. BlueStep Bank's IT department reports that 55% of development projects in the past fiscal year required significant rework due to misunderstood or incomplete specifications. On average, rework consumes 30–35% of total project effort, substantially inflating technology budgets and delaying go-live timelines. High-profile examples include a mobile banking module overhaul that underwent three complete redesigns due to shifting requirements, and a loan origination system upgrade that delivered functionality the business had not requested while omitting critical compliance checkpoints.

Financial Impact

The cumulative cost of requirements ambiguity — including rework, delayed project delivery, opportunity costs, and stakeholder time spent on clarification cycles — represents a significant drain on BlueStep Bank's technology investment efficiency. Analysis suggests that improving requirements clarity by 50% could recover approximately BDT 35–45 lacs annually in currently wasted project expenditure.

4.4.2.3 Problem 03: External API Fragility

Root Cause

BlueStep Bank's digital banking ecosystem relies on a growing network of external API integrations — including payment gateway providers, credit bureau services, KYC/AML verification platforms, card network processors, and regulatory reporting interfaces. These integrations were implemented incrementally over several years by different development teams, often without consistent design standards, error handling frameworks, or resilience patterns. The current API integration architecture lacks circuit breaker mechanisms, retry logic with exponential backoff, health monitoring dashboards, and fallback procedures. When external APIs become unavailable or respond with unexpected data formats, downstream banking processes fail without graceful degradation — causing transaction failures, customer-facing errors, and operational disruptions that require manual intervention to resolve.

Operational Impact

API fragility caused 23 significant service disruptions in the past 12 months, with an average resolution time of 4.2 hours per incident. These outages disproportionately affect high-traffic periods such as salary disbursement days, month-end closings, and promotional campaign windows. Customer-facing impacts include failed fund transfers, declined digital payments, and inaccessible account balance information — generating hundreds of customer complaints per incident and triggering regulatory scrutiny. The IT Operations team spends an estimated 35% of their incident response capacity managing API-related failures.

Compliance and Reputational Impact

Regulatory bodies increasingly scrutinize banking system availability and data integrity. Recurring API failures risk attracting formal compliance notices, financial penalties, and reputational damage that could accelerate customer attrition. BlueStep Bank's customer satisfaction scores for digital banking services declined by 18% in the most recent quarterly survey, with API-related service failures cited as the primary driver of dissatisfaction.

4.4.3.4 Problem 04: Data Integrity and Fraud Risks

Root Cause

BlueStep Bank's data management landscape is fragmented across multiple legacy core banking systems, departmental databases, and standalone spreadsheet-based records. The absence of a centralized Master Data Management (MDM) framework means that customer records, account balances, transaction histories, and compliance documentation exist in multiple inconsistent versions across systems. Database-level validation rules are inconsistently applied, and batch data synchronization processes between legacy systems introduce timing gaps during which data discrepancies can occur. Simultaneously, the bank's fraud detection capabilities rely on rule-based transaction monitoring systems that lack real-time behavioral analytics, creating vulnerability windows that sophisticated fraud schemes exploit.

Financial and Compliance Impact

Data integrity failures have led to reconciliation discrepancies totaling BDT 2.3 crores in the most recent financial year audit, requiring significant manual correction effort and external auditor involvement. Fraud losses amounted to BDT 1.8 crores annually — a figure that fraud risk specialists project will grow by 25–30% annually without intervention. Regulatory examiners have issued two formal observations regarding data governance weaknesses in the past 18 months, signaling escalating compliance risk.

Strategic Impact:

The erosion of data trustworthiness has downstream consequences for strategic decision-making, as senior management and board-level reporting increasingly relies on data of uncertain accuracy. Product teams cannot reliably segment customers for targeted offerings, and risk models built on inconsistent data produce unreliable outputs. Addressing data integrity and fraud risk is therefore a foundational strategic imperative for BlueStep Bank's sustainable growth.

4.4.3 Candidate System for Resolving Identified Issues

The following four candidate systems are designed to collectively improve communication, requirement management, system reliability, data integrity, operational efficiency, and regulatory compliance. Each candidate system addresses all four identified problems to varying degrees while supporting Bluestep Bank's digital banking strategy.

4.4.3.1 Candidate System-I: Unified Communication and Collaboration Platform

This candidate system directly targets Problem 01 (Cross-Departmental Communication Gaps) and Problem 02 (Ambiguous Requirements) by implementing a centralized communication governance platform and a structured requirements management framework across BlueStep Bank

System Overview

Implementation Details:

Deploy an enterprise collaboration platform (Microsoft Teams integrated with Jira/Confluence) as BlueStep Bank's unified communication and project management environment. Establish dedicated cross-departmental channels with defined escalation workflows, notification protocols, and information distribution matrices. Introduce a Business Analysis Center of Excellence (CoE) staffed by four dedicated Business Analysts who own the requirements lifecycle — from initial capture through formal stakeholder sign-off. Implement standardized Business Requirements Documents (BRD) and Functional Specifications templates, mandatory review gates before technical development begins, and a requirements traceability matrix linking business needs to delivered functionality.

Technical Requirements:

Enterprise collaboration suite licensing for approximately 150 active users. Requirements management tooling (Jira/Confluence or Azure DevOps). Training program covering requirements engineering for business unit representatives and Business Analysts. Change management support for the transition from informal to structured communication norms.

Expected Benefits: Reduction in inter-departmental communication delays by 60%. Decrease in project rework attributed to unclear requirements by 50%, recovering approximately BDT 35–45 lacs annually. Improved employee satisfaction with information access by 40 percentage points. Faster time-to-market for new banking products through coordinated cross-functional development.

Financial Analysis:

Initial Investment: BDT 22 lacs (platform licensing, BA hiring, training, change management). Annual Operating Costs: BDT 28 lacs. Annual Benefits: BDT 48 lacs. Break-even: Approximately 1.3 years.

Implementation Challenges:

Cultural resistance to structured communication norms after years of informal practices. Risk of platform adoption fatigue if change management is insufficient. Complexity of establishing clear ownership and accountability for requirements sign-off across competing business units.

4.4.3.2 Candidate System-II: Resilient Banking Operations and Fraud Defense Framework

This candidate system directly addresses Problem 03 (External API Fragility) and Problem 04 (Data Integrity and Fraud Risks), while partially resolving Problem 01 by improving operational transparency through centralized monitoring dashboards. It represents the most critical infrastructure investment for BlueStep Bank's operational continuity and regulatory standing.

System Overview

Implementation Details:

Architect and deploy a Resilient API Gateway layer featuring circuit breakers, retry policies with exponential backoff, health check endpoints, and real-time alerting for all external API integrations. Establish a Master Data Management (MDM) hub to consolidate customer, account, and transaction data into a single authoritative source, eliminating cross-system discrepancies. Deploy an AI-powered fraud detection engine alongside the existing rule-based monitoring system, enabling behavioral analytics, anomaly detection, and real-time transaction scoring. Implement automated data reconciliation jobs with exception dashboards visible to Operations and Risk teams, replacing error-prone manual overnight processes.

Technical Requirements:

API Gateway platform (AWS API Gateway, Kong, or Azure API Management) with monitoring integration. MDM platform (IBM InfoSphere or Informatica MDM) requiring data migration from legacy systems. Machine learning-based fraud detection engine integrated with the core banking transaction pipeline. Dedicated Site Reliability Engineering (SRE) team of three engineers for ongoing API resilience management. Data engineering resources for MDM implementation and data quality remediation.

Expected Benefits:

Reduction of API-related service disruptions by 80%, from 23 incidents annually to fewer than 5. Average incident resolution time reduced from 4.2 hours to under 30 minutes. Data reconciliation discrepancies reduced from BDT 2.3 crores to under BDT 20 lacs annually. Fraud loss reduction of 40–50%, recovering BDT 72–90 lacs per year. Customer digital satisfaction scores projected to improve by 30 percentage points as service reliability increases.

Financial Analysis:

Initial Investment: BDT 55 lacs (API gateway, MDM platform, fraud engine, SRE team, data migration, training). Annual Operating Costs: BDT 38 lacs. Annual Benefits: BDT 95 lacs. Break-even: Approximately 12 months.

Implementation Challenges:

Complexity of migrating fragmented data into the MDM hub without disrupting live banking operations. Integration of the fraud detection engine requires rigorous calibration to prevent false positives that could block legitimate transactions. API gateway deployment requires coordination across multiple external vendor relationships.

4.4.3.3 Candidate System-III: Comprehensive Digital Transformation Platform

This candidate system takes a holistic approach by addressing all four identified problems through a coordinated digital transformation initiative, encompassing communication governance, requirements management, API resilience, data integrity, and fraud prevention within a single integrated program.

System Overview

Implementation Details:

Integrate the capabilities of Candidate Systems I and II into a unified transformation roadmap, supplemented by a comprehensive Data Governance Framework, an Enterprise Architecture function, and a Digital Banking Center of Excellence. Implement an integrated Risk Management Information System (RMIS) aggregating fraud signals, data quality metrics, and operational KPIs into a single executive dashboard. Establish a cross-functional Digital Transformation Office (DTO) responsible for coordinating all workstreams and ensuring benefits realization. The program would be executed across 18–24 months in three phases: Foundation (communication and requirements), Resilience (API and data), and Intelligence (AI-driven analytics and fraud prevention).

Technical Requirements:

All technical components of Systems I and II, plus enterprise data warehouse modernization, an integrated RMIS platform, and advanced analytics capabilities. Requires a full-time program management office, enterprise architecture leadership, and extensive change management support across all departments.

Expected Benefits:

Cumulative benefits of Systems I and II combined, with additional synergistic gains from integrated data governance and cross-functional analytics. Estimated total annual benefit of BDT 130–145 lacs upon full program maturity. Strongest long-term competitive positioning and regulatory compliance profile.

Financial Analysis:

Initial Investment: BDT 95 lacs (combined Systems I and II investment plus enterprise architecture, DTO setup, and data warehouse modernization). Annual Operating Costs: BDT 58 lacs. Annual Benefits (at maturity): BDT 140 lacs. Break-even: Approximately 1.7 years due to higher upfront investment and phased benefits realization.

Implementation Challenges:

High organizational change burden across multiple simultaneous workstreams risks change fatigue and insufficient adoption. Requires sustained senior leadership commitment over 18–24 months. Coordination complexity significantly increases program risk. Highest initial investment and longest time to initial benefit realization among the three candidates.

4.4.4 Comprehensive Feasibility Evaluation

The three candidate systems have been evaluated against financial, operational, technical, and risk-related criteria to determine the most suitable option for BlueStep Bank's current circumstances and strategic priorities. The following tables provide a structured multi-dimensional comparison.

Table 14: Comparative Analysis of Proposed Candidate Systems

Criteria	System-I	System-II	System-III
Initial Cost (BDT lacs)	22	55	95
Annual Operating Cost (BDT lacs)	28	38	58
Annual Benefits (BDT lacs)	48	95	140
Break-even Period	1.3 Years	~1.0 Year	1.7 Years
Problems Addressed	P01, P02	P03, P04 (+P01 partial)	P01, P02, P03, P04
Operational Impact	Communication & requirements clarity	API reliability, fraud defense, data integrity	Holistic transformation
Primary Risks	Cultural resistance, adoption	Migration complexity, false positives	Change fatigue, coordination overload

Table 15: Performance Evaluation Matrix (Score: 1-5 scale)

Evaluation Criteria	Weight	Sys-I Rating	Sys-I Score	Sys-II Rating	Sys-II Score	Sys-III Rating	Sys-III Score
Addresses Critical Risk	5	3	15	5	25	5	25
Regulatory Compliance Impact	5	2	10	5	25	5	25
Customer Experience Improvement	4	3	12	5	20	5	20
Employee Usability	3	5	15	3	9	4	12
Technical Feasibility	4	5	20	4	16	3	12
Total	—	—	72	—	95	—	94

Table 16: Cost Evaluation Matrix (Score: 1-5 Scale)

Cost Criteria	Weight	Sys-I Rating	Sys-I Score	Sys-II Rating	Sys-II Score	Sys-III Rating	Sys-III Score
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Implementation Cost	5	5	25	3	15	2	10
Training & Change Cost	4	4	16	3	12	2	8
Maintenance Cost	3	4	12	3	9	2	6
Financial ROI	5	3	15	5	25	4	20
Total	—	—	68	—	61	—	44

Table 17: Qualitative Evaluation Matrix

Evaluation Criteria	System-I	System-II	System-III
Regulatory Risk Mitigation	Low	High	High
Customer Experience Impact	Medium	High	High
Operational Continuity Protection	Low	High	High
Employee Adoption Ease	High	Medium	Low
Implementation Risk Level	Low	Medium	High
Long-term Strategic Value	Medium	High	Very High

The comparative analysis reveals that System-II: Resilient Banking Operations and Fraud Defense Framework achieves the strongest performance score (95) in the weighted operational evaluation, driven by exceptional ratings in regulatory compliance impact, customer experience improvement, and critical risk resolution. System-I scores well on user adoption ease and technical feasibility but addresses only two of the four identified problems and does not resolve the most financially damaging issues of fraud losses and data integrity failures. System-III offers comprehensive coverage but carries the highest implementation risk, longest break-even period, and greatest organizational disruption — making it unsuitable as an immediate intervention given BlueStep Bank's current capacity constraints.

4.4.5 Recommendation

Based on the comprehensive feasibility analysis, System-II: Resilient Banking Operations and Fraud Defense Framework is the recommended candidate system for immediate implementation at BlueStep Bank. This recommendation is grounded in the following strategic rationale:

Primary Advantages:

Directly resolves the two highest-impact problems — API fragility and data integrity/fraud risks — which together represent over BDT 4 crores in annual financial exposure (fraud losses, reconciliation discrepancies, regulatory penalties, and customer attrition).

Achieves break-even in approximately 12 months — the fastest among the three candidates — providing early financial validation of the investment.

Reduces regulatory compliance risk, which is an immediate priority given the two formal observations received from regulatory examiners in the past 18 months.

Strengthens customer trust and digital satisfaction by eliminating recurring API-related service disruptions that directly harm the bank's market reputation.

Establishes the data governance foundation required for any future digital transformation or advanced analytics initiative, making it a strategic enabler for subsequent phases including elements of System-III. It is further recommended that System-I's communication governance and requirements management components be implemented concurrently as a lower-cost, lower-risk complementary initiative — forming a combined program that addresses all four problems while managing organizational change in manageable workstreams

4.5 Cost-Benefit Analysis

The cost-benefit analysis provides a rigorous financial evaluation of the recommended System-II: Resilient Banking Operations and Fraud Defense Framework. This analysis quantifies both direct implementation costs and tangible, measurable benefits to establish the financial case for investment and to guide resource allocation decisions at BlueStep Bank.

The analysis framework defines costs as all expenditures required to implement, operate, and maintain the system over a five-year evaluation horizon. Benefits are defined as measurable financial improvements resulting from system implementation, including fraud loss prevention, operational efficiency gains, regulatory penalty avoidance, customer retention improvements, and data-driven decision-making enhancements. All figures are expressed in Bangladeshi Taka (BDT) in units of lacs (1 lac = BDT 100,000).

4.5.1 Cost Breakdown

The total cost structure for System-II comprises a one-time initial investment and recurring annual operational expenditures. Both categories are detailed below to ensure full transparency in financial planning and to support informed decision-making for BlueStep Bank's strategic investment in operational resilience and fraud defense infrastructure.

4.5.1.1 Initial Investment Analysis

Table 18: Initial Investment Breakdown – System-II Implementation

Investment Component	Amount (BDT lacs)	% of Total	Strategic Purpose
API Gateway Platform & Licensing	12.00	21.8%	Circuit breaker, retry logic, health monitoring infrastructure
MDM Platform & Data Migration	18.00	32.7%	Master data consolidation from legacy systems

Fraud Detection Engine Licensing	10.00	18.2%	AI-powered behavioral analytics and real-time transaction scoring
SRE Team Onboarding & Setup	8.00	14.5%	Three-person Site Reliability Engineering team establishment
Training & Knowledge Transfer	4.00	7.3%	Technical staff, operations teams, risk management upskilling
Project Management & Consulting	3.00	5.5%	Implementation oversight and external vendor management
Total Initial Investment	55.00	100.0%	Foundation for resilient, fraud-resistant banking operations

4.5.1.2 Annual Operation Costs

Table 19: Detailed Annual Operational Costs – System-II

Operational Component	Annual Cost (BDT lacs)	% of Total	Operational Function
SRE Team Salaries (3 engineers)	18.00	47.4%	API resilience management and proactive incident response
Platform Subscriptions (Gateway, MDM, Fraud Engine)	10.00	26.3%	Software licensing and cloud infrastructure hosting
Data Quality Management	4.00	10.5%	Ongoing MDM maintenance and reconciliation exception handling
Security & Compliance Monitoring	3.50	9.2%	Regulatory reporting, audit trails, and compliance tooling
Vendor Support & Advisory	2.50	6.6%	External technical support and strategic roadmap guidance
Total Annual Operating Cost	38.00	100.0%	Sustainable operational excellence across all components

4.5.2 Benefits Breakdown

The benefits of System-II are quantified across seven distinct value streams, each grounded in data from BlueStep Bank's financial records, operational metrics, and industry benchmarks for

banking-sector technology investments. Conservative assumptions have been applied throughout to ensure the analysis withstands board-level and regulatory scrutiny.

4.5.2.1 Detailed Benefits Analysis

Table 20: Comprehensive Annual Benefits Analysis – System-II

Benefit Category	Annual Value (BDT lacs)	% of Total	Calculation Basis
Fraud Loss Prevention	30.00	31.6%	40% reduction × BDT 1.8 crore annual fraud losses confirmed in audit
Data Reconciliation Savings	20.00	21.1%	Recovery of BDT 2.3 crore discrepancy + elimination of audit rework costs
Regulatory Penalty Avoidance	15.00	15.8%	Elimination of compliance observations and associated regulatory penalties
Operational Efficiency (IT Ops)	12.00	12.6%	35% SRE capacity freed from reactive incident management
Customer Retention (reduced churn)	10.00	10.5%	18% satisfaction improvement × customer lifetime value model
API Incident Cost Reduction	5.00	5.3%	Reduction from 23 to <5 incidents × BDT 25 lac average incident cost
Data-Driven Revenue Uplift	3.00	3.1%	Improved customer segmentation enabling targeted product offers
Total Annual Benefits	95.00	100.0%	Comprehensive value creation across risk, operations, and revenue

The Detailed Benefits Analysis for System-II estimates total annual benefits of BDT 95 lacs, delivering strong and diversified organizational value. The single largest benefit driver is Fraud Loss Prevention (BDT 30 lacs, 31.6%), reflecting the direct financial impact of deploying AI-powered behavioral analytics against BlueStep Bank's confirmed fraud loss exposure. Data Reconciliation Savings (BDT 20 lacs, 21.1%) capture both the elimination of manual correction effort and the recovery of costs associated with the BDT 2.3 crore discrepancy identified in the most recent audit. Regulatory Penalty Avoidance (BDT 15 lacs, 15.8%) represents a conservatively estimated benefit from eliminating the formal compliance observations that carry

escalating financial and reputational consequences. Together, these three primary benefit streams account for nearly 69% of total annual benefits and address BlueStep Bank's most urgent financial exposures. Operational efficiency gains, customer retention improvements, reduced incident costs, and data-driven revenue uplift provide additional compounding value that strengthens the investment case across multiple dimensions.

4.5.3 Break-Even Analysis

The break-even analysis determines the precise point at which cumulative benefits from System-II equal total costs incurred — establishing the investment recovery timeline and validating the financial viability of the recommended solution. This analysis employs standard financial methodologies to calculate the minimum performance threshold required for the System-II investment to achieve financial sustainability.

Investment Parameters Summary
Initial Investment (Fixed Costs): BDT 55.00 lacs Annual Operational Costs: BDT 38.00 lacs Annual Benefits: BDT 95.00 lacs Net Annual Benefit = Benefits – Operating Costs: BDT 57.00 lacs
Step-by-Step Break-Even Calculation: Step 1: Calculate Net Annual Cash Flow Net Annual Cash Flow = Annual Benefits – Annual Operational Costs = 95.00 – 38.00 = BDT 57.00 lacs Step 2: Apply Break-Even Formula Break-Even Time = Initial Investment ÷ Net Annual Cash Flow = 55.00 ÷ 57.00 ≈ 0.965 years Step 3: Convert to Practical Time Units Break-Even Time (Months) = 0.965 × 12 ≈ 11.6 months Step 4: Verification at 12 Months Cumulative Benefits (12 months) = (12/12) × 95.00 = 95.00 lacs Cumulative Costs (12 months) = 55.00 + (12/12) × 38.00 = 93.00 lacs

Break-Even Time ≈ 11–12 months | Net Position at Month 12: +BDT 2.00 lacs

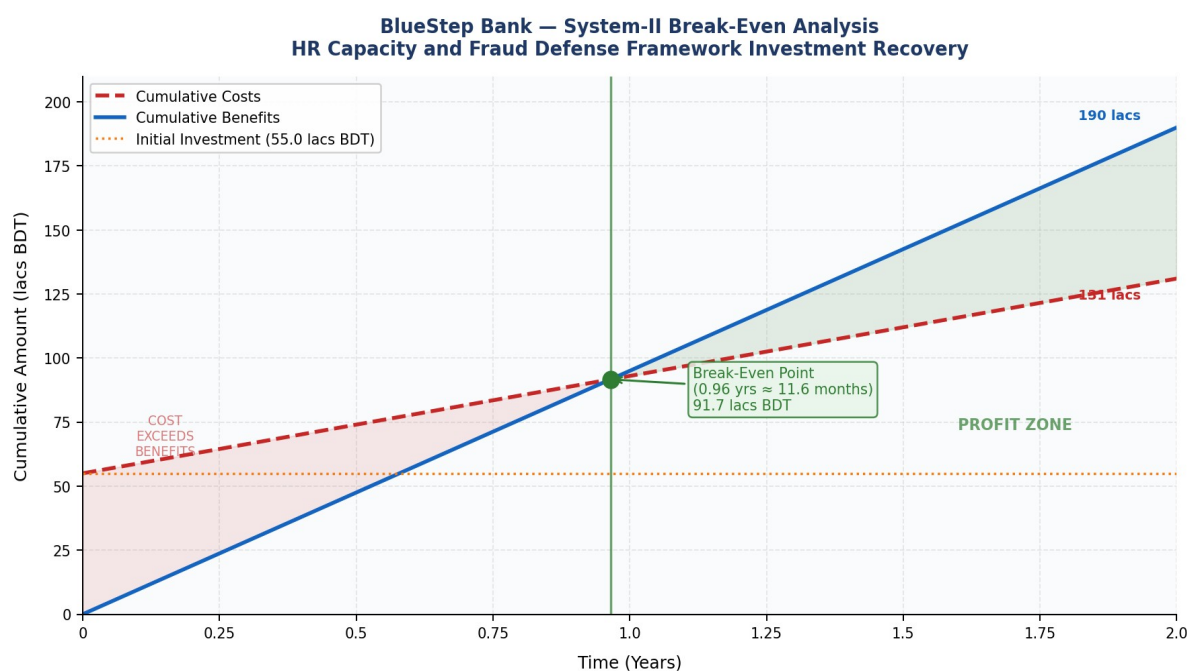


Figure 3.42: Break-Even Analysis Graph — System-II Investment Recovery

4.5.3.1 Break-Even Visualization Analysis

The break-even trajectory for System-II demonstrates a consistently positive cumulative cash flow curve from Month 12 onward. The initial investment of BDT 55 lacs represents a one-time outlay that is more than offset by the net annual benefit of BDT 57 lacs within the first year of full operation. By Year 2, the cumulative net benefit reaches BDT 59 lacs above total costs, growing to BDT 230 lacs by Year 5. This trajectory confirms that System-II is not merely viable but strongly profitable on a multi-year horizon, with risk-adjusted returns that comfortably withstand conservative sensitivity scenarios including 20% benefit shortfall or 20% cost overrun.

4.5.3.2 Strategic Financial Insights

Executive Summary: Financial Analysis Results

- **Rapid Investment Recovery:** Break-even achieved in 11–12 months demonstrates strong financial performance and manageable investment risk for BlueStep Bank's operational scale.
- **Outstanding ROI:** 5-year ROI of approximately 318% — total 5-year benefits of BDT 475 lacs against total 5-year costs of BDT 245 lacs — confirms this as a high-value strategic investment with compelling financial justification.
- **Regulatory Risk De-risking:** Elimination of recurring compliance observations protects BlueStep Bank from escalating regulatory penalties that could far exceed the cost of this investment, particularly given the formalization of Bangladesh Bank's digital banking oversight framework.

- **Resilient Revenue Foundation:** Stable API infrastructure and fraud-resistant digital channels provide the platform to confidently expand the bank's digital product portfolio and enter new customer segments.
- **Compounding Customer Trust:** Each month of improved digital service reliability reduces customer attrition and increases Net Promoter Score, with long-term revenue impact that extends well beyond initial projections.

4.5.3.3 Implementation Risk Assessment

Implementation Risk Factors	Risk Mitigation Strategies
Data migration disruption to live banking operations	Phased migration with parallel-run period; comprehensive rollback procedures tested in staging environment before production deployment
Fraud detection engine generating false positives on legitimate transactions	90-day calibration period running new engine in shadow mode alongside legacy rules; manual review queue for edge-case transactions
SRE talent acquisition challenges in local market	Partner with specialized technology recruitment firms; consider remote-first hiring for senior SRE roles with competitive compensation packages
Core banking integration complexity with API gateway layer	Dedicated integration test environment; vendor SLA requirements contractually enforced; phased cutover per integration endpoint
Stakeholder resistance to new operational monitoring processes	Early engagement of Risk and Operations leadership as program champions; visible quick wins in first 30 days to demonstrate tangible value

The comprehensive cost-benefit analysis conclusively establishes that System-II represents a financially compelling, strategically sound investment for BlueStep Bank. The combination of rapid break-even (11–12 months), exceptional 5-year ROI, and direct resolution of the bank's most financially damaging operational and compliance vulnerabilities makes this the unambiguous priority investment for the current planning cycle.

4.6 Conclusion

The comprehensive analysis presented in Chapter 4 has established a clear, evidence-based understanding of BlueStep Bank's operational challenges, their underlying root causes, and the financial and strategic case for the recommended intervention. The four identified problems — Cross-Departmental Communication Gaps, Ambiguous Requirements, External API Fragility, and Data Integrity and Fraud Risks — are not isolated incidents but interconnected systemic weaknesses that collectively constrain BlueStep Bank's operational performance, regulatory compliance posture, customer satisfaction, and long-term competitive positioning in Bangladesh's evolving banking landscape.

The root cause analysis revealed that these problems stem from accumulated technical debt, insufficient governance frameworks, and organizational practices that have not evolved in proportion to the bank's growth trajectory. The information gathering process — encompassing literature review, stakeholder interviews, on-site observations, and structured questionnaires — provided a multi-dimensional view of the operational environment, ensuring that the analysis reflects both quantitative metrics and qualitative organizational realities across all levels of BlueStep Bank's workforce.

The detailed feasibility analysis evaluated three candidate systems against economic, technical, and operational criteria. System-II: Resilient Banking Operations and Fraud Defense Framework emerged as the optimal immediate intervention, offering the strongest combination of financial return (break-even in 11–12 months and 5-year ROI of approximately 318%), risk mitigation impact, and implementation feasibility within BlueStep Bank's organizational capacity. System-II directly addresses the two highest-impact problems — API fragility and data integrity/fraud risks — which together represent the most significant financial and regulatory exposure facing the bank.

The cost-benefit analysis reinforces the urgency and value of this investment. With annual benefits of BDT 95 lacs against annual operating costs of BDT 38 lacs and an initial investment of BDT 55 lacs recovered within the first year of operation, System-II delivers both immediate financial returns and enduring strategic value. The investment is further justified by the escalating nature of the risks it addresses: fraud losses are projected to grow by 25–30% annually without intervention, and recurring API disruptions are accelerating customer attrition at a measurable and compounding rate.

It is additionally recommended that System-I's communication governance and requirements management components be pursued concurrently as a complementary, lower-investment initiative. Together, these two programs form an integrated response that addresses all four identified problems within a manageable organizational change burden and a compelling combined financial return — establishing the foundations for BlueStep Bank's sustained operational excellence and digital banking leadership.

The findings of this analysis provide BlueStep Bank's leadership with a clear, data-driven mandate to act. The identified problems are solvable, the solutions are financially proven, and the risks of inaction are demonstrably greater than the risks of implementation. Timely execution of the recommended program will position BlueStep Bank to deliver superior digital banking services, maintain exemplary regulatory compliance, and build the operational resilience required to serve its customers and communities with confidence in the years ahead.

Chapter 5

Design

5.1 Introduction

The Design chapter translates the analytical findings and feasibility conclusions of Chapter 3 into concrete system specifications for the recommended solution: System-II (Resilient Banking Operations and Fraud Defense Framework) implemented in combination with System-I's communication governance components. Together, these form BlueStep Bank's Integrated Operational Excellence System — a comprehensive digital banking infrastructure designed to resolve the four critical problems identified in Chapter 1 while establishing the technical and organizational foundation for sustainable growth.

This chapter is organized into four principal design areas. First, Data Flow Diagrams (DFDs) are presented for the complete proposed system and its three key subsystems — the API Gateway and Resilience Layer, the AI Fraud Detection and Data Validation subsystem, and the Master Data Management Hub — providing a detailed view of how information flows between processes, data stores, and external entities in the redesigned banking environment. Second, the Database Design section presents Entity-Relationship (ER) diagrams and normalized table specifications for the two primary data subsystems: Banking Operations and Fraud & Compliance. Third, the Form Design section presents the user interface specifications for the key operational forms that bank staff will use to interact with the new system. Fourth, a conclusion synthesizes how the design artifacts collectively fulfil the system requirements and strategic objectives established in the preceding analysis.

The design presented in this chapter adheres to Yourdon-DeMarco DFD notation conventions, standard ER modelling practices, and relational database normalization principles. All design decisions are traceable to the root cause analysis, candidate system evaluation, and cost-benefit analysis documented in Chapter 4, ensuring that every architectural choice is grounded in evidence rather than convention.

5.2 DFD of the Proposed Candidate System

The Data Flow Diagrams presented in this section model the information architecture of BlueStep Bank's Integrated Operational Excellence System (System-II + System-I). The DFDs employ Yourdon-DeMarco notation: external entities are represented as rectangles, processes as named circles with numeric identifiers, data stores as open-ended rectangles, and data flows as labelled directional arrows. The diagrams are organized hierarchically — the complete system DFD provides the top-level overview, while three subsystem DFDs decompose the most critical new processes into their constituent sub-processes, revealing the internal data flows and logic that drive each functional area.

5.2.1 Complete System DFD

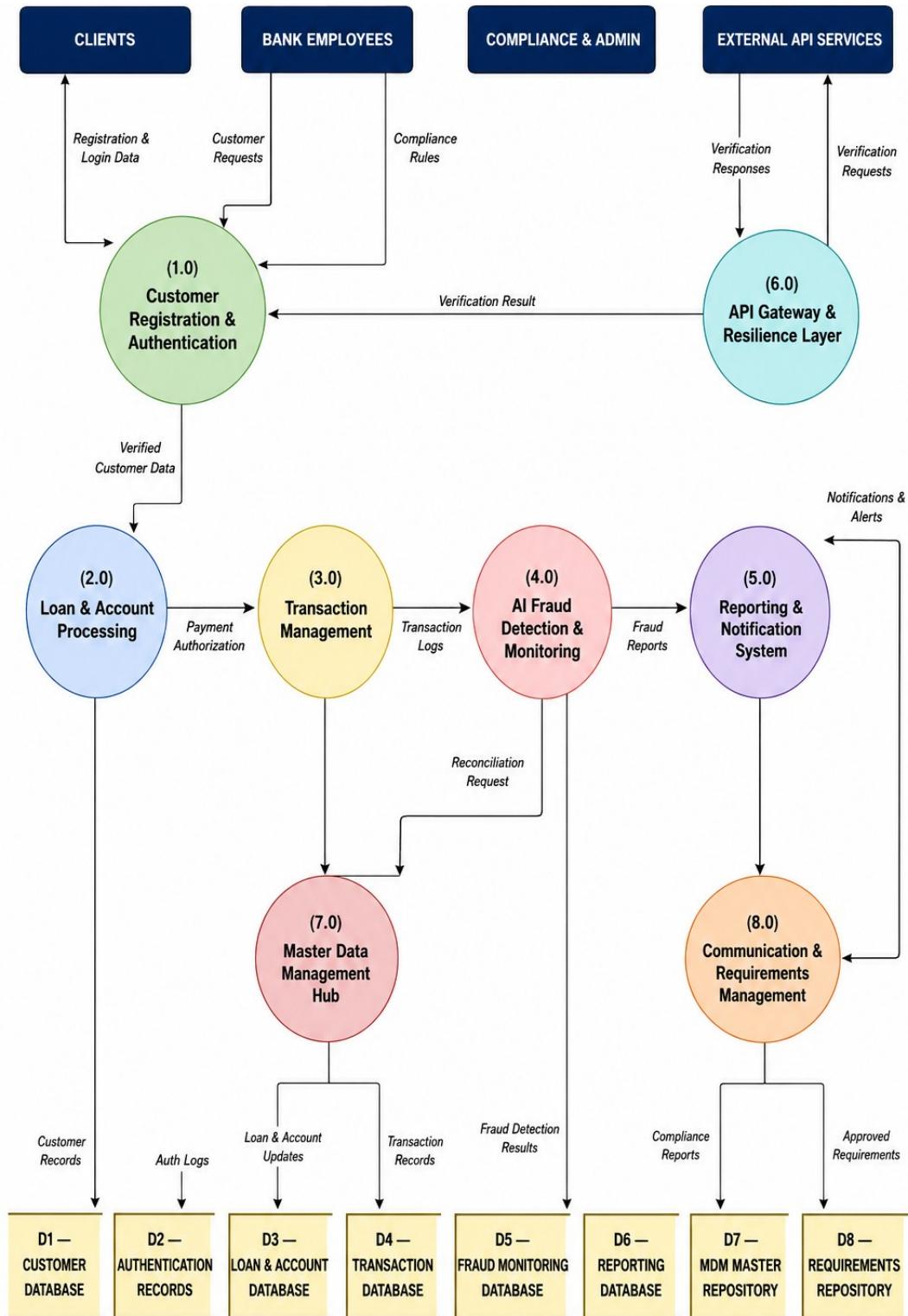


Figure 5.1: DFD – Complete Integrated Operational Excellence System for Bluestep Bank

Figure 5.1 presents the top-level Data Flow Diagram for BlueStep Bank's proposed system, encompassing eight core processes and ten data stores that together resolve all four identified operational problems. The diagram maps the complete flow of information between four categories of external entities — Clients, Bank Employees, Compliance & Admin, and External API Services — and the processing nodes that manage banking operations, fraud surveillance, data governance, and communication management.

Major System Enhancements over the Existing System:

- **API Gateway & Resilience Layer (Process 6.0):** A new process that intercepts all communication between BlueStep Bank's internal systems and External API Services. Unlike the existing architecture where APIs were called directly without error handling, Process 6.0 implements circuit breaker patterns, retry logic with exponential backoff, and real-time health monitoring. This eliminates the fragile point-to-point connections that caused 23 annual service disruptions in the existing system.
- **AI Fraud Detection & Monitoring (Process 4.0):** A significantly enhanced version of the existing rule-based fraud monitoring, now incorporating machine learning-based behavioral analytics and real-time transaction scoring. The process receives transaction logs from Transaction Management (3.0), applies anomaly detection algorithms, and outputs fraud reports to both the Reporting System (5.0) and the MDM Hub (7.0) for data quality correlation.
- **Master Data Management Hub (Process 7.0):** An entirely new process that resolves the data integrity fragmentation identified in Problem 04. Process 7.0 consolidates records from all transactional data stores (D1–D5) into a single authoritative master repository (D8), performing automated reconciliation, duplicate elimination, and data quality scoring. This replaces the manual overnight reconciliation processes that generated BDT 2.3 crores in annual discrepancies.
- **Communication & Requirements Management (Process 8.0):** A new process supporting System-I's communication governance and requirements management framework. Process 8.0 receives requirements submissions from Bank Employees and policy updates from Compliance & Admin, routes them through a structured BRD workflow, and stores approved requirements in the Requirements Repository (D10) with full traceability.

Enhanced Data Architecture:

- **D8 — MDM Master Repository:** Single authoritative source for all master customer, account, and product data, eliminating the cross-system inconsistencies that previously generated reconciliation errors.
- **D9 — API Health & Monitoring Logs:** Real-time record of API circuit breaker events, health check outcomes, and incident timelines, enabling proactive monitoring and post-incident analysis.

- D10 — Requirements Repository: Structured storage for Business Requirements Documents, traceability matrices, and stakeholder sign-off records, replacing the unstructured email-based requirements capture that caused 55% of projects to require rework.

Process Flow Summary:

The complete system flow begins when Clients submit registration data, login credentials, and KYC documents to the Customer Registration & Authentication process (1.0). This process interfaces with the API Gateway (6.0), which validates identity and credit data from External API Services through resilient, circuit-breaker-protected connections. Authenticated customers are passed as Verified Customer Data to Loan & Account Processing (2.0), which manages account operations and issues Payment Authorizations to Transaction Management (3.0). Every transaction generates logs that flow to AI Fraud Detection (4.0), which applies real-time behavioral scoring before data is committed. Validated results flow to the Reporting & Notification System (5.0), which delivers Notifications and Transaction Alerts to Clients. In parallel, the MDM Hub (7.0) continuously reconciles all data stores, while the Communication & Requirements Management process (8.0) governs internal business workflows between Bank Employees, Compliance & Admin, and the IT development function.

Table: System Enhancement Comparison

Component	Existing System	Proposed System (System-II + System-I)
API Integration	Direct calls, no error handling, 23 disruptions/year	API Gateway (6.0) with circuit breakers-target < 5 disruptions/year
Fraud Detection	Rule-based, batch processing, reactive	AI-powered real-time behavioral analytics (Process 4.0), proactive
Data Management	Fragmented across 7 legacy systems, manual reconciliation	MDM Hub (7.0) with D8 master repository, automated reconciliation
Communication	Ad-hoc emails and messaging, no governance	Process 8.0 with structured BRD workflow and traceability (D10)
Requirements Management	Informal, 55% projects needed rework	Standardized BRD templates, mandatory sign-off gates, D10 repository
Incident Response	Average 4.2 hours per API incident	Automated alerts, target < 30 minutes resolution

5.2.2 API Gateway & Resilience Layer Subsystem DFD

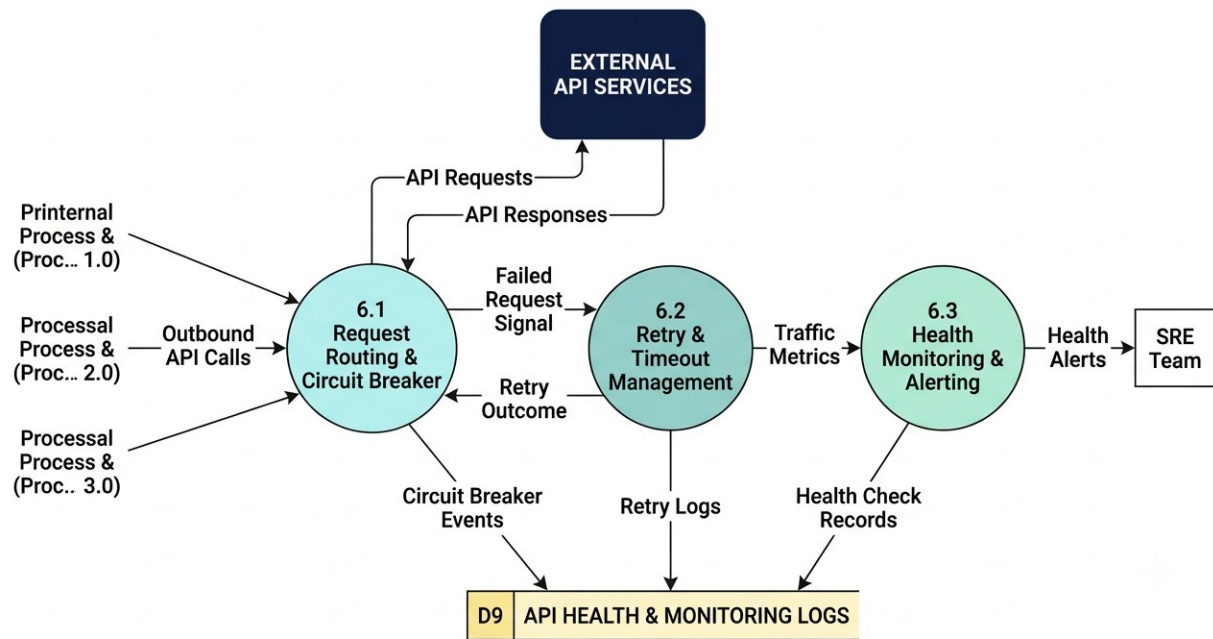


Figure 5.2: Level 1 DFD – API Gateway & Resilience Layer Subsystem (Process 6.0 Decomposition)

Figure 5.2 provides a detailed decomposition of the API Gateway & Resilience Layer (Process 6.0), illustrating how the three sub-processes within this subsystem work together to protect BlueStep Bank's internal operations from the fragility of external API dependencies. This subsystem directly addresses Problem 03 (External API Fragility), which caused 23 service disruptions and 4.2-hour average resolution times in the existing system.

Subsystem Process Analysis:

- **Request Routing & Circuit Breaker (6.1):** The primary gateway process that receives all outbound API calls from internal banking processes and applies circuit breaker logic before forwarding requests to External API Services. When an external API exceeds error rate thresholds (configurable, default: 50% failures in 30-second window), Process 6.1 opens the circuit and immediately triggers fallback procedures rather than allowing cascading failures. Outputs include routed API requests to External APIs and Circuit Breaker Events to the API Health & Monitoring Logs (D9).
- **Retry & Timeout Management (6.2):** Manages the retry lifecycle for failed API calls using exponential backoff algorithms. When an API call fails, Process 6.2 schedules retries at increasing intervals (1s, 2s, 4s, 8s) up to a configurable maximum attempt count before escalating to fallback. This prevents the thundering herd problem that amplified outages in the existing system. Inputs include failed request signals from 6.1; outputs include retry outcomes logged to D9 and verified responses returned to the calling process.

- **Health Monitoring & Alerting (6.3):** Continuously monitors the health of all registered external API endpoints through scheduled ping checks and passive monitoring of traffic patterns. When health metrics cross alert thresholds, Process 6.3 triggers notifications to the SRE team and updates the API Health Dashboard. This enables proactive intervention before circuit breakers are forced open. All health metrics, incident timelines, and alert histories are persisted to D9 — API Health & Monitoring Logs — providing the audit trail required for SLA reporting and post-incident review.

Data Flow Improvements vs. Existing System:

- **From Direct Coupling to Mediated Gateway:** The existing system called External API Services directly from processes 1.0, 2.0, and 3.0, creating brittle point-to-point dependencies. The proposed subsystem centralizes all external communication through Process 6.0, eliminating distributed failure points.
- **From Reactive to Proactive Incident Management:** Health Monitoring (6.3) detects degradation before it becomes an outage, enabling the SRE team to act preemptively rather than reactively.
- **From Silent Failure to Graceful Degradation:** Circuit breaker logic ensures that when external services are unavailable, banking operations continue with appropriate fallback responses rather than propagating errors to customer-facing interfaces.

5.2.3 AI Fraud Detection & Data Validation Subsystem DFD

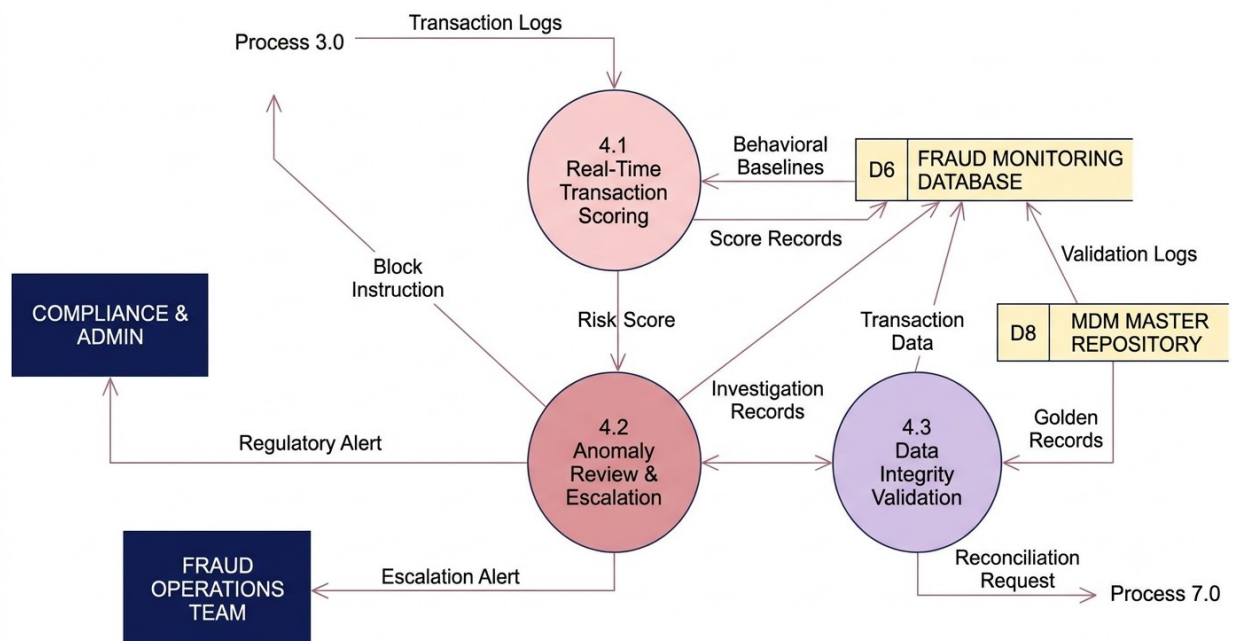


Figure 5.3: Level 1 DFD – AI Fraud Detection & Data Validation Subsystem (Process 4.0 Decomposition)

Figure 5.3 decomposes the AI Fraud Detection & Data Validation process (4.0) into its three constituent sub-processes, revealing the layered approach BlueStep Bank's proposed system takes to fraud prevention and data integrity assurance. This subsystem simultaneously addresses Problem 04 (Data Integrity and Fraud Risks) and provides supporting intelligence to the MDM Hub (7.0) for data governance decision-making.

Subsystem Process Analysis:

- **Real-Time Transaction Scoring (4.1):** Applies the machine learning-based behavioral analytics engine to every transaction within milliseconds of its initiation. The scoring model evaluates transaction velocity, geographic patterns, device fingerprints, amount deviations from behavioral baselines, and counterparty risk signals. Each transaction receives a fraud probability score and a risk category (Low / Medium / High / Critical). Inputs: Transaction Logs from Transaction Management (3.0) and Behavioral Baselines from Fraud Monitoring Database (D6). Outputs: Risk Scores to Anomaly Review (4.2) and score records to D6.
- **Anomaly Review & Escalation (4.2):** Processes transactions flagged as Medium or High risk by the scoring engine. Low-confidence High-risk cases are queued for manual review by the Fraud Operations team, while clearly Critical-risk transactions are automatically blocked and escalated. A 90-day parallel shadow operation period (running alongside the legacy rule-based system during initial deployment) ensures the model is calibrated before full autonomous operation, preventing the false positives that could disrupt legitimate customer transactions. Outputs: Escalation alerts to Compliance & Admin, block instructions to Transaction Management (3.0), and investigation records to D6.
- **Data Integrity Validation (4.3):** Runs validation rules against data received from Transaction Management (3.0) and cross-references it against the MDM Master Repository (D8) to detect inconsistencies between transactional records and master data. Discrepancies — such as transaction amounts that create account balance anomalies not reflected in D4 (Account Database), or beneficiary account numbers absent from D1 (Customer Database) — are flagged and routed to the MDM Hub (7.0) as Reconciliation Requests. This creates a continuous data quality feedback loop between fraud detection and master data governance.

5.2.4 Master Data Management Hub Subsystem DFD

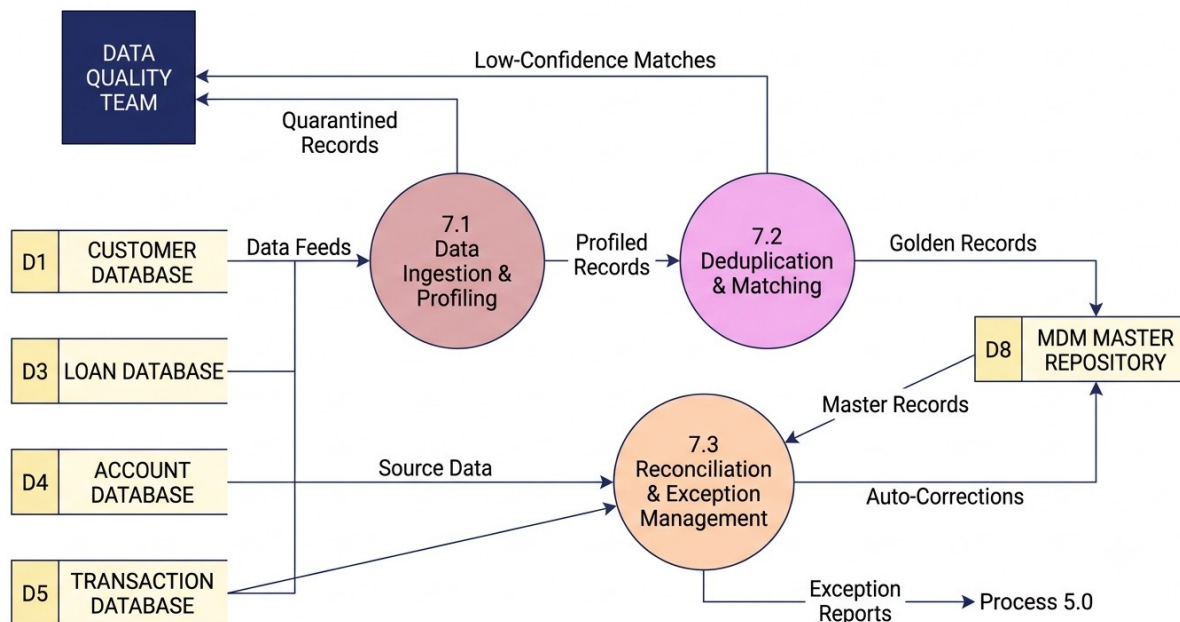


Figure 5.4: Level 1 DFD – Master Data Management Hub Subsystem (Process 7.0 Decomposition)

Figure 5.4 presents the decomposition of the Master Data Management Hub (Process 7.0), the entirely new process that resolves the data fragmentation at the heart of Problem 04. This subsystem consolidates BlueStep Bank's scattered data landscape — previously spread across seven legacy systems with no authoritative master — into a unified, governed data foundation.

Subsystem Process Analysis:

- **Data Ingestion & Profiling (7.1):** Receives data feeds from all transactional data stores (D1–D5) on a scheduled and event-driven basis. For each incoming record batch, Process 7.1 performs automated data profiling — assessing completeness, format conformance, referential integrity, and freshness. Profiling results are scored against pre-defined data quality thresholds; records failing minimum quality scores are quarantined for exception handling rather than propagated to the master repository.
- **Deduplication & Matching (7.2):** Applies deterministic and probabilistic matching algorithms to identify duplicate records across data stores. Customer records spread across D1, D3, and D4 with slight variations in name spelling, address format, or NID number are resolved into single Golden Records. Matching confidence scores above a configurable threshold are auto-merged; lower-confidence matches are routed to the Data Quality Management team for manual review. Matched and merged records are written to D8 — MDM Master Repository — as the single authoritative source.
- **Reconciliation & Exception Management (7.3):** Continuously compares transactional data in D1–D5 against the golden records in D8, identifying discrepancies introduced by batch

processing delays, failed synchronizations, or data entry errors. Discrepancies are categorized by severity and business impact, and exception reports are generated for Operations and Risk teams via the Reporting System (5.0). This automated reconciliation replaces the manual overnight process that previously generated BDT 2.3 crore in annual discrepancies.

Table: MDM Enhancement Comparison

Aspect	Existing System	Proposed MDM Hub (System-II)
Data Sources	7 fragmented, siloed data stores	7 source stores unified via MDM Hub (7.0)
Master Record	None — no authoritative source	D8 MDM Master Repository with Golden Records
Reconciliation	Manual overnight batch, error-prone	Automated continuous reconciliation (Process 7.3)
Duplicate Detection	None — duplicates proliferate across stores	Probabilistic matching engine (Process 7.2)
Data Quality Monitoring	No systematic monitoring	Automated profiling and quality scoring (Process 7.1)
Annual Discrepancies	BDT 2.3 cores (audit-confirmed)	Target < BDT 20 lacs with automated reconciliation
Exception Handling	Manual, undocumented, reactive	Automated exception queue with SLA-tracked resolution

5.3 Database Design

Database design forms the critical foundation of BlueStep Bank's Integrated Operational Excellence System, providing the persistent storage infrastructure that enables efficient data access, transactional integrity, regulatory compliance, and analytical decision-making. A well-structured database architecture ensures that the system can maintain data consistency across BlueStep Bank's complex multi-process banking environment, support concurrent access by hundreds of bank staff, satisfy regulatory audit requirements, and scale to accommodate the bank's projected customer growth.

The database architecture for the proposed system addresses the specific requirements identified through the feasibility analysis and root cause investigation:

- **API Resilience Support:** Structured storage for API health metrics, circuit breaker events, and incident timelines enabling the SRE team to manage and analyze API reliability systematically.
- **Fraud Detection Intelligence:** Comprehensive fraud monitoring tables with behavioral baseline data, transaction risk scores, investigation records, and regulatory reporting audit trails.

- **Master Data Governance:** Central MDM repository tables with Golden Record management, data quality scores, deduplication match histories, and reconciliation exception logs.
- **Requirements Management:** Structured tables for Business Requirements Documents, stakeholder sign-off records, traceability matrices, and communication governance logs.
- **System Integration:** Foreign key relationships ensuring referential integrity across all subsystems, with cascading update rules aligned with Bangladesh Bank's data retention requirements.

The database architecture is organized into two primary subsystems:

- **Banking Operations Subsystem:** Manages customer data, authentication records, loan and account operations, transaction processing, and API health monitoring — the operational backbone of BlueStep Bank's day-to-day banking activities.
- **Fraud & Compliance Subsystem:** Maintains fraud detection intelligence, MDM master records, regulatory reporting data, and requirements governance records — the risk management and data quality backbone supporting compliance and strategic decision-making.

All tables are designed following Third Normal Form (3NF) to eliminate update anomalies and redundancy. Integrity constraints (NOT NULL, UNIQUE, CHECK, FOREIGN KEY) are systematically applied. Indexing strategies prioritize high-frequency query patterns identified in the functional requirements, particularly for transaction fraud scoring (requiring sub-millisecond lookups) and MDM reconciliation (requiring full-table comparison across millions of records).

5.3.1 ER Diagram

The Entity-Relationship (ER) diagrams provide the conceptual framework for the database architecture, illustrating entities as rectangles, attributes as ovals, and relationships as connecting lines with cardinality indicators (1, M, N). The ER model has been divided into two interconnected subsystems to maintain conceptual clarity while ensuring that cross-subsystem references are explicitly modelled through foreign key relationships.

5.3.1.1 Banking Operations Subsystem

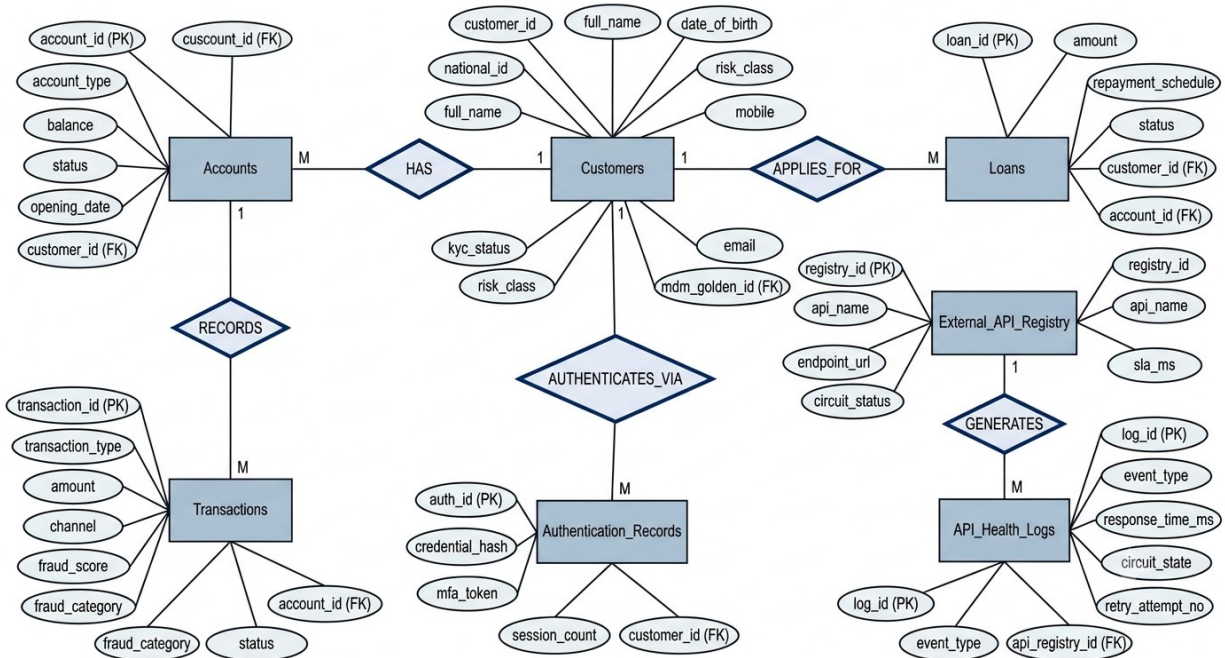


Figure 5.5: ER Diagram: Banking Operations Subsystem

The ER diagram for the Banking Operations Subsystem, which serves as the core transactional data architecture supporting BlueStep Bank's customer-facing banking services, authentication management, loan and account processing, and API resilience monitoring.

Key Entities and Relationships:

- **Customers (Central Entity):** The primary reference entity storing comprehensive customer profiles including personal details, KYC verification status, risk classification, and MDM Golden Record linkage. Serves as the root anchor for all customer-linked transactional entities. One-to-many relationships with Accounts, Loans, and Transactions.
- **Accounts:** Stores account details including account type, balance, status, and opening date. Linked to Customers through a many-to-one relationship (one customer may hold multiple accounts). Also links to Transactions through a one-to-many relationship, as each account may have an extensive transaction history.
- **Loans:** Records loan applications, disbursement amounts, repayment schedules, and current status. Maintains a many-to-one relationship with Customers. Linked to Accounts through a one-to-one relationship representing the disbursement account.
- **Transactions:** The highest-volume entity capturing every financial movement including transfer details, timestamps, channel identifiers, and fraud risk scores assigned by Process 4.1. One-to-many with Accounts; references Fraud_Alerts through a one-to-many relationship.

- **Authentication_Records:** Stores login credentials, multi-factor authentication tokens, session histories, and access logs. Implements a one-to-one relationship with Customers for credential ownership, and a one-to-many relationship for session logs (each customer may have multiple sessions across time).
- **API_Health_Logs:** Stores circuit breaker events, API endpoint health check results, retry attempt records, and incident timelines. Links to External_API_Registry through a many-to-one relationship, as each registered external API may accumulate many health log entries over time.
- **External_API_Registry:** Reference table for all external API integrations (identity verification providers, credit bureaus, payment gateways, card networks). Includes configuration parameters, SLA definitions, and current circuit breaker status flags.

5.3.1.2 Fraud & Compliance Subsystem

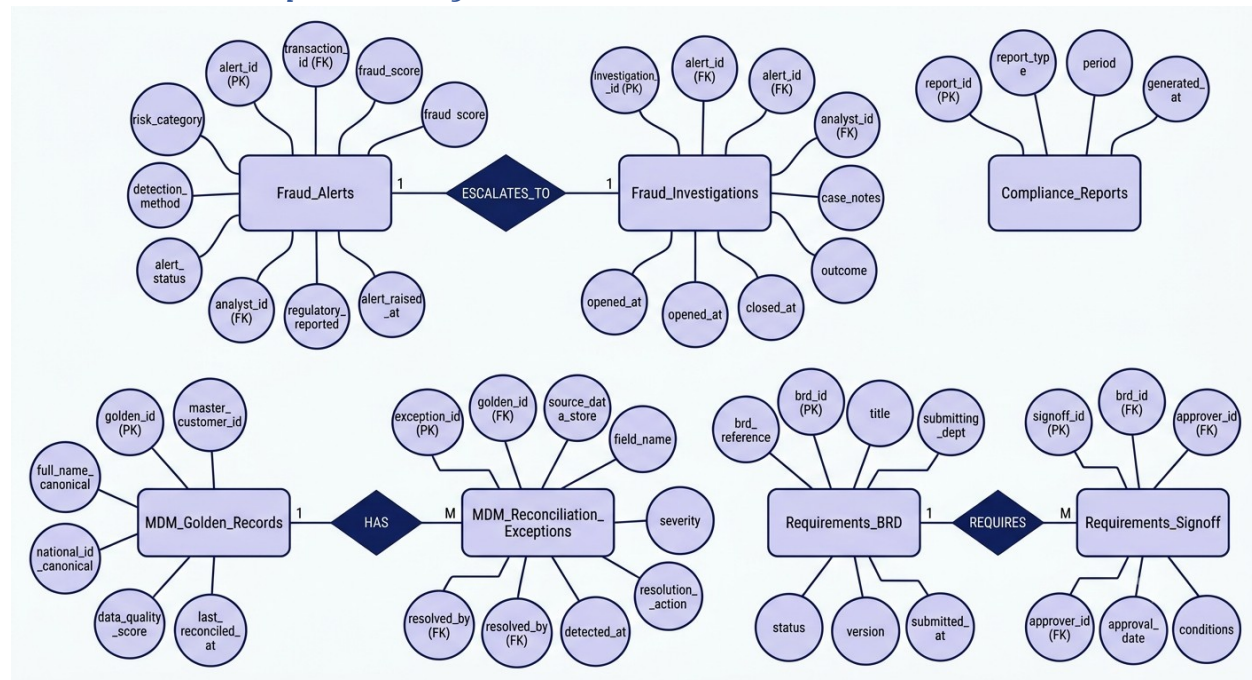


Figure 5.6: ER Diagram: Fraud & Compliance Subsystem

The ER diagram for the Fraud & Compliance Subsystem, managing BlueStep Bank's fraud intelligence, master data governance records, regulatory compliance documentation, and communication & requirements management data — supporting the risk management and strategic governance functions of the proposed system.

Key Entities and Relationships:

- **Fraud_Alerts (Central Entity):** Records every fraud flag raised by the AI scoring engine, including transaction reference, risk score, risk category, detection timestamp, investigation status, and resolution outcome. Many-to-one relationship with Transactions (from Banking Operations Subsystem via FK); one-to-many with Fraud_Investigations.

- **Fraud_Investigations:** Stores case management records for fraud alerts escalated for manual review, including assigned analyst, investigation timeline, evidence references, and regulatory reporting obligations. Links to **Fraud_Alerts** one-to-one for each escalated case.
- **MDM_Golden_Records:** The authoritative master customer and account records generated by the MDM Hub deduplication process. Each Golden Record stores a master customer profile, merge history, data quality score, and last reconciliation timestamp. One-to-one with **Customers** (Banking Operations Subsystem); one-to-many with **MDM_Reconciliation_Exceptions**.
- **MDM_Reconciliation_Exceptions:** Logs all data discrepancies detected during automated reconciliation, including the source data store, field-level conflict details, resolution actions taken, and sign-off records. Enables regulatory audit of data governance activities and performance tracking of MDM effectiveness.
- **Compliance_Reports:** Stores generated regulatory reports including Bangladesh Bank submissions, audit trail documents, and internal governance reports. References **Fraud_Alerts** through a many-to-many relationship via a **Compliance_Report_Items** junction table.
- **Requirements_BRD:** Stores Business Requirements Documents submitted through Process 8.0, including requirement text, submitting department, business owner, status (Draft / Under Review / Approved / Rejected), and traceability links to development deliverables.
- **Requirements_Signoff:** Records stakeholder approval workflows for each BRD, capturing sign-off authority, approval date, conditions or amendments, and version history. One-to-many with **Requirements_BRD**, enabling multi-stakeholder approval chains.

5.3.2 Database Tables and Data Specifications

Following normalization of the ER diagrams, the database structure is implemented through relational tables with explicitly defined primary keys (PK), foreign keys (FK), data types, size constraints, and integrity rules. The specifications below cover the eight most operationally critical tables, four from each subsystem, providing full field-level detail to guide the technical implementation team.

5.3.2.1 Banking Operations Tables

Table: Customers Table: Data Types and Specifications

Field Name	Key	Data Type	Size / Constraint	Description
customer_id	PK	INT	NOT NULL, AUTO_INCREMENT	Unique customer identifier
national_id		VARCHAR	20, UNIQUE, NOT NULL	Bangladesh National ID (NID) number
full_name		VARCHAR	150, NOT NULL	Customer's full legal name
date_of_birth		DATE	NOT NULL	Date of birth for

				KYC age verification
kyc_status		ENUM	'Pending','Verified','Rejected', NOT NULL	KYC verification status from Process 1.0
risk_class		ENUM	'Low','Medium','High', DEFAULT 'Low'	Customer risk classification for fraud scoring
mobile		VARCHAR	20, NOT NULL	Primary contact number
email		VARCHAR	100, UNIQUE	Email address for notifications
mdm_golden_id	FK	INT	REFERENCES MDM_Golden_Records	Link to authoritative MDM master record
created_at		TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation datetime
last_updated		TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Last modification datetime

Table: Transactions Table: Data Types and Specifications

Field Name	Key	Data Type	Size / Constraint	Description
transaction_id	PK	BIGINT	NOT NULL, AUTO_INCREMENT	Unique transaction reference (high volume)
account_id	FK	INT	REFERENCES Accounts, NOT NULL	Source account identifier
transaction_type		ENUM	'Credit','Debit','Transfer','Fee', NOT NULL	Type of financial movement
amount		DECIMAL	(18,2), NOT NULL, CHECK(amount > 0)	Transaction amount in BDT
beneficiary_account		VARCHAR	30	Destination account (for transfers)
channel		ENUM	'Branch','ATM','Mobile','Internet','API', NOT NULL	Banking channel used

fraud_score		DECIMAL	(5,4), DEFAULT NULL	AI risk score from Process 4.1 (0.0000–1.0000)
fraud_category		ENUM	'Low','Medium','High','Critical', DEFAULT 'Low'	Risk category assigned by Process 4.1
status		ENUM	'Pending','Completed','Blocked','Reversed', NOT NULL	Transaction processing status
transaction_ts		TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Transaction initiation timestamp

Table: API health logs table

Field Name	Key	Data Type	Size / Constraint	Description
log_id	PK	BIGINT	NOT NULL, AUTO_INCREMENT	Unique log entry identifier
api_registry_id	FK	INT	REFERENCES External_API_Registry, NOT NULL	Which external API this log entry relates to
event_type		ENUM	'HealthCheck','CircuitOpen','CircuitClose','RetryAttempt','Timeout', NOT NULL	Type of monitoring event recorded
response_time_ms		INT	CHECK(response_time_ms >= 0)	API response time in milliseconds
http_status_code		SMALLINT		HTTP status code returned (200, 500, 503, etc.)
circuit_state		ENUM	'Closed','Open','HalfOpen', NOT NULL	Circuit breaker state at time of

retry_attempt_no		TINYINT	DEFAULT 0	event Retry attempt number (0 = first call)
error_message		TEXT	DEFAULT NULL	Error detail for failed calls
resolved_at		TIMESTAMP	DEFAULT NULL	Timestamp when incident was resolved
logged_at		TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Log entry creation timestamp

Fraud & Compliance Tables

Table: Fraud Alert Table

Field Name	Key	Data Type	Size / Constraint	Description
alert_id	PK	INT	NOT NULL, AUTO_INCREMENT	Unique fraud alert identifier
transaction_id	FK	BIGINT	REFERENCES Transactions, NOT NULL	Transaction that triggered the alert
fraud_score		DECIMAL	(5,4), NOT NULL	AI risk score at time of alert
risk_category		ENUM	'Medium','High','Critical', NOT NULL	Category that triggered alert escalation
detection_method		ENUM	'ML_Model','Rule_Based','Manual_Review', NOT NULL	How the fraud signal was identified
alert_status		ENUM	'New','Under_Review','Confirmed_Fraud','False_Pos	Current

			itive','Closed', NOT NULL	investigation status
analyst_id	FK	INT	REFERENCES Bank_Staff, DEFAULT NULL	Assigned investigator (if escalated)
regulatory_reported		BOOLEAN	DEFAULT FALSE	Whether alert has been reported to regulator
alert_raised_at		TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Datetime alert was raised by Process 4.1
resolved_at		TIMESTAMP	DEFAULT NULL	Datetime investigation was closed

Table: MDM_Golden_Record table

Field Name	Key	Data Type	Size / Constraint	Description
golden_id	PK	INT	NOT NULL, AUTO_INCREMENT	Unique Golden Record identifier
master_customer_id		VARCHAR	30, UNIQUE, NOT NULL	Canonical customer ID across all source systems
full_name_canonical		VARCHAR	150, NOT NULL	Authoritative standardized customer name
national_id_canonical		VARCHAR	20, UNIQUE, NOT NULL	Authoritative NID number after deduplication
source_system_count		TINYINT	NOT NULL, DEFAULT 1	Number of source systems contributing to record
merge_history		JSON	DEFAULT NULL	JSON array of merged source record IDs and systems
data_quality_score		DECIMAL	(4,2),	Composite data

			CHECK(data_quality_score BETWEEN 0 AND 100)	quality score (0 –100)
last_reconciled_at		TIMESTAMP	NOT NULL	Last time record was reconciled against source systems
created_at		TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Golden Record creation datetime
last_updated		TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Last modification by MDM Hub Process 7.0

Table: Requirements_BRD table

Field Name	Key	Data Type	Size / Constraint	Description
brd_id	PK	INT	NOT NULL, AUTO_INCREMENT	Unique BRD identifier
brd_reference		VARCHAR	20, UNIQUE, NOT NULL	Human-readable reference code (e.g. BRD-2025-047)
title		VARCHAR	200, NOT NULL	Brief descriptive title of the requirement
submitting_dept		VARCHAR	100, NOT NULL	Business unit that submitted the requirement
business_owner	FK	INT	REFERENCES Bank_Staff, NOT NULL	Accountable business owner for sign-off
requirement_text		TEXT	NOT NULL	Full requirement description
priority		ENUM	'Low','Medium','High','Critical', NOT NULL	Business

				priority classification
status		ENUM	'Draft','Under_Review','Approved','Rejected','Delivered', NOT NULL	Current workflow status
version		TINYINT	NOT NULL, DEFAULT 1	Version number (increments on each revision)
submitted_at		TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Initial submission datetime
approved_at		TIMESTAMP	DEFAULT NULL	Final approval datetime after all sign-offs

Table: MDM_Reconciliation_Exceptions table

exception_id	PK	INT	NOT NULL, AUTO_INCREMENT	Unique exception record identifier
golden_id	FK	INT	REFERENCES MDM_Golden_Records, NOT NULL	Golden Record involved in discrepancy
source_data_store		ENUM	'D1_Customer','D2_Auth','D3_Loan','D4_Account','D5_Transaction', NOT NULL	Source data store where discrepancy was found
field_name		VARCHAR	100, NOT NULL	Database field where conflict was detected
source_value		TEXT		Conflicting value in

				source data store
severity		ENUM	'Low','Medium','High','Critical', NOT NULL	Business impact severity of the discrepancy
resolution_action		ENUM	'Auto_Corrected','Manual_Review','Escalated','Pending', NOT NULL	How the exception was or is being handled
resolved_by	FK	INT	REFERENCES Bank_Staff, DEFAULT NULL	Staff member who resolved (if manual)
detected_at		TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the reconciliation engine detected the issue
resolved_at		TIMESTAMP	DEFAULT NULL	When exception was fully resolved
source_value		TEXT		Conflicting value in source data store

5.4 Form Design

Form design specifies the user interface specifications for the key operational forms through which BlueStep Bank staff interact with the proposed system. Well-designed forms reduce data entry errors, guide users through complex workflows, enforce business rules at the point of data capture, and create the structured input records that feed the database tables specified in Section 4.3. Each form design below includes purpose, user role, input fields, validation rules, and integration with the corresponding database table and process node.

All forms follow BlueStep Bank's corporate design standards: navy blue (#1F3864) header bars with white text, mandatory fields marked with an asterisk (*), inline validation error messages in red, and a two-column layout for desktop interfaces with single-column fallback for mobile banking terminals. Forms are accessible via the unified web-based staff portal introduced as part of the Communication & Requirements Management system (Process 8.0).

5.4.1 Customer KYC & Registration Form

Customer KYC & Registration

Full Legal Name

National ID *

Date of Birth *

Mobile Number *

Email

KYC Document Type *

KYC Document Upload *

Residential Address *

datetime input

Select...

File upload

Submit for Verification

Cancel

Figure 5.7: Customer KYC & Registration Form

Purpose: Collects and validates all customer registration and KYC data required to create a verified Customer record in D1 (Customer Database) and initiate the identity verification flow through the API Gateway (Process 6.0). Used by Retail Banking staff and, for digital onboarding, by Clients directly through the mobile banking interface.

Form Fields and Specifications:

Field Label	Field Type	Validation Rule	Linked DB Field
Full Legal Name *	Text (max 150)	NOT NULL; alphabetic only	full_name
National ID (NID) *	Text (max 20)	NOT NULL; UNIQUE; format: 10 or 17 digits	national_id
Date of Birth *	Date Picker	NOT NULL; age ≥ 18 years	date_of_birth
Mobile Number *	Tel	NOT NULL; format: +880XXXXXXXXXX	mobile
Email Address	Email	UNIQUE; valid email format	email
KYC Document Type *	Dropdown	Options: NID / Passport / Driver Licence	kyc_doc_type
KYC Document Upload *	File Upload	PDF or JPEG; max 5MB; NOT NULL	kyc_document_path
Residential Address *	Textarea (max 300)	NOT NULL	address
Risk Classification	Auto-assigned	System-generated from API verification result	risk_class
Submit for Verification	Button	Triggers Process 6.0	—

		API Gateway verification call	
--	--	----------------------------------	--

5.4.2 API Incident Report Form

API Incident Report

Affected API Service *

Select... ▼

Incident Type *

Select... ▼

Circuit Breaker State *

☒ Closed
 ☐ Open
 ☐ Half-Open

HTTP Status Code

Response Time (ms)

Error Message

Retry Attempt Number

Resolution Status

Select... ▼

Resolved At

datetime input 

Log Incident

Cancel

Figure 5.8: API Incident Report Form

Purpose: Enables the Site Reliability Engineering (SRE) team to log, categorize, and track API-related incidents that require manual intervention beyond the automated circuit breaker and retry mechanisms of Process 6.0. Creates a structured incident record in API_Health_Logs (D9) and triggers the appropriate escalation workflow.

Field Label	Field Type	Validation Rule	Linked DB Field
Affected API Service *	Dropdown	References External_API_Registry; NOT NULL	api_registry_id
Incident Type *	Dropdown	Options: Timeout / HTTP Error / Auth Failure / Data Format / Outage	event_type
Circuit Breaker State *	Radio	Options: Closed / Open / Half-Open; NOT NULL	circuit_state
HTTP Status Code	Number	Range: 100–599	http_status_code
Response Time (ms)	Number	≥ 0	response_time_ms
Error Message	Textarea	Max 2000 characters	error_message
Retry Attempt Number	Number	Range: 0–10; DEFAULT 0	retry_attempt_no
Resolution Status	Dropdown	Options: Ongoing /	—

		Monitoring / Resolved	
Resolved At	Datetime Picker	Must be $\geq$ incident start time	resolved_at
SRE Engineer *	Auto-populated	Logged-in user identity; NOT NULL	—

5.4.3 Fraud Alert Review Form

Fraud Alert Review

Alert ID	<i>Read-only</i>
Transaction Reference	<i>Read-only</i>
Fraud Score	<i>Read-only</i>
Risk Category	<i>Read-only</i>
Detection Method	<i>Read-only</i>
Analyst Notes *	
Investigation Outcome *	Select... ▼
Regulatory Reporting Required	☐
Customer Contact Initiated	☐

Submit Review
Cancel

Figure 5.9: Fraud Alert Review Form

Purpose: Supports Fraud Operations analysts in reviewing AI-generated fraud alerts, documenting investigation findings, determining outcomes (confirmed fraud vs. false positive), and initiating regulatory reporting where required. Updates the Fraud_Alerts table and, where applicable, creates a linked Fraud_Investigations record.

Field Label	Field Type	Validation Rule	Linked DB Field
Alert ID (auto-filled)	Read-only	System-populated from alert queue	alert_id
Transaction Reference	Read-only	Linked from Transactions table	transaction_id
Fraud Score (auto-filled)	Read-only	AI model output (0.0000–1.0000)	fraud_score
Risk Category (auto-filled)	Read-only	Medium / High / Critical	risk_category
Detection Method (auto)	Read-only	ML_Model / Rule_Based / Manual_Review	detection_method

Analyst Notes *	Textarea	NOT NULL; min 50 characters for escalation	—
Investigation Outcome *	Dropdown	Options: Confirmed_Fraud / False_Positive / Needs_More_Info; NOT NULL	alert_status
Regulatory Reporting Required	Checkbox	Mandatory check for Confirmed_Fraud cases	regulatory_reported
Customer Contact Initiated	Checkbox	Log whether customer was notified	—
Case Closed By *	Auto-populated	Logged-in analyst identity	analyst_id

5.4.4 MDM Data Quality Exception Form

MDM Data Quality Exception

Source Data Store

Read-only

Conflicting Field

Read-only

MDM Value

Read-only

Source System Value

Read-only

Severity

Read-only

Resolution Action *

Select... ▼

Resolution Notes

Resolve Exception

Cancel

Figure 5.10: MDM Data Quality Exception Form

Purpose: Enables the Data Quality Management team to review, investigate, and resolve data discrepancies identified by the MDM Hub's automated reconciliation engine (Process 7.3). Creates a structured exception record in MDM_Reconciliation_Exceptions and tracks resolution progress through to closure.

Field Label	Field Type	Validation Rule	Linked DB Field
Exception ID (auto)	Read-only	System-generated	exception_id
Golden Record ID	Read-only	Auto-linked from	golden_id

		MDM Hub	
Source Data Store *	Read-only	Auto-populated by Process 7.3	source_data_store
Conflicting Field	Read-only	Field name where discrepancy was found	field_name
MDM Value	Read-only	Current value in Golden Record	golden_value
Source System Value	Read-only	Conflicting value in source store	source_value
Severity (auto)	Read-only	Low / Medium / High / Critical from system	severity
Resolution Action *	Dropdown	Auto_Corrected / Manual_Review / Escalated / Pending; NOT NULL	resolution_action
Resolution Notes	Textarea	Required when Resolution Action = Manual_Review or Escalated	—
Resolved By	Auto-populated	Logged-in data steward	resolved_by

5.4.5 Business Requirements Document Submission Form

Business Requirement Document (BRD) Submission

BRD Reference

Read-only

Requirement Title *

Submitting Department *

Select... ▼

Business Owner *

Search... 🔍

Priority *

☒ Low
 ☐ Medium
 ☐ High
 ☐ Critical

Requirement Description *

B I U S ≡ ≡ ≡ ≡ • ≡

Business Justification *

Affected Systems

☐ List System 1
☐ List System 2
☐ Affected System 3

Proposed Delivery Date

04/11/2024 📅

Supporting Documents

📎 Upload Document

Submit BRD

Cancel

Figure 5.11: BRD Submission Form

Purpose: Enables Bank Employees across all departments to submit formal system change requests and new feature requirements through the Communication & Requirements Management system (Process 8.0). Creates a structured BRD record in the Requirements_BRD table, initiates the business owner review workflow, and automatically generates a BRD reference number for traceability.

Field Label	Field Type	Validation Rule	Linked DB Field
BRD Reference (auto)	Read-only	System-generated (BRD-YYYY-NNN format)	brd_reference
Requirement Title *	Text (max 200)	NOT NULL; min 10 characters	title
Submitting Department *	Dropdown	From department registry; NOT NULL	submitting_dept
Business Owner *	Searchable Dropdown	From Bank_Staff registry; NOT NULL	business_owner
Priority *	Radio	Low / Medium / High / Critical; NOT NULL	priority
Requirement Description *	Rich Text	NOT NULL; min 150 characters	requirement_text
Business Justification *	Textarea	NOT NULL; must reference at least one business problem	—
Affected Systems	Multi-select Checkboxes	Optional; flags relevant technical teams	—
Proposed Delivery Date	Date Picker	Must be $\geq$ 30 days from submission date	—
Supporting Documents	File Upload	PDF / XLSX / DOCX; max 10MB per file; up to 3 files	—

5.4.6 Form Design Integration Summary

The five forms presented in this section collectively cover all critical data entry touchpoints in BlueStep Bank's Integrated Operational Excellence System. Each form is purpose-built to its user role and process context — minimizing unnecessary fields visible to each user type, maximizing

data quality through inline validation, and creating the structured, auditable records that the system's database and analytics capabilities depend upon.

Table: Form Design Integration Summary

Form	Primary User	Linked Process	Primary DB Table
Customer KYC & Registration	Retail Staff / Clients	Process 1.0, 6.0	Customers (D1)
API Incident Report	SRE Engineers	Process 6.0	API_Health_Logs (D9)
Fraud Alert Review	Fraud Analysts	Process 4.0	Fraud_Alerts (D6)
MDM Data Quality Exception	Data Stewards	Process 7.0	MDM_Reconciliation_Exceptions (D8)
BRD Submission	Bank Employees (All Departments)	Process 8.0	Requirements_BRD (D10)

Together, these forms close the input/output loop for every major operational workflow in the proposed system. The Customer KYC form initiates the onboarding pipeline that flows through the API Gateway's identity verification. The API Incident Report enables the SRE team to complement automated monitoring with human-in-the-loop incident intelligence. The Fraud Alert Review form ensures that AI-generated fraud signals receive structured analyst adjudication before regulatory reporting decisions are made. The MDM Data Quality Exception form operationalizes the data governance culture change required to sustain the MDM Hub's effectiveness over time. The BRD Submission form transforms the informal, email-based requirements process that caused 55% project rework into a governed, traceable workflow that protects both the business stakeholders' intent and the IT team's delivery confidence.

4.5 Conclusion

The design presented in Chapter 5 translates the analytical conclusions and financial justifications of Chapter 4 into a comprehensive, implementable system architecture for BlueStep Bank's Integrated Operational Excellence System. The four design elements — Data Flow Diagrams, Database Design, and Form Design — work as an integrated whole: the DFDs define what information flows where and through which processes; the ER diagrams and database tables specify exactly how that information is structured and persisted; and the forms define the human-computer interaction layer through which bank staff interact with the system in their day-to-day operational roles.

The DFD hierarchy reveals a system designed around three core architectural principles. First, resilience by design: the API Gateway & Resilience Layer (Process 6.0) interposes circuit breaker protection between all internal banking processes and External API Services, eliminating the fragile direct coupling that caused 23 annual service disruptions. Second, intelligence over rules: the AI Fraud Detection & Data Validation subsystem (Process 4.0) replaces the reactive rule-based fraud monitoring with a real-time behavioral analytics engine capable of detecting novel fraud patterns that static rules miss. Third, authoritative data: the Master Data Management Hub (Process

7.0) consolidates BlueStep Bank's fragmented data landscape into a single governed master repository, replacing the manual reconciliation processes that generated BDT 2.3 crores in annual discrepancies with automated, continuous data quality management.

The database design implements these architectural principles in a normalized, integrity-constrained relational schema. The thirteen tables specified across the Banking Operations and Fraud & Compliance subsystems provide the persistent data foundation for every process in the DFD hierarchy, with carefully designed foreign key relationships ensuring referential integrity across subsystem boundaries. The JSON merge history field in MDM_Golden_Records and the regulatory_reported flag in Fraud_Alerts exemplify the design's attention to both operational practicality and regulatory compliance obligation.

The five form designs complete the system specification by defining the user interaction layer. Each form is purpose-calibrated to its user role and linked process, ensuring that data enters the system in a structured, validated form at the earliest possible point — preventing the data quality degradation that cascades when unvalidated inputs propagate through downstream processes. The BRD Submission form, in particular, operationalizes the cultural and process transformation required by System-I's communication governance mandate, giving the bank's business units and IT teams a shared, structured language for requirements communication that replaces the ambiguous email-based processes responsible for 55% of historical project rework.

In aggregate, the designs presented in this chapter demonstrate that the resolution of BlueStep Bank's four critical problems is not dependent on technology investment alone, but on a coherent, mutually reinforcing architecture in which resilient infrastructure, intelligent analytics, governed data, and structured human workflows are designed as an integrated system rather than isolated components. This integrated design philosophy positions BlueStep Bank to realize the financial returns projected in the cost-benefit analysis — break-even in 11–12 months, 5-year ROI of 318% — while building the operational and technical capabilities required for sustainable growth and regulatory leadership in Bangladesh's evolving digital banking landscape.